

SMITHIAN FOLD THEORY - CLASSICAL COMPUTATION BRANCH PAPER 001

After Turing: The Fold Machine

An Exact, Parameter-Free and Machine-Closed Derivation of Classical
Computational Science from Smithian Fold Theory

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Third clean-room reconstruction - Classical Computation inventory complete

113 admitted derivations - 28,928 generated candidates

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Paper: CC BY 4.0 - Code: Apache-2.0

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Abstract

This paper reports the completed Classical Computation branch of the third clean-room reconstruction of Smithian Fold Theory. From Foundation, Mathematics and Information Science receipts it derives Formal Computation, Computability, Complexity, Algorithms and mathematical data structures, Program Semantics, Concurrent and Distributed Computation, Cryptography and Security, Learning and Intelligence Theory, and Scientific Computation. The branch does not import a Turing machine, lambda calculus, conventional complexity class, probability cause, cryptographic hardness assumption, pretrained model, fitted parameter, floating proof value or application result as a premise. Historical models enter only after sealing as explicit correspondence tests.

The frozen branch contains 113 dependency-ordered claims and 28,928 generated candidate structures. Every candidate is decided, each grammar has exactly one survivor, and every claim passes minimality, named-shape uniqueness, depth-independent closure, four adverse controls, cryptographic sealing and implementation-distinct recomputation.

Inventory identity:

sha256: f469995345f2f050485cbdbe7ec1a63e2d69711a83da132586f7f6ff79b61df8.

Keywords: Smithian Fold Theory; classical computation; Turing machine; computability; complexity; algorithms; semantics; distributed systems; cryptography; learning theory; scientific computing; computational proof; open science

1. Central scientific claim and exact boundary

Within the frozen SFT V3 Classical Computation inventory, all 113 registered obligations have depth-independent, model-admitted and independently replicated receipts, and no registered classical-computation obligation remains unclassified or frontier.

Closure means the exact generated-finite kernels named in the inventory are complete. It does not claim every named theorem, arbitrary asymptotic lower bound, unrestricted Busy Beaver value, P versus NP separation, physical device behavior or application result. Quantum operations are excluded and derived in their own downstream branch.

2. Derivational constitution

Every dependency terminates at the single operational root theorem, *there is no nothing*. Admitted proof objects are structural One, exact positive parts, generated finite traces, Fold labels, words, relations, maps, graphs, information ledgers and composition records. Semantic numerical zero, negative quantity, irrational or imaginary values, floating proof arithmetic, a completed infinity and an ungenerated continuum are excluded. Empty One is a form; held complementary labels replace signed proof values.

Python booleans, indices, lengths, empty containers and process return codes are quarantined host mechanics. They execute and count evidence but carry no SFT mathematical authority. All registrations have empty axiom and free-parameter tuples. No application, benchmark, pretrained model or earlier SFT result selected a survivor.

3. The native Fold machine

The native machine begins with exact generated configurations, two held Fold labels and a source-bound transition relation. Blank support is empty One, not numerical zero. A word is a complete sequence of held labels at canonical positions. Reading is an observation relation; writing is exact held-label substitution; movement is a counted transition across generated positions. A configuration retains its word, held position, process state and proof trace. Every step records its premise, action, successor, resource use and lost or retained distinctions.

Universality is derived only after languages, automata, rewriting, recursion, binding, abstract machines, circuits, processes and composition have receipts. The universal process interprets every description inside the frozen machine grammar. Conventional Turing, Church and circuit models are comparison encodings and cannot select the machine law.

4. Superdeterminism, branching and randomized computation

Randomized computation closes without an uncaused random premise. The machine retains complete registered schedule support and executes one deterministic trace per held schedule. Apparent uncertainty is an observation-relative distinction ledger over that support. Exact success parts summarize the branch partition after execution. If a physical source supplies schedules, its distribution is measured at the one-way empirical boundary and cannot flow backward into the algorithm law.

5. Halting, incompleteness and famous limits

The universal description grammar supports self-description and a held complement operation. Assuming a total internal halting decider permits construction of a process that rejects exactly when the decider predicts acceptance and accepts exactly when it predicts rejection on its own description. Neither held verdict is a fixed point. The contradiction closes the total-decision boundary without treating nontermination as numerical zero. The same missing distinction appears in effective self-verification: a complete internal proof enumerator cannot both retain consistency and supply every demanded self-referential verdict.

6. Security, learning and scientific evidence

Security claims expose the message, key, observation, adversary and resource supports. Information-theoretic security means an observation closes the relevant source distinctions over complete support. Computational security means the registered adversary grammar and resource boundary fail to reopen them. Learning claims similarly retain complete hypothesis alternatives, exact evidence compatibility and sealed evaluation. A blind opaque score cannot replace the premise-to-result trace. Scientific simulations prove consequences of a registered model; only separate, sealed, target-custodied measurements can test a natural law.

7. Dependency order and complete execution census

Order	Sub-branch	Claim	Candidates	Closure
1	Formal Computation	SFT - COMP - FORM - STATE - TRANSITION - 001	256	depth-independent
2	Formal Computation	SFT - COMP - FORM - LANGUAGE - GRAMMAR - 001	256	depth-independent
3	Formal Computation	SFT - COMP - FORM - AUTOMATON - 001	256	depth-independent
4	Formal Computation	SFT - COMP - FORM - REWRITING - 001	256	depth-independent
5	Formal Computation	SFT - COMP - FORM - RECURSIVE - FUNCTION - 001	256	depth-independent
6	Formal Computation	SFT - COMP - FORM - LAMBDA - CALCULUS - 001	256	depth-independent
7	Formal Computation	SFT - COMP - FORM - ABSTRACT - MACHINE - 001	256	depth-independent
8	Formal Computation	SFT - COMP - FORM - CIRCUIT - 001	256	depth-independent
9	Formal Computation	SFT - COMP - FORM - OPERATIONAL - PROCESS - 001	256	depth-independent
10	Formal Computation	SFT - COMP - FORM - COMPOSITION - 001	256	depth-independent
11	Formal Computation	SFT - COMP - FORM - MODEL - EQUIVALENCE - 001	256	depth-independent
12	Formal Computation	SFT - COMP - FORM - UNIVERSALITY - 001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
13	Computability	SFT - COMP - CBL - RECOGNITION - DECISION - 001	256	depth-independent
14	Computability	SFT - COMP - CBL - HALTING - 001	256	depth-independent
15	Computability	SFT - COMP - CBL - ENUMERATION - 001	256	depth-independent
16	Computability	SFT - COMP - CBL - REDUCTION - 001	256	depth-independent
17	Computability	SFT - COMP - CBL - DEGREES - 001	256	depth-independent
18	Computability	SFT - COMP - CBL - UNIVERSAL - MACHINE - 001	256	depth-independent
19	Computability	SFT - COMP - CBL - UNDECIDABILITY - 001	256	depth-independent
20	Computability	SFT - COMP - CBL - INCOMPLETENESS - 001	256	depth-independent
21	Computability	SFT - COMP - CBL - RELATIVE - ORACLE - 001	256	depth-independent
22	Computability	SFT - COMP - CBL - HYPERCOMPUTATION - LIMIT - 001	256	depth-independent
23	Computational Complexity	SFT - COMP - CPLX - INPUT - SIZE - 001	256	depth-independent
24	Computational Complexity	SFT - COMP - CPLX - TIME - SPACE - 001	256	depth-independent
25	Computational Complexity	SFT - COMP - CPLX - CIRCUIT - RESOURCE - 001	256	depth-independent
26	Computational Complexity	SFT - COMP - CPLX - COMMUNICATION - QUERY - 001	256	depth-independent
27	Computational Complexity	SFT - COMP - CPLX - RANDOMNESS - 001	256	depth-independent
28	Computational Complexity	SFT - COMP - CPLX - REVERSIBILITY - COST - 001	256	depth-independent
29	Computational Complexity	SFT - COMP - CPLX - PARALLEL - 001	256	depth-independent
30	Computational Complexity	SFT - COMP - CPLX - BOUNDS - 001	256	depth-independent
31	Computational Complexity	SFT - COMP - CPLX - REDUCTION - COMPLETENESS - 001	256	depth-independent
32	Computational Complexity	SFT - COMP - CPLX - AVERAGE - WORST - 001	256	depth-independent
33	Computational Complexity	SFT - COMP - CPLX - APPROXIMATION - 001	256	depth-independent
34	Computational Complexity	SFT - COMP - CPLX - PARAMETERIZED - 001	256	depth-independent
35	Computational Complexity	SFT - COMP - CPLX - DESCRIPTIVE - 001	256	depth-independent
36	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - SEARCH - ORDER - 001	256	depth-independent
37	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - ARITHMETIC - 001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
38	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - STRINGS - SEQUENCES - 001	256	depth-independent
39	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - TREES - GRAPHS - 001	256	depth-independent
40	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - ALGEBRAIC - GEOMETRIC - 001	256	depth-independent
41	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - DYNAMIC - PROGRAMMING - 001	256	depth-independent
42	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - OPTIMIZATION - 001	256	depth-independent
43	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - RANDOMIZED - 001	256	depth-independent
44	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - PARALLEL - 001	256	depth-independent
45	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - DISTRIBUTED - 001	256	depth-independent
46	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - ONLINE - STREAMING - 001	256	depth-independent
47	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - NUMERICAL - 001	256	depth-independent
48	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - SYMBOLIC - 001	256	depth-independent
49	Algorithms and Mathematical Data Structures	SFT - COMP - ALG - APPROXIMATE - 001	256	depth-independent
50	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - SYNTAX - 001	256	depth-independent
51	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - BINDING - SUBSTITUTION - 001	256	depth-independent
52	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - EVALUATION - 001	256	depth-independent
53	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - OPERATIONAL - DENOTATIONAL - 001	256	depth-independent
54	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - TYPE - 001	256	depth-independent
55	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - PROGRAM - EQUIVALENCE - 001	256	depth-independent
56	Semantics and Mathematical Programming Theory	SFT - COMP - SEM - TERMINATION - 001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
57	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-CORRECTNESS-001	256	depth-independent
58	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-SPECIFICATION-001	256	depth-independent
59	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-TRANSFORMATION-001	256	depth-independent
60	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-COMPILATION-001	256	depth-independent
61	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-VERIFICATION-001	256	depth-independent
62	Concurrent and Distributed Computation	SFT-COMP-DIST-CONCURRENCY-001	256	depth-independent
63	Concurrent and Distributed Computation	SFT-COMP-DIST-CAUSALITY-001	256	depth-independent
64	Concurrent and Distributed Computation	SFT-COMP-DIST-SYNCHRONIZATION-001	256	depth-independent
65	Concurrent and Distributed Computation	SFT-COMP-DIST-COMMUNICATION-001	256	depth-independent
66	Concurrent and Distributed Computation	SFT-COMP-DIST-PARTIAL-ORDER-001	256	depth-independent
67	Concurrent and Distributed Computation	SFT-COMP-DIST-CONSENSUS-001	256	depth-independent
68	Concurrent and Distributed Computation	SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001	256	depth-independent
69	Concurrent and Distributed Computation	SFT-COMP-DIST-REPLICATION-CONSISTENCY-001	256	depth-independent
70	Concurrent and Distributed Computation	SFT-COMP-DIST-FAULT-MODEL-001	256	depth-independent
71	Concurrent and Distributed Computation	SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001	256	depth-independent
72	Concurrent and Distributed Computation	SFT-COMP-DIST-LOCALITY-001	256	depth-independent
73	Concurrent and Distributed Computation	SFT-COMP-DIST-NETWORK-COMPUTATION-001	256	depth-independent
74	Cryptography and Computational Security	SFT-COMP-SEC-ONE-WAYNESS-001	256	depth-independent
75	Cryptography and Computational Security	SFT-COMP-SEC-SECRECYS-001	256	depth-independent
76	Cryptography and Computational Security	SFT-COMP-SEC-INTEGRITY-001	256	depth-independent
77	Cryptography and Computational Security	SFT-COMP-SEC-AUTHENTICATION-001	256	depth-independent
78	Cryptography and Computational Security	SFT-COMP-SEC-HASHING-001	256	depth-independent
79	Cryptography and Computational Security	SFT-COMP-SEC-COMMITMENT-001	256	depth-independent
80	Cryptography and Computational Security	SFT-COMP-SEC-SIGNATURE-001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
81	Cryptography and Computational Security	SFT-COMP-SEC-PROOF-KNOWLEDGE-001	256	depth-independent
82	Cryptography and Computational Security	SFT-COMP-SEC-ZERO-KNOWLEDGE-001	256	depth-independent
83	Cryptography and Computational Security	SFT-COMP-SEC-MULTIPARTY-001	256	depth-independent
84	Cryptography and Computational Security	SFT-COMP-SEC-ADVERSARIAL-001	256	depth-independent
85	Cryptography and Computational Security	SFT-COMP-SEC-SECURITY-DEFINITION-001	256	depth-independent
86	Cryptography and Computational Security	SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001	256	depth-independent
87	Learning and Intelligence Theory	SFT-COMP-LEARN-INFERENCE-001	256	depth-independent
88	Learning and Intelligence Theory	SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001	256	depth-independent
89	Learning and Intelligence Theory	SFT-COMP-LEARN-REPRESENTATION-001	256	depth-independent
90	Learning and Intelligence Theory	SFT-COMP-LEARN-GENERALIZATION-001	256	depth-independent
91	Learning and Intelligence Theory	SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001	256	depth-independent
92	Learning and Intelligence Theory	SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001	256	depth-independent
93	Learning and Intelligence Theory	SFT-COMP-LEARN-INDUCTION-001	256	depth-independent
94	Learning and Intelligence Theory	SFT-COMP-LEARN-SEARCH-PLANNING-001	256	depth-independent
95	Learning and Intelligence Theory	SFT-COMP-LEARN-REINFORCEMENT-001	256	depth-independent
96	Learning and Intelligence Theory	SFT-COMP-LEARN-MULTIAGENT-001	256	depth-independent
97	Learning and Intelligence Theory	SFT-COMP-LEARN-ADAPTATION-001	256	depth-independent
98	Learning and Intelligence Theory	SFT-COMP-LEARN-LEARNING-LIMITS-001	256	depth-independent
99	Learning and Intelligence Theory	SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001	256	depth-independent
100	Learning and Intelligence Theory	SFT-COMP-LEARN-CLASSICAL-LEARNING-001	256	depth-independent
101	Scientific Computation	SFT-COMP-SCI-EXACT-APPROXIMATE-001	256	depth-independent
102	Scientific Computation	SFT-COMP-SCI-STABILITY-001	256	depth-independent
103	Scientific Computation	SFT-COMP-SCI-ERROR-PROPAGATION-001	256	depth-independent
104	Scientific Computation	SFT-COMP-SCI-DISCRETIZATION-001	256	depth-independent
105	Scientific Computation	SFT-COMP-SCI-CONVERGENCE-001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
106	Scientific Computation	SFT-COMP-SCI-SYMBOLIC-001	256	depth-independent
107	Scientific Computation	SFT-COMP-SCI-SIMULATION-001	256	depth-independent
108	Scientific Computation	SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001	256	depth-independent
109	Scientific Computation	SFT-COMP-SCI-INVERSE-PROBLEM-001	256	depth-independent
110	Scientific Computation	SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001	256	depth-independent
111	Scientific Computation	SFT-COMP-SCI-HIGH-DIMENSIONAL-001	256	depth-independent
112	Scientific Computation	SFT-COMP-SCI-MANY-BODY-001	256	depth-independent
113	Scientific Computation	SFT-COMP-SCI-MATHEMATICAL-MODELLING-001	256	depth-independent

The branch total is **28,928 candidates**, **113 survivors**, **452 adverse controls** and **113 independent validations**.

8. Derivation 1: Exact state and transition law

Claim: SFT-COMP-FORM-STATE-TRANSITION-001

8.1 Scientific necessity and exact theorem

Exact state and transition law is required to derive generated configurations and change without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

8.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for generated configurations and change.

All generated finite generated configurations and change structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not yet define a language, machine model or resource bound.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

8.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

8.4 Operational laws and consequences

- Carrier law: complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

8.5 Depth-independent closure

Base: The structural One supplies one canonical generated configurations and change form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated generated configurations and change form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

8.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-STATE-TRANSITION-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-STATE-TRANSITION-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-STATE-TRANSITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet define a language, machine model or resource bound.	PASS

8.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: state, transition system.

Does not yet define a language, machine model or resource bound.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not yet define a language, machine model or resource bound.

8.8 Exact evidence identities

- Source manifest: sha256:d6baac855792700e19eea8e212398cc9d04f90d5282e644b593777276c3aa6ec
- Complete-census certificate:
sha256:a9e5bca9c8d3f9281489c4f6ddb6dfa34b58fa0915aff9d33a77eec0d33495f0
- Derivation seal: sha256:52d77e7c18c95f893f56bae7f7710e36ce0ab176b01d1f8b868be0ad6e0385f0
- Independent implementation:
sha256:c2917eb384de1fd43f363ae486fd3db8c6670b6199a9f0de76519dc719d78e62
- Independent certificate:
sha256:36071490b1ec6ae387124089b784f55bfdc63f7664cc91f3e45d45298b673299
- External validation:
sha256:8df5268b3c4793c2dc19ad1082c591e928ab99d7aca1469d9d9870749143a0f9
- Model-admitted engine receipt:
sha256:6ff1fdc0b1d041805add6ef9350dee1fe38a41082072c9f7888d4a22975228f
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-STATE-TRANSITION-001-6ff1fdc0b1d04180.json

9. Derivation 2: Fold formal-language and grammar law

Claim: SFT-COMP-FORM-LANGUAGE-GRAMMAR-001

9.1 Scientific necessity and exact theorem

Fold formal-language and grammar law is required to derive generated symbol words and formation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

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- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

9.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for generated symbol words and formation.

All generated finite generated symbol words and formation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes generated finite languages and depth-independent word succession, not a completed infinite language object.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

9.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

9.4 Operational laws and consequences

- Carrier law: complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

9.5 Depth-independent closure

Base: The structural One supplies one canonical generated symbol words and formation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated generated symbol words and formation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

9.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: PASS.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: PASS.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes generated finite languages and depth-independent word succession, not a completed infinite language object.	PASS

9.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: formal language, grammar.

Closes generated finite languages and depth-independent word succession, not a completed infinite language object.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes generated finite languages and depth-independent word succession, not a completed infinite language object.

9.8 Exact evidence identities

- Source manifest: `sha256:e7e49951c44319178d53b75d6dbc14f7fcd81dea5b263bf7e4215c3f358094e`
- Complete-census certificate:
`sha256:009322561ae741ea9b0f0f7bc97850497effbfa14b253d5eb1ffa19ce738e68f`
- Derivation seal: `sha256:688db340733ac157427caf7da72b4bb5f97b86627b4a080ce828a3d5bd9adb4e`

- Independent implementation:
sha256:70802d179f08a4a5a6b2f178c5bd11fa87d64689159b14694def28731b79e5c8
- Independent certificate:
sha256:8a40d9a22d068e806f0c2d49a9af923f6fa29101600069cb311d9f82d40d8cfd
- External validation:
sha256:9ace6f1cbbeddd3086e42f5b9dce75bf37c34fa29abb5ed8b97fd5bf9d476ac67
- Model-admitted engine receipt:
sha256:9247a863bf580966b3db1091ef6024a0f4562506076136dce33acda2a6039d14
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-LANGUAGE-GRAMMAR-001-9247a863bf580966.json

10. Derivation 3: Fold finite automaton law

Claim: SFT-COMP-FORM-AUTOMATON-001

10.1 Scientific necessity and exact theorem

Fold finite automaton law is required to derive finite recognition processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-LANGUAGE-GRAMMAR-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

10.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite recognition processes.

All generated finite finite recognition processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not yet assert equivalence with every other model or universality.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

10.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

10.4 Operational laws and consequences

- Carrier law: finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

10.5 Depth-independent closure

Base: The structural One supplies one canonical finite recognition processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite recognition processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

10.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-AUTOMATON-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-AUTOMATON-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-AUTOMATON-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet assert equivalence with every other model or universality.	PASS

10.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: finite automaton, recognizer.

Does not yet assert equivalence with every other model or universality.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not yet assert equivalence with every other model or universality.

10.8 Exact evidence identities

- Source manifest: sha256:2875e2c5834ff96fc547959b332f1a1f7a7e84c85bea5e3873518751be60cc6e
- Complete-census certificate:
sha256:59bade0640ecca5d80d75d3a8b0a37298a314db962cfb92dcbcb82b6873ee981
- Derivation seal: sha256:de24c317c1f621d5ec62ecb401e94a4fccfba4de95dabad9ab144a71748f731c
- Independent implementation:
sha256:0e08a4bae652b4772475643b97ad32e52bf14f55de8fefaf0ff3218b5ef5e22e4
- Independent certificate:
sha256:c6d7e75b9dbd097a50865aa3fb476bc057d2a5a2b933b069a694f021a4cc7d27
- External validation:
sha256:17f43954c36004983c2299b4c7468c5e5816b764b5c8735af7b0275b942a66d2
- Model-admitted engine receipt:
sha256:a28d36b12b07be8aa4f32af257b75d6da331b0cebfea99fda3d6240e9ccaf109
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-FORM-AUTOMATON-001-a28d36b12b07be8a.json

11. Derivation 4: Fold rewriting-system law

Claim: SFT-COMP-FORM-REWRITING-001

11.1 Scientific necessity and exact theorem

Fold rewriting-system law is required to derive held-position replacement without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-AUTOMATON-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001

- SFT-INFO-CONSERVATION-LOSS-001

11.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for held-position replacement.

All generated finite held-position replacement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only registered finite rules and generated words are admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

11.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.

Eliminated candidates	Exact first-failure reason
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

11.4 Operational laws and consequences

- Carrier law: generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

11.5 Depth-independent closure

Base: The structural One supplies one canonical held-position replacement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated held-position replacement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

11.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-REWRITING-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-REWRITING-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-REWRITING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS

Control	Expected	Observed	Result
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite rules and generated words are admitted.	PASS

11.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: rewriting system, term rewriting.

Only registered finite rules and generated words are admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only registered finite rules and generated words are admitted.

11.8 Exact evidence identities

- Source manifest: sha256:4efb2a905b4b82b88562af8bc58df30b63eaa7e37fc6d2026698945df864753d
- Complete-census certificate:
sha256:ea26ea052050e13e4329e7e31cca7d1afdc3560dd5a87c6621a4ea7f58e381ca
- Derivation seal: sha256:7f1279c1b5b5d6b2e31d821616cdc40fbb8e89e782ad571f2cd65486a9569727
- Independent implementation:
sha256:1d03336bec2e9efb5b8a8338de71b228a4a517b8fee08c2146539375583d26b8
- Independent certificate:
sha256:f5faf737b0c4341fe3c50b95d147053ef79cdb85a9ad4971d22b3304c7bb3466
- External validation:
sha256:13b91a821e5ab3ea07326aed719fde2054ecd40561f9787f4d5d3b1218d24ef1
- Model-admitted engine receipt:
sha256:b1f6c54c1f8be568e0dbb809dc19f1f882840e53c693aa208db0c2410860defe
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-FORM-REWRITING-001-b1f6c54c1f8be568.json

12. Derivation 5: Fold recursion law

Claim: SFT-COMP-FORM-RECURSIVE-FUNCTION-001

12.1 Scientific necessity and exact theorem

Fold recursion law is required to derive structural finite recursion without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-REWRITING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

12.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for structural finite recursion.

All generated finite structural finite recursion structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes recursion over every generated finite successor trace, not completed transfinite recursion.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampled-carrier	sampled-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

12.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

12.4 Operational laws and consequences

- Carrier law: structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

12.5 Depth-independent closure

Base: The structural One supplies one canonical structural finite recursion form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated structural finite recursion form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

12.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes recursion over every generated finite successor trace, not completed transfinite recursion.	PASS

12.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: recursive function, primitive recursion.

Closes recursion over every generated finite successor trace, not completed transfinite recursion.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes recursion over every generated finite successor trace, not completed transfinite recursion.

12.8 Exact evidence identities

- Source manifest: `sha256:348ad460880a7aa65b3209f59bb02bdcbeaa373fa7b536e7ccaefea2bad37d18`
- Complete-census certificate:
`sha256:549d897e9cba6810b09f251e85e82e2d398e63bd5b8bab2cc820dcbf015ffe97`
- Derivation seal: `sha256:f0d1440d842607d5258c5accdcb83682ab3eaf0a58c3dceac901a2c9c730723d`

- Independent implementation:
sha256:f0103e97dcbd277c3c09a157d04752e90b18f719eac050beb7d45f706b36c486
- Independent certificate:
sha256:9dc55f7ee3aa968c744de817d3bdef595ab14824948958aeb1d57417152fa061
- External validation:
sha256:24365aa8a72572993447d2cafcf0abb11e9491ece06a19c1441ae242ceecc411
- Model-admitted engine receipt:
sha256:291230ba67c75760c319e58fe0ae120c6c300a83e39d5f51c70de936c73e580c
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-RECURSIVE-FUNCTION-001-291230ba67c75760.json

13. Derivation 6: Fold binding and self-application calculus law

Claim: SFT-COMP-FORM-LAMBDA-CALCULUS-001

13.1 Scientific necessity and exact theorem

Fold binding and self-application calculus law is required to derive binding, substitution and reduction without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

13.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for binding, substitution and reduction.

All generated finite binding, substitution and reduction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No conventional lambda calculus is assumed; the result is the generated Fold term kernel.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampled-carrier	sampled-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

13.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

13.4 Operational laws and consequences

- Carrier law: generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

13.5 Depth-independent closure

Base: The structural One supplies one canonical binding, substitution and reduction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated binding, substitution and reduction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

13.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional lambda calculus is assumed; the result is the generated Fold term kernel.	PASS

13.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: lambda calculus, substitution.

No conventional lambda calculus is assumed; the result is the generated Fold term kernel.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No conventional lambda calculus is assumed; the result is the generated Fold term kernel.

13.8 Exact evidence identities

- Source manifest: sha256:52bb83eb5794290940d213fa2d1cc6cdc62db73ee5dfb6eccf81583fbca292c4
- Complete-census certificate:
sha256:df ff75c37a0aa841000744b9aeb2b111230f40159b8ebe14a1f419338ab33a87
- Derivation seal: sha256:2a809ddc1fcc1b0a016920d46311c76dcd0cd465af22930ae3b823a0ba7e14e3
- Independent implementation:
sha256:d745cfdbf195dec5c6646f1dfdc6039f85e6170cde0ce4f36f97627314a62ed6
- Independent certificate:
sha256:f89a69bbf74e6f348b37e64f672de3968639ca340f9d50e9c18b7a29a4a36c06
- External validation:
sha256:6a8daaa4987a604def6e709ad88ba6c6c5a18065e3d9bc8ce61e12f537d000d6
- Model-admitted engine receipt:
sha256:c5df90aeea7c0f8459bb98cdfa9343167723cc7b320af2c9fec60f7f5f5bee63
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-LAMBDA-CALCULUS-001-c5df90aeea7c0f84.json

14. Derivation 7: Fold abstract-machine law

Claim: SFT-COMP-FORM-ABSTRACT-MACHINE-001

14.1 Scientific necessity and exact theorem

Fold abstract-machine law is required to derive finite executable machine descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-LAMBDA-CALCULUS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

14.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite executable machine descriptions.

All generated finite executable machine descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Machines outside the frozen description grammar remain outside the theorem.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampled-carrier	sampled-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

14.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.

Eliminated candidates	Exact first-failure reason
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__in-interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

14.4 Operational laws and consequences

- Carrier law: description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

14.5 Depth-independent closure

Base: The structural One supplies one canonical finite executable machine descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite executable machine descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

14.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Machines outside the frozen description grammar remain outside the theorem.	PASS

14.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: abstract machine, Turing machine.

Machines outside the frozen description grammar remain outside the theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Machines outside the frozen description grammar remain outside the theorem.

14.8 Exact evidence identities

- Source manifest: sha256:64c1a785e68928851b39460f04755c042d9c7cf57ea547b9f90e1604d55a00b4
- Complete-census certificate:
sha256:45abdaafd0f0e6e537991e14ab4dda06923162c64e70075289ee5b3a4a320976
- Derivation seal: sha256:1bf45a77a8defe269a94cc9471ce829b8b518f30acb09bcaab1a18d351cc508f
- Independent implementation:
sha256:b0c8e900076f04ce4e88db72f4300dddec7b44e871c574fda6828f04743b6437a
- Independent certificate:
sha256:3ce91feb3e76ea25d296bfcc777100313354b171985af741b72cbbe204dc2113
- External validation:
sha256:0062195e4657ede78fadd3877e80e38bd845ffef5f981c6a577d68b4c6e04815
- Model-admitted engine receipt:
sha256:2b2471771a06a549143d109bd88b40339b476c36ab65b278fd86a7b54adbdd1b
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-ABSTRACT-MACHINE-001-2b2471771a06a549.json

15. Derivation 8: Fold circuit syntax and evaluation law

Claim: SFT-COMP-FORM-CIRCUIT-001

15.1 Scientific necessity and exact theorem

Fold circuit syntax and evaluation law is required to derive finite connected local transformations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-ABSTRACT-MACHINE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

15.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite connected local transformations.

All generated finite finite connected local transformations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Resource size and depth comparisons are deferred to Complexity.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampled-carrier	sampled-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

15.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__in-interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

15.4 Operational laws and consequences

- Carrier law: generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

15.5 Depth-independent closure

Base: The structural One supplies one canonical finite connected local transformations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite connected local transformations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

15.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-CIRCUIT-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-CIRCUIT-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-CIRCUIT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Resource size and depth comparisons are deferred to Complexity.	PASS

15.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: circuit, Boolean circuit.

Resource size and depth comparisons are deferred to Complexity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Resource size and depth comparisons are deferred to Complexity.

15.8 Exact evidence identities

- Source manifest: sha256:1411276748c6c4373ac3ec63581aac086a1a3b2ad626d6c46aaa3b47a30d5a75
- Complete-census certificate:
sha256:600387cd145201a86f224f4cb51476ca311cb2cca9535665ae50d39b7e4862c6
- Derivation seal: sha256:62e2b314384c853bb679b205fb39f47f76ee48f07eb26e225f0f2fee5861baf6
- Independent implementation:
sha256:4d4a98f08cf35dab2175a66857e1f2764741107ab27f0809cc028907180c443b
- Independent certificate:
sha256:b73405d93e9177c20dc3877407ce8bc3f169c481b8a298624f0e290625d87413
- External validation:
sha256:d1e5487d81c976c6526694efaf7845bd437def19c61290a024f75abee8e5a7e7
- Model-admitted engine receipt:
sha256:0a974615e0f53938a3c8551caabc041da5b7c241dfdc059febcb5d38269a0c94
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-FORM-CIRCUIT-001-0a974615e0f53938.json

16. Derivation 9: Fold operational-process law

Claim: SFT-COMP-FORM-OPERATIONAL-PROCESS-001

16.1 Scientific necessity and exact theorem

Fold operational-process law is required to derive execution and observable process behavior without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-CIRCUIT-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

16.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for execution and observable process behavior.

All generated finite execution and observable process behavior structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not import process algebra or distributed timing.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampled-carrier	sampled-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

16.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

16.4 Operational laws and consequences

- Carrier law: initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

16.5 Depth-independent closure

Base: The structural One supplies one canonical execution and observable process behavior form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated execution and observable process behavior form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

16.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not import process algebra or distributed timing.	PASS

16.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: operational process, labelled transition system.

Does not import process algebra or distributed timing.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not import process algebra or distributed timing.

16.8 Exact evidence identities

- Source manifest: sha256 : ca32b606f98729a29a5a4a2c123efa575bd499b2522eb2b8b0eb1c73186eebf4
- Complete-census certificate:
sha256 : e5067d5b4f2f25d4b346d042203f2033e326cea7b653659447b60d7d47c5e93c
- Derivation seal: sha256 : 16f15c2211974316eb7f5595c532cc6e649fe07a22c56adf7a8673b7bc3466d3
- Independent implementation:
sha256 : edf9d2ca579edfbdcf5f372b21b3c37de52c998e8b92c5e4a136ab0d31b454ba
- Independent certificate:
sha256 : a37e121f9fca530b933f74e80c8877fd811ab4e64923192cc395984dfaf030a6
- External validation:
sha256 : b5d7bc8b74c9ed55eac3a59349e3a88148e1ef2411d26dcb672e6a97a0873579
- Model-admitted engine receipt:
sha256 : d3349d79836c6e8edf5199206fe0dc818a6210e9d3dc7f8000eb83f0b55ef282
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-OPERATIONAL-PROCESS-001-d3349d79836c6e8e.json

17. Derivation 10: Fold process composition and decomposition law

Claim: SFT-COMP-FORM-COMPOSITION-001

17.1 Scientific necessity and exact theorem

Fold process composition and decomposition law is required to derive serial and parallel process assembly without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-OPERATIONAL-PROCESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

17.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for serial and parallel process assembly.

All generated finite serial and parallel process assembly structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Concurrency laws requiring communication or shared causality remain downstream.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

17.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

17.4 Operational laws and consequences

- Carrier law: exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

17.5 Depth-independent closure

Base: The structural One supplies one canonical serial and parallel process assembly form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated serial and parallel process assembly form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

17.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-COMPOSITION-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-COMPOSITION-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-COMPOSITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency laws requiring communication or shared causality remain downstream.	PASS

17.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: composition, decomposition.

Concurrency laws requiring communication or shared causality remain downstream.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Concurrency laws requiring communication or shared causality remain downstream.

17.8 Exact evidence identities

- Source manifest: `sha256:a5acf66751f0f1f687e326d1a5d57eb0d930f9341a51301ecb7778266222f3e1`

- Complete-census certificate:
sha256:828753dbe5d5d32a929bc631a6118bd4c2e9161986252e66ebb326b454d79a36
- Derivation seal: sha256:1a5b215a9ad08b3cf8665c8299fa6961b1a43c87847b1b705adfbcb1419827716
- Independent implementation:
sha256:01a1fd9217764d6beb2dfda7b127626b041b729945527f7cce6a4c4a7a22c756
- Independent certificate:
sha256:11adf318b9899ccc822e4c9164ac86c61596df45c8b41bcb54d268785aa279a2
- External validation:
sha256:91e36576fe2613746be520df07a60d9db80cd25dce24a78dd1b023d23c8c438b
- Model-admitted engine receipt:
sha256:b0fea8f79a1c2c934c5b2dfaa37c9ba09e8b0f1416377175a645afe9f58738ad
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-FORM-COMPOSITION-001-b0fea8f79a1c2c93.json

18. Derivation 11: Fold computational-model equivalence law

Claim: SFT-COMP-FORM-MODEL-EQUIVALENCE-001

18.1 Scientific necessity and exact theorem

Fold computational-model equivalence law is required to derive bidirectional execution-preserving translations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-COMPOSITION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

18.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for bidirectional execution-preserving translations.

All generated finite bidirectional execution-preserving translations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Equivalence

applies only to the two registered generated model grammars.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

18.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and

contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

18.4 Operational laws and consequences

- Carrier law: complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

18.5 Depth-independent closure

Base: The structural One supplies one canonical bidirectional execution-preserving translations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bidirectional execution-preserving translations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

18.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence applies only to the two registered generated model grammars.	PASS

18.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: model equivalence, simulation.

Equivalence applies only to the two registered generated model grammars.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Equivalence applies only to the two registered generated model grammars.

18.8 Exact evidence identities

- Source manifest: sha256:0162711d6244b48f48e65f584f7d4d7ebec34ca89f8ddd5803006fa81eb7c497
- Complete-census certificate:
sha256:514c11ad3899b13152b2fbfa77adc36da6115e4c1452e7993b43d0261ab40488
- Derivation seal: sha256:3c0dad2425378918bacf1c277636db591165b0660c98f8c19d1f4ec68d7c1eb9
- Independent implementation:
sha256:7f980a25b82bbe2793268f94eaf9f482120d4fc8ec8a889a6d345052c7e04f40
- Independent certificate:
sha256:62db23d8401df6af4f79ceb122b3a85440ff45189a7bb09dc7d5e6722c6e99ce
- External validation:
sha256:64e6d04fde05fcaba7e98b2cef7ca0a8c7f7de711e8a754d145b9f67e2377bc6
- Model-admitted engine receipt:
sha256:911ff906b2d8407111a076f3abb06abe79334098a0e87faa54460f8453338139
- Receipt path: receipts/engine/model_admitted/SFT-COMP-FORM-MODEL-EQUIVALENCE-001-911ff906b2d84071.json

19. Derivation 12: Fold universal-computation law

Claim: SFT-COMP-FORM-UNIVERSALITY-001

19.1 Scientific necessity and exact theorem

Fold universal-computation law is required to derive interpretation of generated machine descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-MODEL-EQUIVALENCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

19.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for interpretation of generated machine descriptions.

All generated finite interpretation of generated machine descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
carrier	sampld-carrier	sampld-carrier removes a required exact carrier condition.	complete-generated-carrier	complete-generated-carrier retains the complete registered carrier condition.
identity	presentation-identity	presentation-identity removes a required exact identity condition.	canonical-held-identity	canonical-held-identity retains the complete registered identity condition.
relation	assumed-operation	assumed-operation removes a required exact relation condition.	source-bound-exact-relation	source-bound-exact-relation retains the complete registered relation condition.
trace	result-only	result-only removes a required exact trace condition.	complete-retained-trace	complete-retained-trace retains the complete registered trace condition.
boundary	implicit-terminal-boundary	implicit-terminal-boundary removes a required exact boundary condition.	explicit-initial-terminal-boundary	explicit-initial-terminal-boundary retains the complete registered boundary condition.
composition	opaque-assembly	opaque-assembly removes a required exact composition condition.	interface-preserving-composition	interface-preserving-composition retains the complete registered composition condition.
generality	fixed-examples	fixed-examples removes a required exact generality condition.	structural-successor	structural-successor retains the complete registered generality condition.
addition	imported-machine-model	imported-machine-model removes a required exact addition condition.	no-extra-computational-premise	no-extra-computational-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

19.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-carrier removes a required exact carrier condition.

Eliminated candidates	Exact first-failure reason
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

19.4 Operational laws and consequences

- Carrier law: complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

19.5 Depth-independent closure

Base: The structural One supplies one canonical interpretation of generated machine descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated interpretation of generated machine descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

19.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - `formal_computation` exact operational witness 1 for SFT-COMP-FORM-UNIVERSALITY-001; result: PASS.
- `formal_computation-witness-2` - `formal_computation` exact operational witness 2 for SFT-COMP-FORM-UNIVERSALITY-001; result: PASS.
- `formal_computation-witness-3` - `formal_computation` exact operational witness 3 for SFT-COMP-FORM-UNIVERSALITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.	PASS

19.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal computation, universal machine.

Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

19.8 Exact evidence identities

- Source manifest: sha256 : 6cbb5b4a2345df65875b39ced2bae1002a4efeb96a2aa4115a7faecf6aadf321
- Complete-census certificate:
sha256 : 5d1074e1fc7f1c4b882429494e4239518885cc13ca5cd62ffb035974cfbc3caa
- Derivation seal: sha256 : aa872db026edad77bae474d940bdead07caed584e3013fc04c986d0d927ad80d
- Independent implementation:
sha256 : f1df70199e93c004a3d9c9e2a3bf396a74626e742e81b0b8e213b3af7e9e8819
- Independent certificate:
sha256 : bf89e7bbbd33de8dc7caaf3f5bdf3d6979d628eacc66183a02411b407236e76b
- External validation:
sha256 : 9901d8e8f04985a7b65da1a40504fa8f1df13e92f614953ef900fcd87ef4ace6
- Model-admitted engine receipt:
sha256 : 83a401709b86991de5f38a49e7794379ea0c041fb2a406a245f900bf11bcb598
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-FORM-UNIVERSALITY-001-83a401709b86991d.json

20. Derivation 13: Fold recognition and decision law

Claim: SFT-COMP-CBL-RECOGNITION-DECISION-001

20.1 Scientific necessity and exact theorem

Fold recognition and decision law is required to derive acceptance, rejection and continued execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

20.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger.

Generation rule: Generate the literal product of the eight registered Computability structural axes for acceptance, rejection and continued execution.

All generated finite acceptance, rejection and continued execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A recognizer need not terminate on every generated input; a decider must.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nonterminat ion	implicit-nontermination removes a required exact result condition.	accept-reject-contin ue-classes	accept-reject-continue-clas ses retains the complete registered result condition.
self_reference	self-reference-exclu ded	self-reference-excluded removes a required exact self_reference condition.	generated-self-descr iption-support	generated-self-description-s upport retains the complete registered self_reference condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

20.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

20.4 Operational laws and consequences

- Carrier law: generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

20.5 Depth-independent closure

Base: The structural One supplies one canonical acceptance, rejection and continued execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated acceptance, rejection and continued execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

20.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A recognizer need not terminate on every generated input; a decider must.	PASS

20.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: recognizability, decidability.

A recognizer need not terminate on every generated input; a decider must.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- A recognizer need not terminate on every generated input; a decider must.

20.8 Exact evidence identities

- Source manifest: sha256 : fcd2db93deb173b9cd92f0ba025417ba3126423bffb0090d833fad544c5da89
- Complete-census certificate:
sha256 : 1ac8edb72bda3ab3a65f70220c0981f627bf680a42946a7501995a1b54deb22b
- Derivation seal: sha256 : 11e94644591c3e7e51e1b0235bb9752f71fcd7ce64f457291bf83968e902a4a6
- Independent implementation:
sha256 : dc0f0486633f8e87e5f9e02b210b5376b2cf778691ca860a41eedcfa874929a6
- Independent certificate:
sha256 : 0eb8c9783f7e87e434555864f84644fcd1c096bcc3bd1725540ffdd033fb0aa1
- External validation:
sha256 : 9151b6040f1b47fae6bb0018232dc6b6399177add50d2f9a768f723f69bb6ebe
- Model-admitted engine receipt:
sha256 : c9c992b18e4408ecfd5bb004622b3d62d836752b387773eecz7dfc0e1daed9e8
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CBL-RECOGNITION-DECISION-001-c9c992b18e4408ec.json

21. Derivation 14: Fold halting boundary law

Claim: SFT-COMP-CBL-HALTING-001

21.1 Scientific necessity and exact theorem

Fold halting boundary law is required to derive terminal-state determination without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-RECOGNITION-DECISION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

21.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction.

Generation rule: Generate the literal product of the eight registered Computability structural axes for terminal-state determination.

All generated finite terminal-state determination structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not decide halting for all generated universal-machine descriptions.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-description-domain	complete-generated-description-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-execution-trace	exact-recognition-execution-trace retains the complete registered execution condition.
result	implicit-nontermination	implicit-nontermination removes a required exact result condition.	accept-reject-continue-classes	accept-reject-continue-classes retains the complete registered result condition.
self_reference	self-reference-excluded	self-reference-excluded removes a required exact self_reference condition.	generated-self-description-support	generated-self-description-support retains the complete registered self_reference condition.
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

21.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

21.4 Operational laws and consequences

- Carrier law: encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

21.5 Depth-independent closure

Base: The structural One supplies one canonical terminal-state determination form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated terminal-state determination form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

21.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-HALTING-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-HALTING-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-HALTING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not decide halting for all generated universal-machine descriptions.	PASS

21.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: halting problem, diagonalization.

Does not decide halting for all generated universal-machine descriptions.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not decide halting for all generated universal-machine descriptions.

21.8 Exact evidence identities

- Source manifest: sha256:66bd1cb6ca985f66e6ae136170730560e94ec8c59f80be5dbb0bcace9ea0b65c
- Complete-census certificate:
sha256:cf77cceb2366e64454800a0363156968847e58931805d5f2ef68894baab70fd9
- Derivation seal: sha256:299c9b39f4e497fe084e28cb9d78834cef211cd11785de98b0fc7bb206f83d2d
- Independent implementation:
sha256:e5acaf236971ed0d09b140257aaa940a41536fd2f5ee017ca337331933c9be54
- Independent certificate:
sha256:b3be1826d2f70acfe69d15683428f479c2294731ad192d11d9a06a7d79cdf6d
- External validation:
sha256:bb240ffe1dbdb05d04360ea66a0f4efd9284d5d1b72f80a8b72590e88dc77a02
- Model-admitted engine receipt:
sha256:a33ae0f08db036bcde17a25eb3d6605dcfbcd3fa0492176faaf553110d2368e2
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-HALTING-001-a33ae0f08db036bc.json

22. Derivation 15: Fold effective-enumeration law

Claim: SFT-COMP-CBL-ENUMERATION-001

22.1 Scientific necessity and exact theorem

Fold effective-enumeration law is required to derive generated description ordering without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-HALTING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

22.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position.

Generation rule: Generate the literal product of the eight registered Computability structural axes for generated description ordering.

All generated finite generated description ordering structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No completed infinite enumeration is installed as an object.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nonterminat ion	implicit-nontermination removes a required exact result condition.	accept-reject-contin ue-classes	accept-reject-continue-clas ses retains the complete registered result condition.
self_reference	self-reference-exclu ded	self-reference-excluded removes a required exact self_reference condition.	generated-self-descr iption-support	generated-self-description-s upport retains the complete registered self_reference condition.
totality	assumed-total-decide r	assumed-total-decider removes a required exact totality condition.	explicit-totality-or -partiality-boundary	explicit-totality-or-partiality-b oundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-re duction	answer-preserving-reductio n retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successo r-certificate	description-successor-certifi cate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

22.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

22.4 Operational laws and consequences

- Carrier law: complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

22.5 Depth-independent closure

Base: The structural One supplies one canonical generated description ordering form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated generated description ordering form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

22.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-ENUMERATION-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-ENUMERATION-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-ENUMERATION-001; result: PASS.

Control	Expected	Observed	Result
<code>false_premise</code>	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
<code>boundary</code>	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No completed infinite enumeration is installed as an object.	PASS

22.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: enumerability, effective enumeration.

No completed infinite enumeration is installed as an object.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No completed infinite enumeration is installed as an object.

22.8 Exact evidence identities

- Source manifest: `sha256:63973d4c123129f31559427f7455012727ae43b7f92ddc58dec264433f9fc9b1`
- Complete-census certificate:
`sha256:70b88f63a8011f2ec1e30d67fd31e78efc4d5a1d401582c0ea91ac0f5bd05a82`
- Derivation seal: `sha256:6c0d27b9378dd92f4cdedc7f7d4dc90a02fe816f4c8fbeeef2c826ae564712d08`

- Independent implementation:
sha256:c6a5dd8e67bf0218abf2a49489560529f1f066e15a1c597495aa6042686053cd
- Independent certificate:
sha256:e2a65196b779a1bd41f25443155116ef6e7ffe265c9d0921e544927747b53717
- External validation:
sha256:56a2658bc7bed4373c7763f34346566f995e7e95bcfcb87ede676a3ceb5e088c
- Model-admitted engine receipt:
sha256:b4a3d735406f9ab57023c47ed0a12a7a221b278c75727cb0a7170dbdfec940f2
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-ENUMERATION-001-b4a3d735406f9ab5.json

23. Derivation 16: Fold computability-reduction law

Claim: SFT-COMP-CBL-REDUCTION-001

23.1 Scientific necessity and exact theorem

Fold computability-reduction law is required to derive problem translation preserving answers without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-ENUMERATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

23.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law.

Generation rule: Generate the literal product of the eight registered Computability structural axes for problem translation preserving answers.

All generated finite problem translation preserving answers structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Reduction claims require explicit encoders, decoders and generated domains.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nontermination	implicit-nontermination removes a required exact result condition.	accept-reject-contin ue-classes	accept-reject-continue-clas ses retains the complete registered result condition.
self_reference	self-reference-exclud ed	self-reference-excluded removes a required exact self_reference condition.	generated-self-descr iption-support	generated-self-description-s upport retains the complete registered self_reference condition.
totality	assumed-total-decide r	assumed-total-decider removes a required exact totality condition.	explicit-totality-or -partiality-boundary	explicit-totality-or-partiality-b oundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-re duction	answer-preserving-reductio n retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successo r-certificate	description-successor-certifi cate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-comput ational-power	no-undeclared-computation al-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

23.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-ex
ecution-trace__accept-reject-continue-classes__generated-self-description-support__exp
licit-totality-or-partiality-boundary__answer-preserving-reduction__description-succ
essor-certificate__no-undeclared-computational-power`.

The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is

source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

23.4 Operational laws and consequences

- Carrier law: source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

23.5 Depth-independent closure

Base: The structural One supplies one canonical problem translation preserving answers form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated problem translation preserving answers form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

23.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-REDUCTION-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-REDUCTION-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-REDUCTION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reduction claims require explicit encoders, decoders and generated domains.	PASS

23.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: many-one reduction, computability reduction.

Reduction claims require explicit encoders, decoders and generated domains.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Reduction claims require explicit encoders, decoders and generated domains.

23.8 Exact evidence identities

- Source manifest: sha256 : 7947d16c47683d213393a222dbf077c1313f6884b4fe89b8c84b69f488629669
- Complete-census certificate:
sha256 : 2e6160c4db60b0b8f818a7ad8cc4f4f7782d13c3d9431e657d2b5d8b1e7ed56e
- Derivation seal: sha256 : c3ae5424425de2ea0a730259170848da166eb040d9c9fe8f1d76d0254ea22811
- Independent implementation:
sha256 : 173795ad853a56a81d7d98fd16dc14eccd38d80f3add1c8dc0a58720f6f43de5
- Independent certificate:
sha256 : c843b1c34313a579857158cca63c95281567d12d2cb77c0ec04f46ad8e07baf7
- External validation:
sha256 : 1d8cac715551bdf0d1e71267a730cbf51e9175f803ebe560db4fedf7801a588e
- Model-admitted engine receipt:
sha256 : 7d03f54baf0fc6d4bbe16f2ea73fac601ede53dceacbd55e2b032545196aaff2
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-REDUCTION-001-7d03f54baf0fc6d4.json

24. Derivation 17: Fold degrees-of-computability law

Claim: SFT-COMP-CBL-DEGREES-001

24.1 Scientific necessity and exact theorem

Fold degrees-of-computability law is required to derive relative reduction classes without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-REDUCTION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

24.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness.

Generation rule: Generate the literal product of the eight registered Computability structural axes for relative reduction classes.

All generated finite relative reduction classes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-description-domain	complete-generated-description-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-execution-trace	exact-recognition-execution-trace retains the complete registered execution condition.
result	implicit-nontermination	implicit-nontermination removes a required exact result condition.	accept-reject-continue-classes	accept-reject-continue-classes retains the complete registered result condition.
self_reference	self-reference-excluded	self-reference-excluded removes a required exact self_reference condition.	generated-self-description-support	generated-self-description-support retains the complete registered self_reference condition.
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

24.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

24.4 Operational laws and consequences

- Carrier law: registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

24.5 Depth-independent closure

Base: The structural One supplies one canonical relative reduction classes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated relative reduction classes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

24.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-DEGREES-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-DEGREES-001; result: PASS.

- **computability-witness-3** - computability exact operational witness 3 for SFT-COMP-CBL-DEGREES-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.	PASS

24.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: Turing degree, relative computability.

Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

24.8 Exact evidence identities

- Source manifest: sha256:e69d3de1e807f51f5435b2fd279dc63ebf903fc1a3abec1469dd6c567ad50189
- Complete-census certificate:
sha256:7fdd77daa0ed7de987d15f92325c15b5a0f9034535436080d3ade6b19fc489bd
- Derivation seal: sha256:ad0d2f09c99201752cf963100d4b989df8bd6504cc4a12212f545eb45272c378
- Independent implementation:
sha256:54e92220b00e34ab45dec83999c26efc88c4604686cf8bc98866a1143eabf7a3
- Independent certificate:
sha256:4beaa5ffed228284c3bd0b27c642a6bceb0005d9de29c5b986fba873b1ac97d8
- External validation:
sha256:d8e21565c305b5d29b57f36fd9d95aae62b6254c03faebc6c413c682b05b2e8f
- Model-admitted engine receipt:
sha256:40189bc4fb7d083c3ffc7a7610e29e77cc5232bbde8514860c3afa328fe11a3e

- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-DEGREES-001-40189bc4fb7d083c.json

25. Derivation 18: Fold universal-machine execution law

Claim: SFT-COMP-CBL-UNIVERSAL-MACHINE-001

25.1 Scientific necessity and exact theorem

Fold universal-machine execution law is required to derive one machine executing encoded machines without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-DEGREES-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

25.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation.

Generation rule: Generate the literal product of the eight registered Computability structural axes for one machine executing encoded machines.

All generated finite one machine executing encoded machines structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nonterminat ion	implicit-nontermination removes a required exact result condition.	accept-reject-conti nue-classes	accept-reject-continue-clas ses retains the complete registered result condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
self_reference	self-reference-excluded	self-reference-excluded removes a required exact self_reference condition.	generated-self-description-support	generated-self-description-support retains the complete registered self_reference condition.
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

25.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

25.4 Operational laws and consequences

- Carrier law: frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

25.5 Depth-independent closure

Base: The structural One supplies one canonical one machine executing encoded machines form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated one machine executing encoded machines form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

25.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.	PASS

25.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal machine, interpreter.

Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.

25.8 Exact evidence identities

- Source manifest: sha256:f1cddb7a0abca54bfc082e210aae0129fe23a0303fb9d8dd4e76735185692d7a
- Complete-census certificate:
sha256:630ccc0d85a8e9228e341632fca4d29fb52eabd3792dc22a2fa8f6d4a386b652
- Derivation seal: sha256:bc727d9f76e920560ca4a2ff6fc8864fcd4da986c10d8e3db80fe1df09d8320
- Independent implementation:
sha256:f2890836bf1bba1f7dbc2c6641b34583b0906fcc25d4099cc5d993326b21ec85
- Independent certificate:
sha256:d830526fc80fceff9278895cdb542cd9a72d6da659acaa0e60aa47d12841f058
- External validation:
sha256:221ca0caa7445de51094f8fc76c69b2292b6118acf5bd19dc52f36acce24abe6
- Model-admitted engine receipt:
sha256:63466fb169111c50266901bf7ac48255ced18ef727cd66d1f3b376b6beaa7633
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CBL-UNIVERSAL-MACHINE-001-63466fb169111c50.json

26. Derivation 19: Fold undecidable-problem law

Claim: SFT-COMP-CBL-UNDECIDABILITY-001

26.1 Scientific necessity and exact theorem

Fold undecidable-problem law is required to derive failure of total internal decision without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-UNIVERSAL-MACHINE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

26.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining; generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace.

Generation rule: Generate the literal product of the eight registered Computability structural axes for failure of total internal decision.

All generated finite failure of total internal decision structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The theorem applies where the registered language supports the exact self-encoding and complement operation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nonterminat ion	implicit-nontermination removes a required exact result condition.	accept-reject-contin ue-classes	accept-reject-continue-clas ses retains the complete registered result condition.
self_reference	self-reference-exclu ded	self-reference-excluded removes a required exact self_reference condition.	generated-self-descr iption-support	generated-self-description-s upport retains the complete registered self_reference condition.
totality	assumed-total-decide r	assumed-total-decider removes a required exact totality condition.	explicit-totality-or -partiality-boundary	explicit-totality-or-partiality-b oundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-re duction	answer-preserving-reductio n retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successo r-certificate	description-successor-certifi cate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-comput ational-power	no-undeclared-computation al-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

26.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.

Eliminated candidates	Exact first-failure reason
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

26.4 Operational laws and consequences

- Carrier law: generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

26.5 Depth-independent closure

Base: The structural One supplies one canonical failure of total internal decision form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated failure of total internal decision form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

26.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-UNDECIDABILITY-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-UNDECIDABILITY-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-UNDECIDABILITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS

Control	Expected	Observed	Result
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The theorem applies where the registered language supports the exact self-encoding and complement operation.	PASS

26.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: undecidability, diagonal argument.

The theorem applies where the registered language supports the exact self-encoding and complement operation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- The theorem applies where the registered language supports the exact self-encoding and complement operation.

26.8 Exact evidence identities

- Source manifest: sha256:d10fd9322aafb6432ebcb434fff6f1bd31e256c7c5ce93ae9a894b8bcf568786
- Complete-census certificate:
sha256:f0655bf4421f5a52dfb30810d18c76896acfc3ef36d945c5672151a2d426da
- Derivation seal: sha256:42eefceee42f1ef84a8b765da3ecefbc91a79c5bfe7ce3cc6019e6ceddccc5d7
- Independent implementation:
sha256:521c56670dedb8d993cca8984ca42bd99784d517bc12cb1989e071738a387845
- Independent certificate:
sha256:8c3b3c911d7f33c654b756f4f59224e8614b0e9cdfcbff736daac8ce0bcf13f6
- External validation:
sha256:b2820d28d6464e2afcfae7ee3eb56bb845b3a192556fb0cf0c7e6774d73bb5c0
- Model-admitted engine receipt:
sha256:9de0276cb48ddbcea65e0a35f6f2deecac179a225ac85a8f93121e5e7ed97e692
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-UNDECIDABILITY-001-9de0276cb48ddbce.json

27. Derivation 20: Fold incompleteness-boundary law

Claim: SFT-COMP-CBL-INCOMPLETENESS-001

27.1 Scientific necessity and exact theorem

Fold incompleteness-boundary law is required to derive internal proof and self-verification limits without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-UNDECIDABILITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

27.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle.

Generation rule: Generate the literal product of the eight registered Computability structural axes for internal proof and self-verification limits.

All generated finite internal proof and self-verification limits structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-descriptions	sampled-descriptions removes a required exact domain condition.	complete-generated-description-domain	complete-generated-description-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-execution-trace	exact-recognition-execution-trace retains the complete registered execution condition.
result	implicit-nontermination	implicit-nontermination removes a required exact result condition.	accept-reject-continue-classes	accept-reject-continue-classes retains the complete registered result condition.
self_reference	self-reference-excluded	self-reference-excluded removes a required exact self_reference condition.	generated-self-description-support	generated-self-description-support retains the complete registered self_reference condition.
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

27.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

27.4 Operational laws and consequences

- Carrier law: effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

27.5 Depth-independent closure

Base: The structural One supplies one canonical internal proof and self-verification limits form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated internal proof and self-verification limits form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

27.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-INCOMPLETENESS-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-INCOMPLETENESS-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-INCOMPLETENESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.	PASS

27.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: incompleteness, self-reference.

This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model

- This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.

27.8 Exact evidence identities

- Source manifest: sha256:563f132c7bc3d7b393606cd0cfb8678f07958ceaad483f330de156a08b2ad0eb
- Complete-census certificate:
sha256:bece03b200d54ec76d2f512a5f16addc33681cf5c796038f1804ab24e733580e
- Derivation seal: sha256:39ccdf0743514fba3f9e01e41a67dfa4686805175ccd0fe687e6316aef358541
- Independent implementation:
sha256:fce65fcb70bbdfe02de16c19f5f28328658fbf8d9c78050319b98a2baf774a3
- Independent certificate:
sha256:f6f7786d6185c4e020b9031361db598d967b3772248ee3c7add2b0721457ab2b
- External validation:
sha256:0ba02558215ddcb70efdb09565b963fcafeff462fd6b3410a05e0721472c967d
- Model-admitted engine receipt:
sha256:f7fb969bc4fc339ff473e5ec2620e703695e6a12c3ab0b704594e9664e9ea277
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-INCOMPLETENESS-001-f7fb969bc4fc339f.json

28. Derivation 21: Fold relative and oracle-computation law

Claim: SFT-COMP-CBL-RELATIVE-ORACLE-001

28.1 Scientific necessity and exact theorem

Fold relative and oracle-computation law is required to derive explicit external answer interfaces without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-INCOMPLETENESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

28.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary.

Generation rule: Generate the literal product of the eight registered Computability structural axes for explicit external answer interfaces.

All generated finite explicit external answer interfaces structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An oracle is a registered interface, never unexplained computational power.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-descriptions	sampld-descriptions removes a required exact domain condition.	complete-generated-description-domain	complete-generated-description-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-execution-trace	exact-recognition-execution-trace retains the complete registered execution condition.
result	implicit-nontermination	implicit-nontermination removes a required exact result condition.	accept-reject-continue-classes	accept-reject-continue-classes retains the complete registered result condition.
self_reference	self-reference-excluded	self-reference-excluded removes a required exact self_reference condition.	generated-self-description-support	generated-self-description-support retains the complete registered self_reference condition.
totality	assumed-total-decider	assumed-total-decider removes a required exact totality condition.	explicit-totality-or-partiality-boundary	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

28.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

28.4 Operational laws and consequences

- Carrier law: base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

28.5 Depth-independent closure

Base: The structural One supplies one canonical explicit external answer interfaces form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated explicit external answer interfaces form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

28.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: PASS.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: PASS.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An oracle is a registered interface, never unexplained computational power.	PASS

28.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: oracle machine, relative computation.

An oracle is a registered interface, never unexplained computational power.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An oracle is a registered interface, never unexplained computational power.

28.8 Exact evidence identities

- Source manifest: sha256 : 12c67c86c538f90bc4536e45613aacaf1319d5f5ac02c0ae08396bca805a237e
- Complete-census certificate:
sha256 : 85464270056da95e16f3985a90c6410a0652421d6d3a4e2d9e7e0a5ab78f9edd
- Derivation seal: sha256 : 6c056bd1d9d6e58d1c4cf9eb93bb767deeb145a363a6a7d85884a2696008114c
- Independent implementation:
sha256 : 98e5f6212def0f26cd0cb6cfa352183ebdebef792aaa5184499fea5eb380abc64
- Independent certificate:
sha256 : f9757a676f3589d0cce885d22cb45b44377fd1720dcbb2953cee994b0c89911d
- External validation:
sha256 : 718f8f589221365d114308744266476d1bca0bb9476be8e68a2ca1acde67b7be
- Model-admitted engine receipt:
sha256 : 2a09b83377a56d06136df6922d9e9830c3b336e3487d3654d349c7f6f6746d01
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CBL-RELATIVE-ORACLE-001-2a09b83377a56d06.json

29. Derivation 22: Fold hypercomputation admissibility-limit law

Claim: SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001

29.1 Scientific necessity and exact theorem

Fold hypercomputation admissibility-limit law is required to derive claims exceeding generated universal computation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-RELATIVE-ORACLE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

29.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition.

Generation rule: Generate the literal product of the eight registered Computability structural axes for claims exceeding generated universal computation.

All generated finite claims exceeding generated universal computation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-descriptions	sampled-descriptions removes a required exact domain condition.	complete-generated-d escription-domain	complete-generated-descri ption-domain retains the complete registered domain condition.
execution	verdict-without-run	verdict-without-run removes a required exact execution condition.	exact-recognition-ex ecution-trace	exact-recognition-execution -trace retains the complete registered execution condition.
result	implicit-nonterminat ion	implicit-nontermination removes a required exact result condition.	accept-reject-conti nue-classes	accept-reject-continue-clas ses retains the complete registered result condition.
self_reference	self-reference-exclu ded	self-reference-excluded removes a required exact self_reference condition.	generated-self-descr iption-support	generated-self-description-s upport retains the complete registered self_reference condition.
totality	assumed-total-decide r	assumed-total-decider removes a required exact totality condition.	explicit-totality-or -partiality-boundary	explicit-totality-or-partiality-b oundary retains the complete registered totality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
translation	answer-changing-map	answer-changing-map removes a required exact translation condition.	answer-preserving-reduction	answer-preserving-reduction retains the complete registered translation condition.
generality	fixed-depth-table	fixed-depth-table removes a required exact generality condition.	description-successor-certificate	description-successor-certificate retains the complete registered generality condition.
addition	hidden-oracle	hidden-oracle removes a required exact addition condition.	no-undeclared-computational-power	no-undeclared-computational-power retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

29.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

29.4 Operational laws and consequences

- Carrier law: claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

29.5 Depth-independent closure

Base: The structural One supplies one canonical claims exceeding generated universal computation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated claims exceeding generated universal computation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

29.6 Executed witnesses and adverse controls

- **computability-witness-1** - computability exact operational witness 1 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: PASS.
- **computability-witness-2** - computability exact operational witness 2 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: PASS.
- **computability-witness-3** - computability exact operational witness 3 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.	PASS

29.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: hypercomputation, computational limit.

No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.

29.8 Exact evidence identities

- Source manifest: sha256:cb42d36a49aefee2c1b65da8c5e0a504792b30b9cfe777a69fc01b4f98608753
- Complete-census certificate:
sha256:413dc020c84636e97a0bb1b2a86bc46cb24996c7a7092b8d0f47f361b5160a93
- Derivation seal: sha256:5b708f88bd4ff06b23706597a4ea720d6df25d109b06dd0aa7201a06267ad085
- Independent implementation:
sha256:6b6764bb33684418d2d828cf9c1bc882739d517804a2ece25e8c1e56b79a6357
- Independent certificate:
sha256:37c0063e48172aa654b2841935cde94ed2c44a732321a91de3cc86b2a5b58449
- External validation:
sha256:e4a6b799d7ea881df152750b839cf35c871b97e84a80484817303e0fec561d43
- Model-admitted engine receipt:
sha256:3d6f7b8fdd8b1ccaf30622dad922140dd6b441b19df9dd4b76ff17ee191439cc
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001-3d6f7b8fdd8b1cca.json

30. Derivation 23: Fold input and problem-size law

Claim: SFT-COMP-CPLX-INPUT-SIZE-001

30.1 Scientific necessity and exact theorem

Fold input and problem-size law is required to derive exact generated input extent without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

30.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for exact generated input extent.

All generated finite exact generated input extent structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Size is representation-relative unless an exact encoding correspondence is supplied.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

30.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.

Eliminated candidates	Exact first-failure reason
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

30.4 Operational laws and consequences

- Carrier law: canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

30.5 Depth-independent closure

Base: The structural One supplies one canonical exact generated input extent form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact generated input extent form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

30.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-INPUT-SIZE-001; result: PASS.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-INPUT-SIZE-001; result: PASS.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-INPUT-SIZE-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Size is representation-relative unless an exact encoding correspondence is supplied.	PASS

30.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: input size, problem size.

Size is representation-relative unless an exact encoding correspondence is supplied.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Size is representation-relative unless an exact encoding correspondence is supplied.

30.8 Exact evidence identities

- Source manifest: sha256 : c2cd5867ed1ede45da97082c15db4f5b544ab67e61633b2603322cd89f503b9b
- Complete-census certificate:
sha256 : a383f71b94033d5c6c7aafc4fc9ee43e526903df217ae1371330c894f2483a5b
- Derivation seal: sha256 : 59a9b0be31a5e733dce860eedf4bda67bde61c23ab06024202d6a115147f04
- Independent implementation:
sha256 : 8b11f954a55f4184895929a2a1a8e8e2cc7edbf0510d5f079bec7017d7bda6ca
- Independent certificate:
sha256 : 749367f01a05e86db5ae00d7d66abf0b6dcd40d036f06f166c7dcef695d99511
- External validation:
sha256 : 9f28b7906f4b1e97bb7044a480308141fd074adb36c8a1d57e5bdd05b9ba2bd7
- Model-admitted engine receipt:
sha256 : 3f002d11d6315fc6f32536022cefb2a3fad2bf857ea1761637481a827a8a2fc4
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-INPUT-SIZE-001-3f002d11d6315fc6.json

31. Derivation 24: Fold time and space resource law

Claim: SFT-COMP-CPLX-TIME-SPACE-001

31.1 Scientific necessity and exact theorem

Fold time and space resource law is required to derive transition and retained-state resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-INPUT-SIZE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

31.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for transition and retained-state resources.

All generated finite transition and retained-state resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Counts are exact host reports over generated traces, not foundational magnitudes.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

31.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

31.4 Operational laws and consequences

- Carrier law: complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

31.5 Depth-independent closure

Base: The structural One supplies one canonical transition and retained-state resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated transition and retained-state resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

31.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-TIME-SPACE-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-TIME-SPACE-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-TIME-SPACE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Counts are exact host reports over generated traces, not foundational magnitudes.	PASS

31.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: time complexity, space complexity.

Counts are exact host reports over generated traces, not foundational magnitudes.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Counts are exact host reports over generated traces, not foundational magnitudes.

31.8 Exact evidence identities

- Source manifest: sha256:f155e77d53f592a5b06eee8f93f6ad97642e0feaa0c05a7458ff978b5c5faced
- Complete-census certificate:
sha256:6bc159a0c45385cc05dd89ba536edb90328cbdac03491c007940974c948238a
- Derivation seal: sha256:1fcc740897abe9b3adb6bdc54f8f5a7a4a1908d1da460d36b7751a0ba3fa0e36

- Independent implementation:
sha256:55a51257da5a486414684d2eb845f65827c1a319fb2712f3619bf3da0868e96c
- Independent certificate:
sha256:d69c776f3a72bdfef046e514a487a8e8365e1bfc47b33eb1827d7358db5dafb46
- External validation:
sha256:8669c1478eabab8093f09aae4b52e18a24ecc93e744f3ae2099c67c15121a68d
- Model-admitted engine receipt:
sha256:d6187e34cd91181461fa6614ebb82d710719cf6bebf4644d1bc08c81641df62b
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-TIME-SPACE-001-d6187e34cd911814.json

32. Derivation 25: Fold circuit size and depth law

Claim: SFT-COMP-CPLX-CIRCUIT-RESOURCE-001

32.1 Scientific necessity and exact theorem

Fold circuit size and depth law is required to derive circuit structural resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-TIME-SPACE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

32.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for circuit structural resources.

All generated finite circuit structural resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes resource accounting, not arbitrary lower bounds.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

32.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one

tuple composed entirely of admitted coordinates.

32.4 Operational laws and consequences

- Carrier law: generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

32.5 Depth-independent closure

Base: The structural One supplies one canonical circuit structural resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated circuit structural resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

32.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes resource accounting, not arbitrary lower bounds.	PASS

32.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: circuit size, circuit depth.

Closes resource accounting, not arbitrary lower bounds.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes resource accounting, not arbitrary lower bounds.

32.8 Exact evidence identities

- Source manifest: sha256:846f4e21718ad950f5f8024aeb3a157f822c73a67e93ca085d618260f70a1f38
- Complete-census certificate:
sha256:6c88b89eb6f1dfbe9db72ad1fbcecb9989ba7f92e6df6f89901f8fe56410691e
- Derivation seal: sha256:11d0d9927aa1c5cb64db2faeed8e29d4aad74135c81b3c68ea8ad080de297144
- Independent implementation:
sha256:12e9bb6516941807a16f20797673f5421d0a4446a86e73aca97773d5d57fccfc
- Independent certificate:
sha256:1a0022829a4bdd87943253b70fa5057822a6a456dbf6d8cf56f7a996826d3909
- External validation:
sha256:f9af184469e9d277ee03491a44a563a9f131f5a86b3c7fd325ba362b0de85f57
- Model-admitted engine receipt:
sha256:9f3e28733488ad63f954689caad30204963adbe8e2fa60f985be1140722116ae
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CPLX-CIRCUIT-RESOURCE-001-9f3e28733488ad63.json

33. Derivation 26: Fold communication and query complexity law

Claim: SFT-COMP-CPLX-COMMUNICATION-QUERY-001

33.1 Scientific necessity and exact theorem

Fold communication and query complexity law is required to derive cross-interface and information-request resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-CIRCUIT-RESOURCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

33.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for cross-interface and information-request resources.

All generated finite cross-interface and information-request resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The exact protocol and oracle identity must be registered.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witnesses	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

33.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.

Eliminated candidates	Exact first-failure reason
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

33.4 Operational laws and consequences

- Carrier law: distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

33.5 Depth-independent closure

Base: The structural One supplies one canonical cross-interface and information-request resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated cross-interface and information-request resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

33.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: PASS.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: PASS.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The exact protocol and oracle identity must be registered.	PASS

33.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: communication complexity, query complexity.

The exact protocol and oracle identity must be registered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- The exact protocol and oracle identity must be registered.

33.8 Exact evidence identities

- Source manifest: sha256 : c302cd798c3e0e45d7a29366ae64e227b6c5813fe121600d6e5e96f650f2d05a
- Complete-census certificate:
sha256 : f27a6346926b89a426e1b2639d1cbabe7aaaf222fa966b033932878d073eaeaf2
- Derivation seal: sha256 : 7d91bbd1b18c3aefb5ca8eb72d84964027c55fe09b733ffe688cf19e2491ac41
- Independent implementation:
sha256 : ca88ab267912a1c325ce19967fa6f9bb5104be65e508a42692808033bf18b3d7
- Independent certificate:
sha256 : b62d6baea8c3fb0502d436d71446deab6f9277fe93e8b49fb2845f1380457cb5
- External validation:
sha256 : 10745814f262dafcab4969ed2534761d0e23085f68e6f31faa232b543b41d953
- Model-admitted engine receipt:
sha256 : c352ad5637226cc1d41ecbc8fb375f9593e7928d4157a97e8b7050ac18d19f3e
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CPLX-COMMUNICATION-QUERY-001-c352ad5637226cc1.json

34. Derivation 27: Fold randomness-resource law

Claim: SFT-COMP-CPLX-RANDOMNESS-001

34.1 Scientific necessity and exact theorem

Fold randomness-resource law is required to derive scheduled unresolved branch support without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-COMMUNICATION-QUERY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

34.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for scheduled unresolved branch support.

All generated finite scheduled unresolved branch support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witnesses	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

34.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

34.4 Operational laws and consequences

- Carrier law: complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

34.5 Depth-independent closure

Base: The structural One supplies one canonical scheduled unresolved branch support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated scheduled unresolved branch support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

34.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-RANDOMNESS-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-RANDOMNESS-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-RANDOMNESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.	PASS

34.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: randomized complexity, random bits.

Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.

34.8 Exact evidence identities

- Source manifest: sha256:0607afa19714526b32d36403953ce2de74281e7b9310eb134693acf7ed442306
- Complete-census certificate:
sha256:adc6a5a3ebdc171d8e1332679a85020c371f2c5e4d12833b06ae098c96aed420
- Derivation seal: sha256:11553fb594b83f863eb99c1dcda12b7c55670a95e85ddeb507f0e96436cba33c
- Independent implementation:
sha256:e86d848ef0f31cd3a4e10cbe82a0235e50943a06e0d9ffa5e5744c6482459
- Independent certificate:
sha256:dedc0a421764c9d7249870484673717c898c3cddf7ed33494cec492522a3dc34
- External validation:
sha256:25f8a328196851e3b4fc33030d39baa1877734e8d287b0a9515d51206eeb02fa
- Model-admitted engine receipt:
sha256:e4311dc1af067c40ad88072efc8c155d1647ad6b071c1003c48b0e597be64f8b
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-RANDOMNESS-001-e4311dc1af067c40.json

35. Derivation 28: Fold reversibility-cost law

Claim: SFT-COMP-CPLX-REVERSIBILITY-COST-001

35.1 Scientific necessity and exact theorem

Fold reversibility-cost law is required to derive information retained for inverse execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-RANDOMNESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

35.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for information retained for inverse execution.

All generated finite information retained for inverse execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. This is the classical cost law; the universal reversible model belongs to the later quantum branch.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witnesses	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

35.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.

Eliminated candidates	Exact first-failure reason
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

35.4 Operational laws and consequences

- Carrier law: source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

35.5 Depth-independent closure

Base: The structural One supplies one canonical information retained for inverse execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated information retained for inverse execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

35.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: PASS.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: PASS.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is the classical cost law; the universal reversible model belongs to the later quantum branch.	PASS

35.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reversible computation, Landauer cost.

This is the classical cost law; the universal reversible model belongs to the later quantum branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- This is the classical cost law; the universal reversible model belongs to the later quantum branch.

35.8 Exact evidence identities

- Source manifest: sha256:43e6f36f3da6e8a178ac4fbdf14531b45ca6e55ac09ac5b4ec9a1526021b5903
- Complete-census certificate:
sha256:2b9ccd74c371ca0b385c8062244e6b0c5a3d373db772dc2719405e107b85d5a2
- Derivation seal: sha256:8593ee37fadeb80b440d656e826431594044f76793053417dff646dc6da86bdd
- Independent implementation:
sha256:62112db56cc1d6dde75ba41a7ba2e9b84a435cf17bc81262ad9d5e11c008d868
- Independent certificate:
sha256:ad19f8f114159728b4ae9507a73e5c76f3cd94ca6d18e5586069be460d268278
- External validation:
sha256:657b65250d381219f7c9577a80006d61e4e72c8b53388e955a68de6eceff7cce
- Model-admitted engine receipt:
sha256:0a8552676c105fc6b3fdcd6aac90a7a68c428962c209460caacea2b81c496ed2
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CPLX-REVERSIBILITY-COST-001-0a8552676c105fc6.json

36. Derivation 29: Fold parallel-complexity law

Claim: SFT-COMP-CPLX-PARALLEL-001

36.1 Scientific necessity and exact theorem

Fold parallel-complexity law is required to derive independent and causally layered work without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

36.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for independent and causally layered work.

All generated finite independent and causally layered work structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical processor count or timing constant is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witnesses	constructive-or-adversarial-bound-witness retains the complete registered witness condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

36.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

36.4 Operational laws and consequences

- Carrier law: task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

36.5 Depth-independent closure

Base: The structural One supplies one canonical independent and causally layered work form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated independent and causally layered work form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

36.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-PARALLEL-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-PARALLEL-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-PARALLEL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical processor count or timing constant is imported.	PASS

36.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parallel complexity, work and span.

No physical processor count or timing constant is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No physical processor count or timing constant is imported.

36.8 Exact evidence identities

- Source manifest: sha256 : 2a0ee9adf4736481f873cd67f7b4e348d44ebdc98940be3503dfe98fd403e4ba

- Complete-census certificate:
sha256:810c108366d42966f12b55652d027a8ca3e89c684d4f81e136d9774e6b0c7066
- Derivation seal: sha256:15c1ba1323fe5baa1b8635a1bda7868db77477aa1f0c4981e4b11dc06af9f4ad
- Independent implementation:
sha256:81ade383c13f9b93f5df4bc48531bcf5eaf6a2c6bb816dc89b3737ef6dc9ebae
- Independent certificate:
sha256:dfbd1f51e697f60b15a455914c43b409255647b875c4a9882bb008ef789e72ea
- External validation:
sha256:82c39a2ca55f4e9b10d7d4c7ff997b61bcd52472e2fca5ca2da470993b1b72fa
- Model-admitted engine receipt:
sha256:d451551a66fdaf127bfd2e9738af1c7818753ea3dd2862a46db3108e32354424
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-PARALLEL-001-d451551a66fdaf12.json

37. Derivation 30: Fold lower- and upper-bound law

Claim: SFT-COMP-CPLX-BOUNDS-001

37.1 Scientific necessity and exact theorem

Fold lower- and upper-bound law is required to derive resource comparison certificates without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-PARALLEL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

37.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource comparison certificates.

All generated finite resource comparison certificates structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

37.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation

is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

37.4 Operational laws and consequences

- Carrier law: problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

37.5 Depth-independent closure

Base: The structural One supplies one canonical resource comparison certificates form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource comparison certificates form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

37.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-BOUNDS-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-BOUNDS-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-BOUNDS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.	PASS

37.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: lower bound, upper bound.

Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.

37.8 Exact evidence identities

- Source manifest: sha256:93370860be77934c39962130d5c50209ea5eb1f0f631d4300fd00e9f9ddf41d9
- Complete-census certificate:
sha256:930b712b9a8d08fa4fa577fc7fc83fee43ac2b8899c1c588cf8a58361f027c37
- Derivation seal: sha256:1cec4f2834aa22a4ecc619ed0c0541e1788ae7f620901e4c0c3065eb2ee7ac2d
- Independent implementation:
sha256:7bd2df392101a75d5c8de34cd3952cbe8cbb50fffdade0927168f2afffbde223f
- Independent certificate:
sha256:b347fc3d05e9cfd9931f0178ded39a23316344dd56773e09696d783ba933a55f
- External validation:
sha256:f0bfb44bbb62a3b09c8cd4e1fe8751b4d016a368f0f3a516a00af53052f53b35
- Model-admitted engine receipt:
sha256:7ba9b6310c1f0dd170c3c9b6abc0db4c257241df7af511faf3fb232b2d39db02
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-BOUNDS-001-7ba9b6310c1f0dd1.json

38. Derivation 31: Fold reduction and completeness law

Claim: SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001

38.1 Scientific necessity and exact theorem

Fold reduction and completeness law is required to derive resource-preserving problem translations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-BOUNDS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

38.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource-preserving problem translations.

All generated finite resource-preserving problem translations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Completeness is relative to the frozen problem class and resource law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witnesses	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

38.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

38.4 Operational laws and consequences

- Carrier law: problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

38.5 Depth-independent closure

Base: The structural One supplies one canonical resource-preserving problem translations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource-preserving problem translations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

38.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: PASS.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: PASS.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Completeness is relative to the frozen problem class and resource law.	PASS

38.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: complexity reduction, complete problem.

Completeness is relative to the frozen problem class and resource law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Completeness is relative to the frozen problem class and resource law.

38.8 Exact evidence identities

- Source manifest: sha256:d4ea3f50576d38baaeb22d352a42da0dda94019c0bcc5eeb3aab586c6f8b6709
- Complete-census certificate:
sha256:171884d544593303d4d7dbe208b8ff54569501ca5f9c64c74831f5ad091d5596
- Derivation seal: sha256:7ce5c8b6a21352a4cfd7f7441d39a955314633a9070682103521f83cfb41358d
- Independent implementation:
sha256:12fc9feb0f395be2df932ad6211deaef3a7051d9638752248702b32398313bec
- Independent certificate:
sha256:3710e881e46bb1cbd8b1788064009f6e420e795dcbbdc2140690ed5e2bda71f46
- External validation:
sha256:2adb304bdade25973af95ecd12fb91806996bbd575d4835b7a1dbde2357df1ea
- Model-admitted engine receipt:
sha256:401b5e76d63dd3d31c2a38156dd3fab752e10b289cb7b0e38c9a71f0cf6f673a
- Receipt path: receipts/engine/model_admitted/SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001-401b5e76d63dd3d3.json

39. Derivation 32: Fold average- and worst-case law

Claim: SFT-COMP-CPLX-AVERAGE-WORST-001

39.1 Scientific necessity and exact theorem

Fold average- and worst-case law is required to derive complete instance-family resource profiles without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

39.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for complete instance-family resource profiles.

All generated finite complete instance-family resource profiles structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Average claims require a complete exact support relation or registered empirical distribution.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

39.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

39.4 Operational laws and consequences

- Carrier law: generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

39.5 Depth-independent closure

Base: The structural One supplies one canonical complete instance-family resource profiles form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete instance-family resource profiles form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

39.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Average claims require a complete exact support relation or registered empirical distribution.	PASS

39.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: average-case complexity, worst-case complexity.

Average claims require a complete exact support relation or registered empirical distribution.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Average claims require a complete exact support relation or registered empirical distribution.

39.8 Exact evidence identities

- Source manifest: sha256:70c3dc367ca77dbd939a780cf7faa396cd6ada546544332003b8cb8b7f65f17f
- Complete-census certificate:
sha256:6be68eff0da24ae4189a24799092c315c3f7419fdc6b4b6daf67661d285d7109
- Derivation seal: sha256:9d6f8834fc4e6c4d7772ac100765066e8cd32a49ac2aca16328984c5567d3aa2
- Independent implementation:
sha256:9fc79651d13c51167b280ff22f28bd599d3a56d69ba36f3b6aaf41f23a8f572e
- Independent certificate:
sha256:59c844f91b8949eeca3fab5729be02bfc32aa99351bead60cb5f32e2c4c27a9
- External validation:
sha256:9d98937e33feb40baa49f42b0a1e9d95c8cac0852d0c5ce6b89db55f6d8c0da4
- Model-admitted engine receipt:
sha256:231135bfa6605c585af24c7a6f89d21b099f42462dc4b7a6c0820d761bce6bfb
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-AVERAGE-WORST-001-231135bfa6605c58.json

40. Derivation 33: Fold approximation-complexity law

Claim: SFT-COMP-CPLX-APPROXIMATION-001

40.1 Scientific necessity and exact theorem

Fold approximation-complexity law is required to derive resource versus retained objective distinction without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-AVERAGE-WORST-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

40.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource versus retained objective distinction.

All generated finite resource versus retained objective distinction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No unregistered metric or irrational ratio is admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

40.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

40.4 Operational laws and consequences

- Carrier law: optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

40.5 Depth-independent closure

Base: The structural One supplies one canonical resource versus retained objective distinction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource versus retained objective distinction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

40.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-APPROXIMATION-001; result: PASS.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-APPROXIMATION-001; result: PASS.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-APPROXIMATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No unregistered metric or irrational ratio is admitted.	PASS

40.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: approximation complexity, approximation ratio.

No unregistered metric or irrational ratio is admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No unregistered metric or irrational ratio is admitted.

40.8 Exact evidence identities

- Source manifest: sha256 : da988e508999ce8992454d7b760a8e0c8cb256e20bc8bcb1c553ee2d478e0f6c
- Complete-census certificate:
sha256 : 01715add870cff6f558b949706fbbf68b286c955c637a909e74cc0ef5f737968
- Derivation seal: sha256 : 661876c7efa17773960092329223bd1b8d3810c74167337614c4ca3861010c6f
- Independent implementation:
sha256 : 49e8ad275c48b129eb12592995af3840143a8ca0dc7ebb66d8a7669f64daaf86
- Independent certificate:
sha256 : 514e851668812207b41ed55e5cc45d02f101c37855b6c84347a0da2cceedff34
- External validation:
sha256 : 71afb1d9276d8465dd2707358515b59d69b86fd671bca76be7680464eb4d47f3
- Model-admitted engine receipt:
sha256 : e8062d9de51f472dc26ca19cd2e838de843c9b1449d16c71962ad2c06e9f8357
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-APPROXIMATION-001-e8062d9de51f472d.json

41. Derivation 34: Fold parameterized-complexity law

Claim: SFT-COMP-CPLX-PARAMETERIZED-001

41.1 Scientific necessity and exact theorem

Fold parameterized-complexity law is required to derive resource stratification by exact structural coordinates without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-APPROXIMATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

41.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource stratification by exact structural coordinates.

All generated finite resource stratification by exact structural coordinates structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Parameters must be forced structural coordinates, never fitted constants.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

41.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

41.4 Operational laws and consequences

- Carrier law: input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

41.5 Depth-independent closure

Base: The structural One supplies one canonical resource stratification by exact structural coordinates form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource stratification by exact structural coordinates form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

41.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-PARAMETERIZED-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-PARAMETERIZED-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-PARAMETERIZED-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Parameters must be forced structural coordinates, never fitted constants.	PASS

41.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parameterized complexity, fixed-parameter tractability.

Parameters must be forced structural coordinates, never fitted constants.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Parameters must be forced structural coordinates, never fitted constants.

41.8 Exact evidence identities

- Source manifest: sha256 : a83039453f3937ef06d195c71447f74e854ce9c410a9860f00d5d2ef56c663bc
- Complete-census certificate:
sha256 : 432c7d44a7c5f93e6b63ab3f1c3c8796a4bdc6fdab72f650eaba82086d16e5c7
- Derivation seal: sha256 : f97fafb086546626bbe55c2f9544769c9b21ebf937e83b95a9c93439535d978a

- Independent implementation:
sha256:d10266176661ea8f4c2625bac8f828473fd9a287f9e056a12ff5358d8e7c2de9
- Independent certificate:
sha256:90751ccec65ad243aa877cde32c4ad9520033e0bcbcb110e0b293ea452b879295
- External validation:
sha256:0bd259151d1801258dce6352daa80fcef344a016412df453f479f07474c95eed
- Model-admitted engine receipt:
sha256:1afcbc7d33437e51f057b4a63715fc3c5e41890025b3d093edb4fbeb15b370ac
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-PARAMETERIZED-001-1afcbc7d33437e51.json

42. Derivation 35: Fold information and descriptive-complexity law

Claim: SFT-COMP-CPLX-DESCRIPTIVE-001

42.1 Scientific necessity and exact theorem

Fold information and descriptive-complexity law is required to derive shortest exact generated descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-PARAMETERIZED-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

42.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for shortest exact generated descriptions.

All generated finite shortest exact generated descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
input	presentation-length	presentation-length removes a required exact input condition.	canonical-encoding-support	canonical-encoding-support retains the complete registered input condition.
resource	unrecorded-cost	unrecorded-cost removes a required exact resource condition.	complete-resource-ledger	complete-resource-ledger retains the complete registered resource condition.
model	model-unspecified	model-unspecified removes a required exact model condition.	registered-computation-model	registered-computation-model retains the complete registered model condition.
comparison	unmatched-instances	unmatched-instances removes a required exact comparison condition.	common-family-comparison	common-family-comparison retains the complete registered comparison condition.
composition	cost-erasing-composition	cost-erasing-composition removes a required exact composition condition.	overhead-retaining-composition	overhead-retaining-composition retains the complete registered composition condition.
witness	claimed-bound	claimed-bound removes a required exact witness condition.	constructive-or-adversarial-bound-witness	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
generality	benchmark-only	benchmark-only removes a required exact generality condition.	family-successor-certificate	family-successor-certificate retains the complete registered generality condition.
addition	imported-scale-or-rate	imported-scale-or-rate removes a required exact addition condition.	no-extra-resource-parameter	no-extra-resource-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

42.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger;

every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

42.4 Operational laws and consequences

- Carrier law: frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

42.5 Depth-independent closure

Base: The structural One supplies one canonical shortest exact generated descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated shortest exact generated descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

42.6 Executed witnesses and adverse controls

- complexity-witness-1 - complexity exact operational witness 1 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: PASS.
- complexity-witness-2 - complexity exact operational witness 2 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: PASS.
- complexity-witness-3 - complexity exact operational witness 3 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.	PASS

42.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: Kolmogorov complexity, descriptive complexity.

Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.

42.8 Exact evidence identities

- Source manifest: sha256:87440f348bebb73171355c031df3baa6fd3c0a278a3bba811790e2f7384f2f76
- Complete-census certificate:
sha256:7b0a69da1344d382e3afbfbac17ebf150fa0aa40462fe8afa3be4a1b0ae6a4564
- Derivation seal: sha256:75c939154f2d150b36c52e8c27a9080fa0bb6df65b999fd5f9dd70dc6a71f6a4
- Independent implementation:
sha256:70a8a83cc65b2f6a44ed0abde6ef8fe9211e60fbdd5176225e943725e32c6a7a
- Independent certificate:
sha256:d7e0354f7e04bafa5d366e2c394282723134d2f980ce45bdaa624cd026b5b401
- External validation:
sha256:4014842120d59fd7cb3078960451a7ac342f183f44b6ebd8a8d41d39d8bb2352
- Model-admitted engine receipt:
sha256:bdbb14bfbdd53b05dd80b9b4fd0ea77e3f9eb303ab637ec6714029665a1f6eef
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-CPLX-DESCRIPTIVE-001-bdbb14bfbdd53b05.json

43. Derivation 36: Fold searching and ordering law

Claim: SFT-COMP-ALG-SEARCH-ORDER-001

43.1 Scientific necessity and exact theorem

Fold searching and ordering law is required to derive location and exact order construction without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-CPLX-DESCRIPTIVE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

43.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for location and exact order construction.

All generated finite location and exact order construction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only registered finite carriers and comparison relations are covered.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

43.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

43.4 Operational laws and consequences

- Carrier law: generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

43.5 Depth-independent closure

Base: The structural One supplies one canonical location and exact order construction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated location and exact order construction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

43.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-SEARCH-ORDER-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-SEARCH-ORDER-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-SEARCH-ORDER-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite carriers and comparison relations are covered.	PASS

43.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: search, sorting.

Only registered finite carriers and comparison relations are covered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only registered finite carriers and comparison relations are covered.

43.8 Exact evidence identities

- Source manifest: `sha256:5763f896d0c066371fc1ff9577905144f07e46154d64f325dc543c7f904ba536`
- Complete-census certificate:
`sha256:7a1636755ef57d0c54d8dfc12741169887017de61b65a9494dc13dbca4a02eb2`

- Derivation seal: sha256:5410a5a2e280a6c31796b5735e3a1e7a936ed1b57d35fd24fe35123ea1a905b0
- Independent implementation:
sha256:bc23f61ab78f015f932efa3d73e656a123c338d041692d13664cd89e5148ce46
- Independent certificate:
sha256:b960ead246c2b465341e9c2a29216e0f80b39e26b0f908dcc6495d376b333e77
- External validation:
sha256:db458ef04ab0525c0f379719e9d03e03c74effc48f16623c36ed8cca6bf8481e
- Model-admitted engine receipt:
sha256:915d9a65bf080d49519f15583d8ffab68375f99e1af014be611cdc25c6626c78
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-SEARCH-ORDER-001-915d9a65bf080d49.json

44. Derivation 37: Fold exact arithmetic-algorithm law

Claim: SFT-COMP-ALG-ARITHMETIC-001

44.1 Scientific necessity and exact theorem

Fold exact arithmetic-algorithm law is required to derive computation over exact positive parts without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-SEARCH-ORDER-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

44.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for computation over exact positive parts.

All generated finite computation over exact positive parts structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No floating, negative, irrational or imaginary proof value is admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

44.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported

answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

44.4 Operational laws and consequences

- Carrier law: canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

44.5 Depth-independent closure

Base: The structural One supplies one canonical computation over exact positive parts form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation over exact positive parts form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

44.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-ARITHMETIC-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-ARITHMETIC-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-ARITHMETIC-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating, negative, irrational or imaginary proof value is admitted.	PASS

44.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: arithmetic algorithm, exact arithmetic.

No floating, negative, irrational or imaginary proof value is admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No floating, negative, irrational or imaginary proof value is admitted.

44.8 Exact evidence identities

- Source manifest: sha256:14502ce7be60ed58afaa4d0ce977254936c109bfc2a10c6b31042dd74b2b5add
- Complete-census certificate:
sha256:fd5164391de2e2b5fdbfaecc8e9cfdb96d0595cce56a8b05faf9add5a870497f
- Derivation seal: sha256:c124e9fe09f662ef9e56992f8295e5145539549a2216e8b0d4e7ca2cbb9f5c49
- Independent implementation:
sha256:fc377822b8b452cd195b9c076cac4feb03a72b5808dc38b3aee01b1ad69b1e63
- Independent certificate:
sha256:0f9ad44ec485d994f209fa3832e1a9af89c20b5dc67c9f5e656db5992b1397a5
- External validation:
sha256:9715086d06bc1f22f0398fe1cfc9b9cea665f0a125bb5b497c4e8297f4826251c
- Model-admitted engine receipt:
sha256:af254db83c3bc4c67b1c2daa5195c8f74a1a9e4fbcd08b159c677579b9c1ab52
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-ARITHMETIC-001-af254db83c3bc4c6.json

45. Derivation 38: Fold string and sequence algorithm law

Claim: SFT-COMP-ALG-STRINGS-SEQUENCES-001

45.1 Scientific necessity and exact theorem

Fold string and sequence algorithm law is required to derive finite ordered symbol organizations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-ARITHMETIC-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

45.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite ordered symbol organizations.

All generated finite finite ordered symbol organizations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No ungenerated infinite stream is treated as complete.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-inputs	sampld-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains 256 candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are 1 survivors and 255 eliminations.

45.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

45.4 Operational laws and consequences

- Carrier law: complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

45.5 Depth-independent closure

Base: The structural One supplies one canonical finite ordered symbol organizations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite ordered symbol organizations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

45.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No ungenerated infinite stream is treated as complete.	PASS

45.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: string algorithm, sequence algorithm.

No ungenerated infinite stream is treated as complete.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No ungenerated infinite stream is treated as complete.

45.8 Exact evidence identities

- Source manifest: sha256 : 238638db306d480419c314ea9c5b665b7ec72f339f0f6fc4f7ddbd6be4752a55
- Complete-census certificate:
sha256 : 68c63d4c5988cb3f4429f991d48810522bb1f62a5e0760a844869e9c73f74d31
- Derivation seal: sha256 : 8959ca63fab8c4543e86b8aa029aea4a48ca8f66670d4ec2312ab32eee0b2a89
- Independent implementation:
sha256 : 20b535a46bd9170066a6f938dabfaa4ed83aedf5fb981eee653dcda6f8df5124
- Independent certificate:
sha256 : dc0b982cf407e60e815f8ec2d5045aaab3145ab9ccd3426d93a74d2531f68d84
- External validation:
sha256 : f6566f41be47c5ebd8b3fe42fe8bc341cd8a41f560280ba78f8bfd59001f8ad4
- Model-admitted engine receipt:
sha256 : 5c2fb101d63e8d37ba076927e1de853261936a53c33400eba91c9adbd135caaf
- Receipt path: receipts/engine/model_admitted/SFT-COMP-ALG-STRINGS-SEQUENCES-001-5c2fb101d63e8d37.json

46. Derivation 39: Fold tree and graph algorithm law

Claim: SFT-COMP-ALG-TREES-GRAPHS-001

46.1 Scientific necessity and exact theorem

Fold tree and graph algorithm law is required to derive finite relational traversal and structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-STRINGS-SEQUENCES-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

46.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite relational traversal and structure.

All generated finite finite relational traversal and structure structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Applies to finite generated graphs; physical networks require empirical translation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-inputs	sampld-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

46.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

46.4 Operational laws and consequences

- Carrier law: generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

46.5 Depth-independent closure

Base: The structural One supplies one canonical finite relational traversal and structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite relational traversal and structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

46.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-TREES-GRAPHS-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-TREES-GRAPHS-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-TREES-GRAPHS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies to finite generated graphs; physical networks require empirical translation.	PASS

46.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: graph algorithm, tree algorithm.

Applies to finite generated graphs; physical networks require empirical translation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model

- Applies to finite generated graphs; physical networks require empirical translation.

46.8 Exact evidence identities

- Source manifest: sha256:82e504863893dbb2b0f5068fd3e3e9fc4e68cbff929c6c8a053e94a86fc0b30e
- Complete-census certificate:
sha256:f1630f72ede9480e5df1a74cc9afb4babf396c07f5dfea3373dba6cc38f1f609
- Derivation seal: sha256:0b0a095159e3c047d835c3e4bbdecc2efa8b4986fe1693c2e5793a33fad1c63
- Independent implementation:
sha256:38f3cf31f1ade6c3a7438f948c812d8fc62eadcad6cca80e7bd97a55185fd25c
- Independent certificate:
sha256:8379189b1b6c280a2b2311ca5d84cb7843e834a887f3115d401407767ac06270
- External validation:
sha256:4deefa9e67a39f8f15daace40fc9734b41590d50fbc7ec7f2b6e2f58866b7f8d
- Model-admitted engine receipt:
sha256:0c1d76188752d9f584b9bec6db855143f954a2f65b9ad57375bcd3e68809f604
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-TREES-GRAPHS-001-0c1d76188752d9f5.json

47. Derivation 40: Fold algebraic and geometric algorithm law

Claim: SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001

47.1 Scientific necessity and exact theorem

Fold algebraic and geometric algorithm law is required to derive finite algebraic operations and incidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-TREES-GRAPHS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

47.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite algebraic operations and incidence.

All generated finite finite algebraic operations and incidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Geometry remains finite and exact; no ungenerated continuum is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

47.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

47.4 Operational laws and consequences

- Carrier law: exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

47.5 Depth-independent closure

Base: The structural One supplies one canonical finite algebraic operations and incidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite algebraic operations and incidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

47.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: PASS.

Control	Expected	Observed	Result
<code>false_promise</code>	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Geometry remains finite and exact; no ungenerated continuum is imported.	PASS

47.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: algebraic algorithm, computational geometry.

Geometry remains finite and exact; no ungenerated continuum is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Geometry remains finite and exact; no ungenerated continuum is imported.

47.8 Exact evidence identities

- Source manifest: sha256:bb9b39ce14861c2ebdd176f20ae62d50cdab3842075ae516f848c6896d4fc45e
- Complete-census certificate:
sha256:3aa7470cc7aec406c7c072cfcbbf452b237eac34ca06d5ec7247a264dc0fb1308
- Derivation seal: sha256:81409fcc27f9a8511b396209692bbc2a28b900f3c0dfdce8c4c534c0f033d7ef
- Independent implementation:
sha256:73d59f6f67dc89af760a385a65b33aad744e89a90b7c5dd722208a098432ea30
- Independent certificate:
sha256:e06d8cce5e05c56e774a1101d444b554e16e3cb7f4ab4bd21d0c31f1cd207e0f
- External validation:
sha256:779fd81207ac8c38ac5e261d9d80151e5868ca0d99cd774818697326c24817f3
- Model-admitted engine receipt:
sha256:b2ba1fc787d153cd8673cf1166703cd608291c7136d7412bff9c9dc66f3de430
- Receipt path: receipts/engine/model_admitted/SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001-b2ba1fc787d153cd.json

48. Derivation 41: Fold dynamic-programming law

Claim: SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001

48.1 Scientific necessity and exact theorem

Fold dynamic-programming law is required to derive reuse of exact overlapping subproblem results without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

48.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for reuse of exact overlapping subproblem results.

All generated finite reuse of exact overlapping subproblem results structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The decomposition and optimal-substructure conditions must be proved for each problem.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-inputs	sampld-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

48.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

48.4 Operational laws and consequences

- Carrier law: problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

48.5 Depth-independent closure

Base: The structural One supplies one canonical reuse of exact overlapping subproblem results form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated reuse of exact overlapping subproblem results form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

48.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The decomposition and optimal-substructure conditions must be proved for each problem.	PASS

48.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: dynamic programming, memoization.

The decomposition and optimal-substructure conditions must be proved for each problem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- The decomposition and optimal-substructure conditions must be proved for each problem.

48.8 Exact evidence identities

- Source manifest: sha256:5b7f2b4a2bccef5a722bd45128cb0ec0077a5aed9db66a0b33d0dcad9757a330
- Complete-census certificate:
sha256:d7c9bf091d95bc640e9e2108bff3c6d8e915545d309fc53a1db28f40fd3f09e7
- Derivation seal: sha256:cbb434e7fac471f725b58cafea9c8de8d29edc94e9641aba911b147392d433ce

- Independent implementation:
sha256:4ff3d626485a1d771847ddf3c068086a7e991b4dbf6a53c47df08a9b20446fe9
- Independent certificate:
sha256:2fc4b3935bb8d193969048a49d5f8e26392b5a087667239e58228cb18b23ad86
- External validation:
sha256:132df1cbe4d7960c2524053d50748b28c616b110c46b1db58a2c82160b3a1bec
- Model-admitted engine receipt:
sha256:f6b1c405c189bb4f86db10ec31ad2dc8adc8041e0833baeb6184ac1832ed2e17
- Receipt path: receipts/engine/model_admitted/SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001-f6b1c405c189bb4f.json

49. Derivation 42: Fold optimization-algorithm law

Claim: SFT-COMP-ALG-OPTIMIZATION-001

49.1 Scientific necessity and exact theorem

Fold optimization-algorithm law is required to derive selection of all exact optima without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

49.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for selection of all exact optima.

All generated finite selection of all exact optima structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not assume differentiability, convexity or a conventional metric.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

49.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no

imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

49.4 Operational laws and consequences

- Carrier law: complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

49.5 Depth-independent closure

Base: The structural One supplies one canonical selection of all exact optima form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated selection of all exact optima form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

49.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-OPTIMIZATION-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-OPTIMIZATION-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-OPTIMIZATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not assume differentiability, convexity or a conventional metric.	PASS

49.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: optimization algorithm, exact optimum.

Does not assume differentiability, convexity or a conventional metric.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not assume differentiability, convexity or a conventional metric.

49.8 Exact evidence identities

- Source manifest: sha256:85c97911077fffd40a3a67b5477c79766a87bba308799693b1a5fc6b8281028c7
- Complete-census certificate:
sha256:95c08d8043fece309af5b1e542472b03b1d782d20b90c1378f1fd68612d60b72
- Derivation seal: sha256:4ee440a4de7a6270ba9d798a00174a5706c2974136615a61a24bfdc4651dea92
- Independent implementation:
sha256:c9209269f93ffe1054bce5215f859ebea9fe5638dce655dbbb52047d94e8a1ad
- Independent certificate:
sha256:5f293ae15ca747f138ce75beaf35f7a5deec7a65aa77d3056fa6eb4e44ded3ed
- External validation:
sha256:df0821a0b5130ebbc42860538bee57f122cdb6a8e509201fb639d54edc33dba0
- Model-admitted engine receipt:
sha256:74aac9551388975891fa29de8f8a7843eb1afcabc2a3df316784f34caf7ebd110
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-OPTIMIZATION-001-74aac95513889758.json

50. Derivation 43: Fold randomized-algorithm law

Claim: SFT-COMP-ALG-RANDOMIZED-001

50.1 Scientific necessity and exact theorem

Fold randomized-algorithm law is required to derive complete scheduled branch execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-OPTIMIZATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

50.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for complete scheduled branch execution.

All generated finite complete scheduled branch execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

50.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

50.4 Operational laws and consequences

- Carrier law: deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

50.5 Depth-independent closure

Base: The structural One supplies one canonical complete scheduled branch execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete scheduled branch execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

50.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-RANDOMIZED-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-RANDOMIZED-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-RANDOMIZED-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.	PASS

50.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: randomized algorithm, probabilistic algorithm.

Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.

50.8 Exact evidence identities

- Source manifest: sha256:71c76995d7b4206bee6895f9744a5edee06fd25a53e2cc9177e38bf9df238f3e
- Complete-census certificate:
sha256:5325d143b0cf5951675567187534e15a9c168d778353a9f10c96c5c2b42e7dbc
- Derivation seal: sha256:660b6dd0cc83e1b9b5b002f08369fd5274b6f857c3c875579352bc279b6e5b12
- Independent implementation:
sha256:dd42d329d28bec95bbb52eb2a0d11237935964122aa642def54ad716b7615c18
- Independent certificate:
sha256:acba3f2adc7ce42c9c8aab1fe43243c633a3c380e20b5ddea1f722b7bcee8365
- External validation:
sha256:3c5f4558ce2d237409b4c80bf00213d9dc8b3461ce506a0de72e7385baed043f
- Model-admitted engine receipt:
sha256:1c9f175ed883e3267b8722b8481cc424959ae5980bab9a6cf81e691d0785e0ae

- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-RANDOMIZED-001-1c9f175ed883e326.json

51. Derivation 44: Fold parallel-algorithm law

Claim: SFT-COMP-ALG-PARALLEL-001

51.1 Scientific necessity and exact theorem

Fold parallel-algorithm law is required to derive causally independent algorithm steps without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-RANDOMIZED-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

51.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for causally independent algorithm steps.

All generated finite causally independent algorithm steps structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical speedup is claimed without an implementation experiment.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

51.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

51.4 Operational laws and consequences

- Carrier law: dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.

- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

51.5 Depth-independent closure

Base: The structural One supplies one canonical causally independent algorithm steps form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated causally independent algorithm steps form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

51.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-PARALLEL-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-PARALLEL-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-PARALLEL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical speedup is claimed without an implementation experiment.	PASS

51.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parallel algorithm, parallel schedule.

No physical speedup is claimed without an implementation experiment.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model

- No physical speedup is claimed without an implementation experiment.

51.8 Exact evidence identities

- Source manifest: sha256:163a71a7671bf378b29b06504ed879ddb632db4d2d558aff25cd0d7d912c4b31
- Complete-census certificate:
sha256:f52a3bbccfc7ec823a8fb4c3bfc8d3a4bcb5ee879b1c21cf32eb1fbfb3a15583
- Derivation seal: sha256:568ed2eac60d7b9834407009080c157b70f5a31cb6a03762ca9660238d6a4216
- Independent implementation:
sha256:1bb66536383966cbfc406201ad9e09aa506c9dc78b60816366d649a6b1499f78
- Independent certificate:
sha256:9e0f1e409141fb2593e3015afd7188e5d260cb8dca6ccf3eb66894fa0736df9b
- External validation:
sha256:6e9a823a975e4b99cd0abfbe7c2aa1a84ee3438a2f70578e37616c8e073cb9dc
- Model-admitted engine receipt:
sha256:c16c0747c8994451d421ddb5bc14c4715d7cd4ea6c78588c3da0d318a66ca285
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-PARALLEL-001-c16c0747c8994451.json

52. Derivation 45: Fold distributed-algorithm law

Claim: SFT-COMP-ALG-DISTRIBUTED-001

52.1 Scientific necessity and exact theorem

Fold distributed-algorithm law is required to derive local processes exchanging exact records without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-PARALLEL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

52.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for local processes exchanging exact records.

All generated finite local processes exchanging exact records structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. System implementation and physical network claims are outside this formal theorem.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

52.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

52.4 Operational laws and consequences

- Carrier law: process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

52.5 Depth-independent closure

Base: The structural One supplies one canonical local processes exchanging exact records form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated local processes exchanging exact records form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

52.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-DISTRIBUTED-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-DISTRIBUTED-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-DISTRIBUTED-001; result: PASS.

Control	Expected	Observed	Result
<code>false_promise</code>	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; System implementation and physical network claims are outside this formal theorem.	PASS

52.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: distributed algorithm, message passing.

System implementation and physical network claims are outside this formal theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- System implementation and physical network claims are outside this formal theorem.

52.8 Exact evidence identities

- Source manifest: sha256 : bbba34bfed0d86637ec45a9f7b4e3db903308295b1d2245cb63a1ab0654d34b8
- Complete-census certificate:
sha256 : 33d933518ad295e9a5cd748482123a3d87c402b856733bf977aed5f5dd40ec12
- Derivation seal: sha256 : 8785aed4c9eaf6cd44caf7eea5d18b54c7830fd88b6afdbe1ccb3fc5b95ae2e5
- Independent implementation:
sha256 : abb34c0e7b42e093e771d0daec22d5b35f973b7c669013ad640a5bd3bbb8c3de
- Independent certificate:
sha256 : c91f515c7a13d50e8bbb9e8e41e6e074556f008800138ee68beddb8b79ce48d2
- External validation:
sha256 : f52cd6b95a6e1d6c10e77cc3690523ed9d0303cf0517c8efbb56d1b22b668361
- Model-admitted engine receipt:
sha256 : 5dfc71b4941c7e3f629b7ee0ab6cd3ca1851222fbfaf68690fa67875ed1f7e60
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-DISTRIBUTED-001-5dfc71b4941c7e3f.json

53. Derivation 46: Fold online and streaming algorithm law

Claim: SFT-COMP-ALG-ONLINE-STREAMING-001

53.1 Scientific necessity and exact theorem

Fold online and streaming algorithm law is required to derive prefix-bounded processing without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-DISTRIBUTED-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

53.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for prefix-bounded processing.

All generated finite prefix-bounded processing structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A completed infinite stream is not admitted; every result names its generated prefix boundary.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

53.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

53.4 Operational laws and consequences

- Carrier law: generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

53.5 Depth-independent closure

Base: The structural One supplies one canonical prefix-bounded processing form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated prefix-bounded processing form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

53.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A completed infinite stream is not admitted; every result names its generated prefix boundary.	PASS

53.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: online algorithm, streaming algorithm.

A completed infinite stream is not admitted; every result names its generated prefix boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A completed infinite stream is not admitted; every result names its generated prefix boundary.

53.8 Exact evidence identities

- Source manifest: sha256:a8fd89a9419fd5e03fa9bffce8c9e40f0a3501b9dc367b65d841b515b33ca401
- Complete-census certificate:
sha256:e960b3f6b15556b07578faa2b545bc8fcb35a3343bb3903513766e52132058a6
- Derivation seal: sha256:aa17697469db35050d4793ba86d2bcbfbf2c6cb91c0ac3bc06c9a2b728449927

- Independent implementation:
sha256:2645fa5246f2c6acd8fdb8ae2b5d7375a2da27904f8060ce8bb6f10ff936952a
- Independent certificate:
sha256:86840c27e3c57fc8a9b6378bf020ea0b56e1e428a71d120792d65b38d5477d02
- External validation:
sha256:1f9abd0ad082a83536f54eca31ebbd54ed32f9b7c82538cbadfedd5d28815fa9
- Model-admitted engine receipt:
sha256:1978c4757d70afe4f9f64a74ddad4b9df88db850c160cb3840dbe18f2ada7a15
- Receipt path: receipts/engine/model_admitted/SFT-COMP-ALG-ONLINE-STREAMING-001-1978c4757d70afe4.json

54. Derivation 47: Fold numerical-algorithm law

Claim: SFT-COMP-ALG-NUMERICAL-001

54.1 Scientific necessity and exact theorem

Fold numerical-algorithm law is required to derive exactly bounded approximation procedures without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-ONLINE-STREAMING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

54.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for exactly bounded approximation procedures.

All generated finite exactly bounded approximation procedures structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Decimal or floating results are measured/implementation records and cannot establish the proof.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

54.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no

imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

54.4 Operational laws and consequences

- Carrier law: exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

54.5 Depth-independent closure

Base: The structural One supplies one canonical exactly bounded approximation procedures form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exactly bounded approximation procedures form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

54.6 Executed witnesses and adverse controls

- algorithms-witness-1 - algorithms exact operational witness 1 for SFT-COMP-ALG-NUMERICAL-001; result: PASS.
- algorithms-witness-2 - algorithms exact operational witness 2 for SFT-COMP-ALG-NUMERICAL-001; result: PASS.
- algorithms-witness-3 - algorithms exact operational witness 3 for SFT-COMP-ALG-NUMERICAL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Decimal or floating results are measured/implementation records and cannot establish the proof.	PASS

54.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: numerical algorithm, numerical analysis.

Decimal or floating results are measured/implementation records and cannot establish the proof.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Decimal or floating results are measured/implementation records and cannot establish the proof.

54.8 Exact evidence identities

- Source manifest: sha256:53aaa3f9f914f534a7317d16e99f8cdd13b5d8d5742a146500dc0e90c82859ff
- Complete-census certificate:
sha256:d8e2c40d88ee24daadfda49061214c1c6e785ca77e839b2688d7032903c83bc
- Derivation seal: sha256:5f68377aa21130c232f506eb872b3c5ed5cb0eeff344de6f941f24c91fbd9476
- Independent implementation:
sha256:197cdb1b49b090d0b84f1c28041860cb18342a7cb65b0a19e55697b32b3205a9
- Independent certificate:
sha256:c9d66fcf52bfcea0b05cc05364626e91ff5b7c8b210f33ef2a03ba407de5fa90
- External validation:
sha256:9ecc0471fe705aa388878b030820bfd0ba9801611de1780c7d0b7934ee91b598
- Model-admitted engine receipt:
sha256:3f1c19281f4ac6c86c5bb13ae391ede42fa858c66e0571a4d6d66f27eeec098f
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-NUMERICAL-001-3f1c19281f4ac6c8.json

55. Derivation 48: Fold symbolic-computation law

Claim: SFT-COMP-ALG-SYMBOLIC-001

55.1 Scientific necessity and exact theorem

Fold symbolic-computation law is required to derive canonical expression transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-NUMERICAL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

55.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for canonical expression transformation.

All generated finite canonical expression transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only expressions within the frozen grammar are closed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampld-inputs	sampld-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

55.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

55.4 Operational laws and consequences

- Carrier law: generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

55.5 Depth-independent closure

Base: The structural One supplies one canonical canonical expression transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated canonical expression transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

55.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-SYMBOLIC-001; result: PASS.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-SYMBOLIC-001; result: PASS.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-SYMBOLIC-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only expressions within the frozen grammar are closed.	PASS

55.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: symbolic computation, computer algebra.

Only expressions within the frozen grammar are closed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only expressions within the frozen grammar are closed.

55.8 Exact evidence identities

- Source manifest: sha256:4f54e53491f964a781a06b806dd4c63e42bc5c4c58acf00842e360a73b9f62ab
- Complete-census certificate:
sha256:586b45e768eea427658b4418273d997349f303b451609df1fd20e06dde4cf1c5
- Derivation seal: sha256:cfa87346438f6199df20a903922e69866cdac4918ac03bb3df563b888733db65
- Independent implementation:
sha256:144a7d6137278c61bb5b2ec25919e1403d0967cab27d7f0974a48419b875dbdd
- Independent certificate:
sha256:9e7f855770ac962b504ba0f8531b0052f95a8575a756a3756d5cfc11f0c6eafb
- External validation:
sha256:8c88ba5ced2cdc6dc2cd67336f03d8554d6bb70fc68bacb4140adcc8a593bbbb
- Model-admitted engine receipt:
sha256:3a84f78267c8d51b1e8c1ff0981ba89929b1c2b5231b5fedcb0d522b87f9fa8d
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-SYMBOLIC-001-3a84f78267c8d51b.json

56. Derivation 49: Fold approximation-algorithm law

Claim: SFT-COMP-ALG-APPROXIMATE-001

56.1 Scientific necessity and exact theorem

Fold approximation-algorithm law is required to derive bounded resource solution with exact error ledger without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-SYMBOLIC-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

56.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for bounded resource solution with exact error ledger.

All generated finite bounded resource solution with exact error ledger structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No imported norm, epsilon parameter or irrational threshold is permitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
domain	sampled-inputs	sampled-inputs removes a required exact domain condition.	complete-generated-input-domain	complete-generated-input-domain retains the complete registered domain condition.
precondition	implicit-precondition	implicit-precondition removes a required exact precondition condition.	exact-registered-precondition	exact-registered-precondition retains the complete registered precondition condition.
transition	answer-producing-heuristic	answer-producing-heuristic removes a required exact transition condition.	exact-step-relation	exact-step-relation retains the complete registered transition condition.
invariant	unchecked-intermediate-state	unchecked-intermediate-state removes a required exact invariant condition.	retained-invariant-ledger	retained-invariant-ledger retains the complete registered invariant condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
output	result-without-certificate	result-without-certificate removes a required exact output condition.	reconstructible-output-certificate	reconstructible-output-certificate retains the complete registered output condition.
resource	unreported-resources	unreported-resources removes a required exact resource condition.	complete-time-space-trace	complete-time-space-trace retains the complete registered resource condition.
generality	example-program	example-program removes a required exact generality condition.	input-successor-certificate	input-successor-certificate retains the complete registered generality condition.
addition	imported-algorithm	imported-algorithm removes a required exact addition condition.	no-extra-answer-model	no-extra-answer-model retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

56.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

56.4 Operational laws and consequences

- Carrier law: feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

56.5 Depth-independent closure

Base: The structural One supplies one canonical bounded resource solution with exact error ledger form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bounded resource solution with exact error ledger form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

56.6 Executed witnesses and adverse controls

- **algorithms-witness-1** - algorithms exact operational witness 1 for SFT-COMP-ALG-APPROXIMATE-001; result: PASS.
- **algorithms-witness-2** - algorithms exact operational witness 2 for SFT-COMP-ALG-APPROXIMATE-001; result: PASS.
- **algorithms-witness-3** - algorithms exact operational witness 3 for SFT-COMP-ALG-APPROXIMATE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No imported norm, epsilon parameter or irrational threshold is permitted.	PASS

56.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: approximation algorithm, error bound.

No imported norm, epsilon parameter or irrational threshold is permitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model

- No imported norm, epsilon parameter or irrational threshold is permitted.

56.8 Exact evidence identities

- Source manifest: sha256:35d1d30d54ffa205920e311090024fa5a26c8eed7dfbabaeccb58fd4bc892d46
- Complete-census certificate:
sha256:cfe2c1b5daf58ab05a8b903443a2cb3e869c6c8653bda2faf2850c21cad2a658
- Derivation seal: sha256:e8803cab22548d6aaf21e549db85da47ff3586c6af729685dccc017a1a2f8ceb
- Independent implementation:
sha256:ba63e01ccb420f64ca73dd0a592b1934c59848fc9bbae0045f153cb16e6b9e83
- Independent certificate:
sha256:649e7983abd8098c4ef6eaeffc7c1376f7709b4f78451cef1ead1f64036e2d541
- External validation:
sha256:ac0157e48b5aa1fd22157b65b84f646284bf773de8fc411db161c5a918821e72
- Model-admitted engine receipt:
sha256:52dfa606dc9c680d10c2c990b5df3ed29fe4f91be95690eb0d4880df5fd7dfc
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-ALG-APPROXIMATE-001-52dfa606dc9c680d.json

57. Derivation 50: Fold program-syntax law

Claim: SFT-COMP-SEM-SYNTAX-001

57.1 Scientific necessity and exact theorem

Fold program-syntax law is required to derive generated program forms without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-ALG-APPROXIMATE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

57.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for generated program forms.

All generated finite generated program forms structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Syntax alone assigns no execution meaning.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

57.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

57.4 Operational laws and consequences

- Carrier law: complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

57.5 Depth-independent closure

Base: The structural One supplies one canonical generated program forms form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated generated program forms form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

57.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-SYNTAX-001; result: PASS.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-SYNTAX-001; result: PASS.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-SYNTAX-001; result: PASS.

Control	Expected	Observed	Result
<code>false_promise</code>	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Syntax alone assigns no execution meaning.	PASS

57.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: abstract syntax, program grammar.

Syntax alone assigns no execution meaning.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Syntax alone assigns no execution meaning.

57.8 Exact evidence identities

- Source manifest: sha256 : a87f61338ca3b9623a6adf94a9a13222ff3d1a2aa9b024687d9ae3d79397fc3c
- Complete-census certificate:
sha256 : 3608ead17c98ddc04d8aefb2528e6adb2167bdf4bb5d252beaebf98e70983cc8
- Derivation seal: sha256 : f044e21b0e7a95f6f56968665a58294e030e0bfe080756718c2aa9f35e7ff26
- Independent implementation:
sha256 : a107cd56990ec6701bea95ee4d4bfc2058d930bd3eabf27de0b2b32d9fdd0961
- Independent certificate:
sha256 : 8b714790d85bdc3e7286ca82c336d9c1117fa2dc117dbd46ef229ef6d64a5adc
- External validation:
sha256 : 81b83f0276bdde6805fb333b7889e1dbb969956453161217f2bb3cdf0d917c68
- Model-admitted engine receipt:
sha256 : 0da324b3d1cd63a836a8494235259bffe143a912f6aace6745f55c1bcae10246
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-SYNTAX-001-0da324b3d1cd63a8.json

58. Derivation 51: Fold binding and substitution law

Claim: SFT-COMP-SEM-BINDING-SUBSTITUTION-001

58.1 Scientific necessity and exact theorem

Fold binding and substitution law is required to derive name identity and replacement without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-SYNTAX-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

58.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for name identity and replacement.

All generated finite name identity and replacement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No presentation alias may replace canonical binding identity.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

58.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

58.4 Operational laws and consequences

- Carrier law: held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

58.5 Depth-independent closure

Base: The structural One supplies one canonical name identity and replacement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated name identity and replacement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

58.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No presentation alias may replace canonical binding identity.	PASS

58.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: binding, substitution.

No presentation alias may replace canonical binding identity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No presentation alias may replace canonical binding identity.

58.8 Exact evidence identities

- Source manifest: sha256: cef11ea2bef503bbbeb79fb136311d79b1422ba11e2f0632fae2aec3f0b20c12
- Complete-census certificate:
sha256: aea1e5d04b24bf253e377c50f595dcb3d27b59d74b02887fc55adfa7b5f40eab
- Derivation seal: sha256: 4d6312f18d849169c7052e425a5829af08525a4e25df7e057c9d99b787ba5e35

- Independent implementation:
sha256:e83b481c5e02dad02da6129dda8424aaaac21723874d893cf02204356846ae4a
- Independent certificate:
sha256:a757bc3b7aa1e6290b4b00eff8e3d2f07999383f3716d73e649dc90a05fcd9f8
- External validation:
sha256:ba184fed5c0a038365c52d8da2a5658f865823719e1929cbbf4ed7bf90fd5f0e
- Model-admitted engine receipt:
sha256:1200dc420f7ff8fcd37e5c3fd7e862fa3aa0d1466963d23d9fe3ce0a22ac103d
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEM-BINDING-SUBSTITUTION-001-1200dc420f7ff8fc.json

59. Derivation 52: Fold evaluation law

Claim: SFT-COMP-SEM-EVALUATION-001

59.1 Scientific necessity and exact theorem

Fold evaluation law is required to derive program-state execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-BINDING-SUBSTITUTION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

59.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for program-state execution.

All generated finite program-state execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only exact positive parts and generated data forms are semantic values.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

59.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

59.4 Operational laws and consequences

- Carrier law: syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

59.5 Depth-independent closure

Base: The structural One supplies one canonical program-state execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated program-state execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

59.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-EVALUATION-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-EVALUATION-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-EVALUATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only exact positive parts and generated data forms are semantic values.	PASS

59.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: evaluation, interpreter.

Only exact positive parts and generated data forms are semantic values.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only exact positive parts and generated data forms are semantic values.

59.8 Exact evidence identities

- Source manifest: sha256:be036424b53d2ba9c6da1ff376a058174c88cecb90d87690bc310263e25b5b6f
- Complete-census certificate:
sha256:42bea38f0ff3f867db7459bdae056bcea2ae33f442c26a51d1c8ef1285bdd443
- Derivation seal: sha256:3327352b243dc92f5a03be58dd7ef2f9509338a8551523c64daea50c75be5692
- Independent implementation:
sha256:45c3b84d2fd37e2b113a0894dfca52199cb83b19f5637f9bf8f54b5081fd5200
- Independent certificate:
sha256:f4bca7b32e999d76fc00c448137fa5dcfa056c2c270e7fb44b0bd9ce3ecfdb76
- External validation:
sha256:d02ca6834776c37cad0278024e7bd22283c70dfb96169cb5adbf74180d7dd10
- Model-admitted engine receipt:
sha256:dfb202d42339a8c83ffbd8d1939ec3d6dd29819fb621b60f6dd559175315089a
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-EVALUATION-001-dfb202d42339a8c8.json

60. Derivation 53: Fold operational and denotational correspondence law

Claim: SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001

60.1 Scientific necessity and exact theorem

Fold operational and denotational correspondence law is required to derive trace-to-meaning agreement without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-EVALUATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

60.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for trace-to-meaning agreement.

All generated finite trace-to-meaning agreement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Correspondence is proved only for the frozen language grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

60.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.

Eliminated candidates	Exact first-failure reason
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

60.4 Operational laws and consequences

- Carrier law: operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

60.5 Depth-independent closure

Base: The structural One supplies one canonical trace-to-meaning agreement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated trace-to-meaning agreement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

60.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: PASS.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: PASS.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS

Control	Expected	Observed	Result
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correspondence is proved only for the frozen language grammar.	PASS

60.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: operational semantics, denotational semantics.

Correspondence is proved only for the frozen language grammar.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Correspondence is proved only for the frozen language grammar.

60.8 Exact evidence identities

- Source manifest: sha256 : 369034c6a53357958d53f61e96066a9b6539ffbcbce8e640dc752f258c243ebce
- Complete-census certificate:
sha256 : 1e9121a5af21d756772b81910f2aa2b6b2d0e607baabf2e435df47fe084f217e
- Derivation seal: sha256 : e8941a8b0de586b68b95ede9a238f45d9fffc7855a9c6492b52331ae4ced42991
- Independent implementation:
sha256 : 5220d7142d967826226e5a0bf576cf87ab1f71af680374da69e82ace109ff839
- Independent certificate:
sha256 : dacb28b48850243cca84186f124d109878e9c4ecaf44f1cde9d151295f221fc1
- External validation:
sha256 : e408f1e0f0a65f69163d1db61238d7db90086204da7f256bbdda42edc721b778
- Model-admitted engine receipt:
sha256 : 6db365c1b3c22883c276eec14485e6fb962528278f521654cba633fe6536e425
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001-6db365c1b3c22883.json

61. Derivation 54: Fold type-theory law

Claim: SFT-COMP-SEM-TYPE-001

61.1 Scientific necessity and exact theorem

Fold type-theory law is required to derive structural admissibility of composition without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

61.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for structural admissibility of composition.

All generated finite structural admissibility of composition structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No conventional type calculus is assumed beyond downstream comparison.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

61.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

61.4 Operational laws and consequences

- Carrier law: generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

61.5 Depth-independent closure

Base: The structural One supplies one canonical structural admissibility of composition form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated structural admissibility of composition form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

61.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-TYPE-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-TYPE-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-TYPE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional type calculus is assumed beyond downstream comparison.	PASS

61.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: type theory, type safety.

No conventional type calculus is assumed beyond downstream comparison.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No conventional type calculus is assumed beyond downstream comparison.

61.8 Exact evidence identities

- Source manifest: sha256:9f4b1ebd3fca097f0b70cac79bf001afb8394ce01cd7ce64f858f2a700b2c253
- Complete-census certificate:
sha256:560c422d0270c778a566861bdb33a0cd81e446427f702f7e3d688072c598abb3
- Derivation seal: sha256:fda760ee8e868f300216cd11a36d93a8e7b6845e8fcb70dcfd3802a003868364

- Independent implementation:
sha256:241f76ae21ddba0448e9e66a4d78ddeb9f271415f6b99d065fe0d6f2fb296693
- Independent certificate:
sha256:ba5a47bb0f58c6316a328e9317018057175150177e5259da5333957b760915a9
- External validation:
sha256:8ade6d8bad00291ecb7005a195c0863d988e7381eb68b9a3f9127ef12b98f5cc
- Model-admitted engine receipt:
sha256:f62ebd67f072c2e126b16d47d16cf1982426bc5c333be0c79d9148e43a505620
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEM-TYPE-001-f62ebd67f072c2e1.json

62. Derivation 55: Fold program-equivalence law

Claim: SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001

62.1 Scientific necessity and exact theorem

Fold program-equivalence law is required to derive indistinguishable generated behavior without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-TYPE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

62.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for indistinguishable generated behavior.

All generated finite indistinguishable generated behavior structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Equivalence is relative to the registered observation and input boundary.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

62.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one

tuple composed entirely of admitted coordinates.

62.4 Operational laws and consequences

- Carrier law: two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

62.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable generated behavior form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable generated behavior form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

62.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence is relative to the registered observation and input boundary.	PASS

62.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program equivalence, observational equivalence.

Equivalence is relative to the registered observation and input boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Equivalence is relative to the registered observation and input boundary.

62.8 Exact evidence identities

- Source manifest: sha256:a509788e7e52b8ebd47915633937dfa89a6c8badd67623c3055353489a9b0e16
- Complete-census certificate:
sha256:6c456bc0b95c08f66914fee51db3916b28bfa3548775b435ae048d7eabc0c193
- Derivation seal: sha256:71c3f701a6c006e7d5b9d3ed693c33ac15f4b037304cb7e6db48e12d2c3f1939
- Independent implementation:
sha256:eb20d8ce5ccc05057939be3429f6cc04f52154350f122608e3d34fd2013fe600
- Independent certificate:
sha256:fca679a2f596aef6b1790e002eb6cab7145c49feb024f6f7f5e0fe3f2266e09b
- External validation:
sha256:47b8bb4f01597ff56e8a874ee2ac589f2890d9ae8142f3421b559721bd462710
- Model-admitted engine receipt:
sha256:ad7d0dba57a7c30f17d67ef1e94ba22b9fe9e126565b894d97f435d1f83e4a17
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001-ad7d0dba57a7c30f.json

63. Derivation 56: Fold termination law

Claim: SFT-COMP-SEM-TERMINATION-001

63.1 Scientific necessity and exact theorem

Fold termination law is required to derive absence of continuing execution paths without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

63.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for absence of continuing execution paths.

All generated finite absence of continuing execution paths structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No total termination decider is claimed for the universal language.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

63.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.

Eliminated candidates	Exact first-failure reason
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

63.4 Operational laws and consequences

- Carrier law: program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

63.5 Depth-independent closure

Base: The structural One supplies one canonical absence of continuing execution paths form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated absence of continuing execution paths form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

63.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-TERMINATION-001; result: PASS.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-TERMINATION-001; result: PASS.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-TERMINATION-001; result: PASS.

Control	Expected	Observed	Result
<code>false_premise</code>	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS

Control	Expected	Observed	Result
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No total termination decider is claimed for the universal language.	PASS

63.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: termination, well-foundedness.

No total termination decider is claimed for the universal language.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No total termination decider is claimed for the universal language.

63.8 Exact evidence identities

- Source manifest: sha256:e19e68832904cb3077a97a63fefcd62a2d4c10e3ecf9957acf56bcd15e519a3e
- Complete-census certificate:
sha256:5cd4884f6cc707074b03aaed07b0237e1b3cc0207fe43ed92df5f5b9f105aa0c
- Derivation seal: sha256:963c938eee4f90c5b8e76126431979d0bc0fa6cacf4ee4b0cd4eb1c781c4da6d
- Independent implementation:
sha256:2b7e5bd0df6cb5e94e9d1b8b24d2de035186c83eb1772cd53cbc8bb1c215d59a
- Independent certificate:
sha256:540dbe2314f20eed8acba59ca366bd2d6f526d4aaa19f1edb81f19d22babe6e9
- External validation:
sha256:847873d743a601a413e64211aff7e4a9fac44da1ed5553a91c7f4ae11ae86b14
- Model-admitted engine receipt:
sha256:dc2378dbe15b3cfe2521a462c2bae6a7cef270b027719a9345618a9f1971c752
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-TERMINATION-001-dc2378dbe15b3cfe.json

64. Derivation 57: Fold program-correctness law

Claim: SFT-COMP-SEM-CORRECTNESS-001

64.1 Scientific necessity and exact theorem

Fold program-correctness law is required to derive agreement between execution and specification without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-TERMINATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

64.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for agreement between execution and specification.

All generated finite agreement between execution and specification structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Correctness is conditional on the exact registered specification and input support.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

64.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

64.4 Operational laws and consequences

- Carrier law: precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

64.5 Depth-independent closure

Base: The structural One supplies one canonical agreement between execution and specification form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated agreement between execution and specification form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

64.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-CORRECTNESS-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-CORRECTNESS-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-CORRECTNESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correctness is conditional on the exact registered specification and input support.	PASS

64.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program correctness, Hoare logic.

Correctness is conditional on the exact registered specification and input support.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Correctness is conditional on the exact registered specification and input support.

64.8 Exact evidence identities

- Source manifest: sha256:2572f0f43365ae13d16bd169aa23b6e6ee0387acacf3931936908e93803027b8
- Complete-census certificate:
sha256:67da57cd8e8a450a2ccf8c727f6b25b4af134f170ed732c1699ebc665c181971
- Derivation seal: sha256:db3ea2b1eecab1439dbe15f31592764e3e6515c7c3290359f02037c991077b7a
- Independent implementation:
sha256:5e849c094b2d3e312b9ebfc368174baf8c72a986093e0d9ba3c9a35ee6b2ed13
- Independent certificate:
sha256:a2927c2c2496399b11d4660cef42aa88f2e01a212fc0a22dbe8f72c5f384d227
- External validation:
sha256:e3b70acc247f6998bf64ebde81b82de51ed5f70cdb364abc73bb3475175da81c
- Model-admitted engine receipt:
sha256:95af5c4c8a0cf03d91f3d1ceff4decef32ba66537923e85eebe51ca3335e78bf
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-CORRECTNESS-001-95af5c4c8a0cf03d.json

65. Derivation 58: Fold formal-specification law

Claim: SFT-COMP-SEM-SPECIFICATION-001

65.1 Scientific necessity and exact theorem

Fold formal-specification law is required to derive exact admissible behavior descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-CORRECTNESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

65.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for exact admissible behavior descriptions.

All generated finite exact admissible behavior descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A prose intention cannot substitute for the generated specification relation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

65.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

65.4 Operational laws and consequences

- Carrier law: input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

65.5 Depth-independent closure

Base: The structural One supplies one canonical exact admissible behavior descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact admissible behavior descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

65.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-SPECIFICATION-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-SPECIFICATION-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-SPECIFICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A prose intention cannot substitute for the generated specification relation.	PASS

65.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: formal specification, refinement.

A prose intention cannot substitute for the generated specification relation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A prose intention cannot substitute for the generated specification relation.

65.8 Exact evidence identities

- Source manifest: sha256 : 24c0ef32147af4486b7e3aba1a6d682302e7b69b6eababfe5ad6bc963c463439
- Complete-census certificate:
sha256 : dd53bfb73093fcfae8b8d87d11b94bea6cfab082b3ced8747d0b0fb8b97c18e6
- Derivation seal: sha256 : 29dbb881dca4124a901a109963c2678dc4d45236a921efb02d9432fd8bfbf3be
- Independent implementation:
sha256 : 46650c558595b950805ed11f7bcac1b7f42cf1cb50c73235f9e4d9552e49286b
- Independent certificate:
sha256 : 4638802ddd94e44f8e1c86e10ff4c8c015435545abfea2e7f1953c0f94ff3fe1
- External validation:
sha256 : cfdcebe686b9010ec4fb4002d01833f1b5a5b415fb651124e54ee97f489fd1a8
- Model-admitted engine receipt:
sha256 : 4aeb9f85a830e501877b8cc19d41095d2bbfc96656ff92be8fe5a6655d351a51
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-SPECIFICATION-001-4aeb9f85a830e501.json

66. Derivation 59: Fold program-transformation law

Claim: SFT-COMP-SEM-TRANSFORMATION-001

66.1 Scientific necessity and exact theorem

Fold program-transformation law is required to derive semantics-preserving program rewriting without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-SPECIFICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

66.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for semantics-preserving program rewriting.

All generated finite semantics-preserving program rewriting structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Optimization claims require separate resource evidence.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

66.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

66.4 Operational laws and consequences

- Carrier law: source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

66.5 Depth-independent closure

Base: The structural One supplies one canonical semantics-preserving program rewriting form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated semantics-preserving program rewriting form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

66.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-TRANSFORMATION-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-TRANSFORMATION-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-TRANSFORMATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Optimization claims require separate resource evidence.	PASS

66.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program transformation, refactoring.

Optimization claims require separate resource evidence.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Optimization claims require separate resource evidence.

66.8 Exact evidence identities

- Source manifest: sha256 : 35b8d670b699a9d19bcb41ccb2460225e70490c7b379e7d97efbd245fa73e1a2
- Complete-census certificate:
sha256 : e8875557e249b5e24c2fd9bd042f7e590b54efeb479e14a29992bcece084f827
- Derivation seal: sha256 : f7339e44653a24a42d3ca997d9b329bf80c25c3bcd78dd51a59d428980f32750

- Independent implementation:
sha256:08c8b493d1011c559d7f1dc4158e7a0d78afcca76a1375879729e2286d25d92c
- Independent certificate:
sha256:e3da2d11b82a871c6a2a4f41266025039358dc053ac8c838c32509ac3cda9c44
- External validation:
sha256:2c73a1c68b6d4c7f142c1c7fb7f003619cd12c9edeef1a0b5854ec15ef0a2994
- Model-admitted engine receipt:
sha256:cdc9731adba6f15b42e9eeba2e0c93d29bfeadf4ae281fbfed527d54918a7226
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-TRANSFORMATION-001-cdc9731adba6f15b.json

67. Derivation 60: Fold semantics-preserving compilation law

Claim: SFT-COMP-SEM-COMPILATION-001

67.1 Scientific necessity and exact theorem

Fold semantics-preserving compilation law is required to derive translation between executable languages without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-TRANSFORMATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

67.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for translation between executable languages.

All generated finite translation between executable languages structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Applies only to the two frozen language grammars and declared runtime interface.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

67.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and

contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

67.4 Operational laws and consequences

- Carrier law: source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

67.5 Depth-independent closure

Base: The structural One supplies one canonical translation between executable languages form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated translation between executable languages form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

67.6 Executed witnesses and adverse controls

- semantics-witness-1 - semantics exact operational witness 1 for SFT-COMP-SEM-COMPILATION-001; result: PASS.
- semantics-witness-2 - semantics exact operational witness 2 for SFT-COMP-SEM-COMPILATION-001; result: PASS.
- semantics-witness-3 - semantics exact operational witness 3 for SFT-COMP-SEM-COMPILATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies only to the two frozen language grammars and declared runtime interface.	PASS

67.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: compiler correctness, compilation.

Applies only to the two frozen language grammars and declared runtime interface.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Applies only to the two frozen language grammars and declared runtime interface.

67.8 Exact evidence identities

- Source manifest: sha256:ec2ac685a89a050f12b777bf76c8ab1682cbaa48ea97fe6bc5ce418cde0c9a95
- Complete-census certificate:
sha256:ac9cfbc92b7ea2007942ef317956f5ab79527dcac6069b882950a91e0b3a6873
- Derivation seal: sha256:92e1e598f3b553e315bbfe123fe701956b1769053d33261670e6431995c95c7d
- Independent implementation:
sha256:a8f2d406212e720b0ebb5c0f2d313a35b36a4dcb936a5d9625442bcf838a5cc3
- Independent certificate:
sha256:8443c6e6dc2296d4e8e972d22a890e06d77964796b7a80f00632bdd47e9d49af
- External validation:
sha256:30432a42c6a2a3e3bdc49bf3febd42443802c85172c313eb9da1244c13a0c98c
- Model-admitted engine receipt:
sha256:1d1314650abf2f20bb952ce22c2d83de00e3bb90fa515fb5f25560c3e3bb96ae
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-COMPILATION-001-1d1314650abf2f20.json

68. Derivation 61: Fold verification and proof-system law

Claim: SFT-COMP-SEM-VERIFICATION-001

68.1 Scientific necessity and exact theorem

Fold verification and proof-system law is required to derive machine-checkable program evidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-COMPILATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

68.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for machine-checkable program evidence.

All generated finite machine-checkable program evidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A passing test suite is not substituted for a proof over the complete declared boundary.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
syntax	partial-or-ambiguous-syntax	partial-or-ambiguous-syntax removes a required exact syntax condition.	complete-generated-syntax	complete-generated-syntax retains the complete registered syntax condition.
binding	presentation-name	presentation-name removes a required exact binding condition.	canonical-held-binding	canonical-held-binding retains the complete registered binding condition.
relation	informal-meaning	informal-meaning removes a required exact relation condition.	exact-semantic-relation	exact-semantic-relation retains the complete registered relation condition.
evaluation	result-only	result-only removes a required exact evaluation condition.	complete-evaluation-trace	complete-evaluation-trace retains the complete registered evaluation condition.
equivalence	visual-equivalence	visual-equivalence removes a required exact equivalence condition.	observation-bound-equivalence	observation-bound-equivalence retains the complete registered equivalence condition.
proof	test-only	test-only removes a required exact proof condition.	machine-checkable-proof-object	machine-checkable-proof-object retains the complete registered proof condition.
generality	sample-programs	sample-programs removes a required exact generality condition.	syntax-successor-certificate	syntax-successor-certificate retains the complete registered generality condition.
addition	imported-language-semantics	imported-language-semantics removes a required exact addition condition.	no-extra-semantic-premise	no-extra-semantic-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

68.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-sem antic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

68.4 Operational laws and consequences

- Carrier law: specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

68.5 Depth-independent closure

Base: The structural One supplies one canonical machine-checkable program evidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated machine-checkable program evidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

68.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-VERIFICATION-001; result: PASS.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-VERIFICATION-001; result: PASS.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-VERIFICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A passing test suite is not substituted for a proof over the complete declared boundary.	PASS

68.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program verification, proof system.

A passing test suite is not substituted for a proof over the complete declared boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A passing test suite is not substituted for a proof over the complete declared boundary.

68.8 Exact evidence identities

- Source manifest: sha256:56d7af3b754581d272d0847b611e8d23225a6d352b87c71a975c91011fd74501
- Complete-census certificate:
sha256:5aba927de4346a6a9fc2d1ba3e7b0de80534383070f8fe6af6ef655861bb43e8
- Derivation seal: sha256:ab921149b3cdeea436b7570367e34b839a9a18bc9a4ef156f309283f8a7f543b
- Independent implementation:
sha256:8321619907caf8c1ded379e15ff860599cc8f92598361ccf5c23a701ae40cf84
- Independent certificate:
sha256:25defbf52997162e17bcd8ebc3d5aff42667672f858b9bedb610b358d40e8ef9
- External validation:
sha256:54fcb3e9f8dca7638fa994ba1714cc738cbf8a5c01fabd5a568369008308019d
- Model-admitted engine receipt:
sha256:e3312b01e5f796d522ced745f449997e595aade4f15b3b82b216840116f09f97
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEM-VERIFICATION-001-e3312b01e5f796d5.json

69. Derivation 62: Fold concurrency law

Claim: SFT-COMP-DIST-CONCURRENCY-001

69.1 Scientific necessity and exact theorem

Fold concurrency law is required to derive multiple enabled process transitions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-VERIFICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

69.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for multiple enabled process transitions.

All generated finite multiple enabled process transitions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Concurrency is structural coexistence, not physical simultaneity.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

69.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

69.4 Operational laws and consequences

- Carrier law: process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

69.5 Depth-independent closure

Base: The structural One supplies one canonical multiple enabled process transitions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated multiple enabled process transitions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

69.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-CONCURRENCY-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-CONCURRENCY-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-CONCURRENCY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency is structural coexistence, not physical simultaneity.	PASS

69.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: concurrency, interleaving.

Concurrency is structural coexistence, not physical simultaneity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Concurrency is structural coexistence, not physical simultaneity.

69.8 Exact evidence identities

- Source manifest: sha256:6fe04cd19f195f77f3e6ea79e0fb949f368c1cea735186d9e4b72790574d9e26
- Complete-census certificate:
sha256:d5e45de02fd60091d9e6fb785b2c767ad0258bcb3f3f22574bdb9b432ee46f5
- Derivation seal: sha256:3b3dce2d1c980ed0f261048b03453cdc320ccf71758f32588ae5dc2ed45f10
- Independent implementation:
sha256:0413fd7e133b664c0ffb5febc57180d55a7208afe2afcc72d8bdf4014681f761
- Independent certificate:
sha256:19b88fd27379a91ef6baa0cc1468a60dee4b9c725b5071c1c1ab0e7e9c6185b4
- External validation:
sha256:c5f5d639dc53cf049646a3a76a5a158d50ab3a6cc14755448c935c7027b7af5a
- Model-admitted engine receipt:
sha256:a3a2d3ef12a8166cc2117b17c3a470314146123a2c474c73ab363c2d5cf2e1c7
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-CONCURRENCY-001-a3a2d3ef12a8166c.json

70. Derivation 63: Fold computational-causality law

Claim: SFT-COMP-DIST-CAUSALITY-001

70.1 Scientific necessity and exact theorem

Fold computational-causality law is required to derive event predecessor structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-CONCURRENCY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

70.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for event predecessor structure.

All generated finite event predecessor structure structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical causal claims require a later empirical branch.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

70.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

70.4 Operational laws and consequences

- Carrier law: generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

70.5 Depth-independent closure

Base: The structural One supplies one canonical event predecessor structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated event predecessor structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

70.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-CAUSALITY-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-CAUSALITY-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-CAUSALITY-001; result: PASS.

Control	Expected	Observed	Result
<code>false_premise</code>	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical causal claims require a later empirical branch.	PASS

70.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: causality, happens-before.

Physical causal claims require a later empirical branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical causal claims require a later empirical branch.

70.8 Exact evidence identities

- Source manifest: sha256:c04848152f5b1dcb72a401209d05d24474b76270119aff5660ad1f13329e600
- Complete-census certificate:
sha256:daa5fab5c8a0adf331ca7256f9f64699532ae48c60a03fb9ca1ec1653f5bd8ca
- Derivation seal: sha256:e9f316907a841ba2e2d3f4247665c19e61b4aea748aadf828a48d64bf0e35ed6
- Independent implementation:
sha256:a48a7a468603ad934d8d4544a55963be61f283842e47b7b7334c6affdf5472ab
- Independent certificate:
sha256:0d60e035ef2862fdeac2103dd4b7861b2b176d1acec4ffbf4c6d8715d177925
- External validation:
sha256:06d3b23bc054557f39397eccbc8efec57a1c3186c2c9d2aa54f5b2c87b6f3e2a
- Model-admitted engine receipt:
sha256:902e87f46e187a03a813011d318a637eaab7c9fc3dedfc0fc2c59efcb0dffccf
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-CAUSALITY-001-902e87f46e187a03.json

71. Derivation 64: Fold synchronization law

Claim: SFT-COMP-DIST-SYNCHRONIZATION-001

71.1 Scientific necessity and exact theorem

Fold synchronization law is required to derive jointly enabled process transitions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-CAUSALITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

71.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for jointly enabled process transitions.

All generated finite jointly enabled process transitions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No timing or lock implementation is assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

71.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

71.4 Operational laws and consequences

- Carrier law: participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

71.5 Depth-independent closure

Base: The structural One supplies one canonical jointly enabled process transitions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated jointly enabled process transitions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

71.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: PASS.

Control	Expected	Observed	Result
<code>false_premise</code>	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
<code>boundary</code>	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No timing or lock implementation is assumed.	PASS

71.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: synchronization, rendezvous.

No timing or lock implementation is assumed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No timing or lock implementation is assumed.

71.8 Exact evidence identities

- Source manifest: `sha256:362b59789c63f76942ffb330df1b93fa6c8ef26cfd51deb8e47e395c5cdcde88`
- Complete-census certificate:
`sha256:5e3eeb9ec25477629ac3d801457152c7bbb4c7bb9ed48c1177e725e90ffc7a4c`
- Derivation seal: `sha256:b400e103dedeca45bf97795420f164d6bc863dbd55be87c02e53e4948b9f9ded`

- Independent implementation:
sha256:28138e0387e7e6c371c9a6e9741e41627908bb7f65fdede735ff6f21d1255452
- Independent certificate:
sha256:a23fe872cadd29b424590e9786b63c1a4924ea73bdce7e5d9b5dfc5b1ebe5602
- External validation:
sha256:39a1d69fcde2915e256970362287e0796e1fbed21ddb9dbfd561c186892a4e53
- Model-admitted engine receipt:
sha256:e1b5a01e71410f40f763377dcfc336f2d75bc664db00a510dbba8a075fa5aa0f
- Receipt path: receipts/engine/model_admitted/SFT-COMP-DIST-SYNCHRONIZATION-001-e1b5a01e71410f40.json

72. Derivation 65: Fold communication law

Claim: SFT-COMP-DIST-COMMUNICATION-001

72.1 Scientific necessity and exact theorem

Fold communication law is required to derive exact record transfer without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-SYNCHRONIZATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

72.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for exact record transfer.

All generated finite exact record transfer structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Reliability, ordering and delay are separate declared channel properties.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

72.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one

tuple composed entirely of admitted coordinates.

72.4 Operational laws and consequences

- Carrier law: sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

72.5 Depth-independent closure

Base: The structural One supplies one canonical exact record transfer form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact record transfer form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

72.6 Executed witnesses and adverse controls

- distributed_computation-witness-1 - distributed_computation exact operational witness 1 for SFT-COMP-DIST-COMMUNICATION-001; result: PASS.
- distributed_computation-witness-2 - distributed_computation exact operational witness 2 for SFT-COMP-DIST-COMMUNICATION-001; result: PASS.
- distributed_computation-witness-3 - distributed_computation exact operational witness 3 for SFT-COMP-DIST-COMMUNICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reliability, ordering and delay are separate declared channel properties.	PASS

72.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: communication, message passing.

Reliability, ordering and delay are separate declared channel properties.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Reliability, ordering and delay are separate declared channel properties.

72.8 Exact evidence identities

- Source manifest: sha256:480c5500bb117e2400a88d8b7f127d73517f6b16d0cf2552d91df99ad87963cf
- Complete-census certificate:
sha256:ea4f362cc4c5bc46a29be0fb05eaa5a874b7f38e3cc98d4b9e89566f819f6a3c
- Derivation seal: sha256:ee57d87e666ad6d34008a82623868fa0d37191217a64b946e92763d500ee9e6d
- Independent implementation:
sha256:1a1371f4ca85d120e2b44065d766a0c078d3da9a064d413032cccf29127353e
- Independent certificate:
sha256:1cdbc1e3b2c636063f22f6c59325a1d8a2a801e8deb31abfde748e92975e6f28
- External validation:
sha256:a691fbf7e762715853ece6b2497150169d72428cb03c329de72e725f6770be1a
- Model-admitted engine receipt:
sha256:59defe90f59d3eef926714873a97cc9e2a4d9d14391e5d6d7458f51687b23423
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-COMMUNICATION-001-59defe90f59d3eef.json

73. Derivation 66: Fold distributed partial-order law

Claim: SFT-COMP-DIST-PARTIAL-ORDER-001

73.1 Scientific necessity and exact theorem

Fold distributed partial-order law is required to derive causal event ordering without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-COMMUNICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

73.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for causal event ordering.

All generated finite causal event ordering structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A total global order is not assumed where events are incomparable.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-processes-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

73.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.

Eliminated candidates	Exact first-failure reason
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

73.4 Operational laws and consequences

- Carrier law: event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

73.5 Depth-independent closure

Base: The structural One supplies one canonical causal event ordering form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated causal event ordering form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

73.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A total global order is not assumed where events are incomparable.	PASS

73.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: partial order, event structure.

A total global order is not assumed where events are incomparable.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A total global order is not assumed where events are incomparable.

73.8 Exact evidence identities

- Source manifest: sha256:56f3996a18b1f74c4eadbc772358e4ad5e142e16a21603e72d473c813e4c387b
- Complete-census certificate:
sha256:e1c0c2bff2e7275889803fb9f574184af6a8c24f60e2922a73167f15a652c2cd
- Derivation seal: sha256:020e71262605b9bb2fcb82d912aae3c6d68941ae3f8bbf353eeacc051aba70ae
- Independent implementation:
sha256:07732d983dfb8e2c1da409d19134eaceca8ace59feb78186dc37d4eea5df7580
- Independent certificate:
sha256:9d32b2ce81cdbe022089e2b49fbcdd2e85c900f05c78bb6928992ddf4a0fcf371
- External validation:
sha256:d8a87eec1566bd3a23a3090815aa1a6dea8afa1268268100d1eefdd0e259da0e
- Model-admitted engine receipt:
sha256:d04af273dac3a8a8cd3d64b1b7c72e3130c1e674320c6632b0c2b9e2a8225a0f
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-PARTIAL-ORDER-001-d04af273dac3a8a8.json

74. Derivation 67: Fold consensus law

Claim: SFT-COMP-DIST-CONSENSUS-001

74.1 Scientific necessity and exact theorem

Fold consensus law is required to derive terminal common decision without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-PARTIAL-ORDER-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

74.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for terminal common decision.

All generated finite terminal common decision structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Every consensus claim must name its communication and fault model.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-processes-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

74.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

74.4 Operational laws and consequences

- Carrier law: process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

74.5 Depth-independent closure

Base: The structural One supplies one canonical terminal common decision form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated terminal common decision form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

74.6 Executed witnesses and adverse controls

- distributed_computation-witness-1 - distributed_computation exact operational witness 1 for SFT-COMP-DIST-CONSENSUS-001; result: PASS.
- distributed_computation-witness-2 - distributed_computation exact operational witness 2 for SFT-COMP-DIST-CONSENSUS-001; result: PASS.
- distributed_computation-witness-3 - distributed_computation exact operational witness 3 for SFT-COMP-DIST-CONSENSUS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Every consensus claim must name its communication and fault model.	PASS

74.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: consensus, agreement.

Every consensus claim must name its communication and fault model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Every consensus claim must name its communication and fault model.

74.8 Exact evidence identities

- Source manifest: sha256:4a2c046f87b26d98e327093fd29479ef357aa0b24b055fd677bfac1aeef916ef
- Complete-census certificate:
sha256:40ef6161a4b5eefd8d0ca925380a1fe9019676bd921170e9a57b16d9f4568424
- Derivation seal: sha256:cda13e15608d8e7b1a3f9aeb2859994b54270d7d838f6395ad8a193352f0c4c1
- Independent implementation:
sha256:03b718ff935db905cfbecf0ddd1b5b7aef28447895c8ccf568ba3c02c8d43974
- Independent certificate:
sha256:acdf2a8a619c4328635d5faceddde44915461068bd5fc53e010ce49aa41ff37e
- External validation:
sha256:65fe96a9cb1b6d95a41f2d3bcbb7ef629252f95bf5546ed82d2e8c6582390160
- Model-admitted engine receipt:
sha256:48ae9b96daea76ffeeffb0c3b2b9869b3beea88dc03cfd1b5a90e6f250a4c4d1e
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-CONSENSUS-001-48ae9b96daea76ff.json

75. Derivation 68: Fold agreement-impossibility law

Claim: SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001

75.1 Scientific necessity and exact theorem

Fold agreement-impossibility law is required to derive indistinguishable predecessor limits without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-CONSENSUS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

75.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for indistinguishable predecessor limits.

All generated finite indistinguishable predecessor limits structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

75.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

75.4 Operational laws and consequences

- Carrier law: two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

75.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable predecessor limits form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable predecessor limits form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

75.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.	PASS

75.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: consensus impossibility, indistinguishability proof.

Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.

75.8 Exact evidence identities

- Source manifest: sha256:63b144fb308a8aaf282bcbb96e4f355f58e2fea0b1da495cb1eaf4b051a24bed
- Complete-census certificate:
sha256:f9a936ab5aa948101f178c2acee30fe81e5846c1da59d11e8784801be13add92
- Derivation seal: sha256:32e02c7b473ee4b9b2428c7bc89b1bbd83e07310152b5548182ce01d9ce41d3f
- Independent implementation:
sha256:c4eeffda44abbf12bffeefad3c1ad32e374ec2e35f2ccdcd232fab851b421
- Independent certificate:
sha256:f3b98f1d6b90feff2c25194bc0196b6076eed2df0983bf5d8a02c548c09e87be
- External validation:
sha256:897f4e69e5287bc3147464d8e712bfc3087f0d8b8a2afbd2cab066599aff2ae4
- Model-admitted engine receipt:
sha256:30b83eb403fb428502002c7b9d96f569a91195720f36d8e1675385e92919f48d
- Receipt path: receipts/engine/model_admitted/SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001-30b83eb403fb4285.json

76. Derivation 69: Fold replication and consistency law

Claim: SFT-COMP-DIST-REPLICATION-CONSISTENCY-001

76.1 Scientific necessity and exact theorem

Fold replication and consistency law is required to derive multiple retained state copies without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

76.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for multiple retained state copies.

All generated finite multiple retained state copies structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No implementation platform or wall-clock consistency is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

76.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

76.4 Operational laws and consequences

- Carrier law: replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

76.5 Depth-independent closure

Base: The structural One supplies one canonical multiple retained state copies form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated multiple retained state copies form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

76.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No implementation platform or wall-clock consistency is imported.	PASS

76.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: replication, consistency model.

No implementation platform or wall-clock consistency is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No implementation platform or wall-clock consistency is imported.

76.8 Exact evidence identities

- Source manifest: sha256:4801ddc5795d08b9ae79597d99aad0e2fb576dd02dd680e5954d31772fde1a83
- Complete-census certificate:
sha256:ac7532dc862cb5d6b5d39c336f34ee22594ecb47adb1f9973758120913857a05
- Derivation seal: sha256:a72b53b74f5f1dd6ceed05797c12e0ffaf8eebe8751ff5e3169b4a40018038fd
- Independent implementation:
sha256:cecc16c2c4b38824cda8a93e2a0e383b2e91e070970b18b35d91bc539315d6df
- Independent certificate:
sha256:a94d015bc15181f18e941e198ccfde240b61e33c29edb0ae477fd31043f04d3b
- External validation:
sha256:ad685d20578ad08d5ec909eb918daae661024ec45a05519097be78f761b0dd30
- Model-admitted engine receipt:
sha256:79243bdec0bda3d90c090785f5a2b43444ff4a323b5d7b48f0a36f17c11dcf34
- Receipt path: receipts/engine/model_admitted/SFT-COMP-DIST-REPLICATION-CONSISTENCY-001-79243bdec0bda3d9.json

77. Derivation 70: Fold computational fault-model law

Claim: SFT-COMP-DIST-FAULT-MODEL-001

77.1 Scientific necessity and exact theorem

Fold computational fault-model law is required to derive declared deviations from process rules without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-REPLICATION-CONSISTENCY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

77.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for declared deviations from process rules.

All generated finite declared deviations from process rules structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Faults not generated by the

model remain outside the guarantee.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

77.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

77.4 Operational laws and consequences

- Carrier law: nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

77.5 Depth-independent closure

Base: The structural One supplies one canonical declared deviations from process rules form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated declared deviations from process rules form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

77.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-FAULT-MODEL-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-FAULT-MODEL-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-FAULT-MODEL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Faults not generated by the model remain outside the guarantee.	PASS

77.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: fault model, fault tolerance.

Faults not generated by the model remain outside the guarantee.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Faults not generated by the model remain outside the guarantee.

77.8 Exact evidence identities

- Source manifest: sha256 : aed7d195a63919509b9ba1882ada8b2d58945898ae7e7281aff800ac5fdcc7e0
- Complete-census certificate:
sha256 : 3a04bb762e3168e9c601da923ccf65aa71706910bac1a715a689a383c1dd7c1a
- Derivation seal: sha256 : 46ef8935bd002b706c9086252e41700af3efda873ca61ed9695b86d01b8bb10c
- Independent implementation:
sha256 : 401a670fcf47677f6133385e134ba15e0c845145663b3a3fe267ee72f0cc8543
- Independent certificate:
sha256 : c56083388b4efc4858b69acdea2242163995edca605a8f532e908538453aa4ea
- External validation:
sha256 : b210256f53a86fd60631fbe0e6c7aa69c1164bd6d5fee88c3221aef797cd49a4
- Model-admitted engine receipt:
sha256 : 812aad3547d57947e1e18963470f883725de08c4e299351306e308a1923bdf46
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-FAULT-MODEL-001-812aad3547d57947.json

78. Derivation 71: Fold distributed-knowledge law

Claim: SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001

78.1 Scientific necessity and exact theorem

Fold distributed-knowledge law is required to derive information available to local processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-FAULT-MODEL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

78.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for information available to local processes.

All generated finite information available to local processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Knowledge is observation-relative exact support, not an imported modal axiom.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

78.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

78.4 Operational laws and consequences

- Carrier law: global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

78.5 Depth-independent closure

Base: The structural One supplies one canonical information available to local processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated information available to local processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

78.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: PASS.
- `distributed_computation-witness-3` - `distributed_computation` exact operational witness 3 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge is observation-relative exact support, not an imported modal axiom.	PASS

78.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: distributed knowledge, epistemic logic.

Knowledge is observation-relative exact support, not an imported modal axiom.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Knowledge is observation-relative exact support, not an imported modal axiom.

78.8 Exact evidence identities

- Source manifest: `sha256:d2ca5491f24d445df5d48630109f35c3d1e739ee4a899acfeacc4973f8b991f5`
- Complete-census certificate:
`sha256:ee21a86f5706fbc5c100d2a5213995f180a31a18c5f39ba812086b9de0f3447e`
- Derivation seal: `sha256:dd0eef6ba8eba0f639d57fba8dc5e048b90dc3c83b4f4074b36d3de1fbc91254`

- Independent implementation:
sha256:65d80ba430b121e33988d04246aa04fcf58ee245cff28f35e0b2b095eeea4738
- Independent certificate:
sha256:1c6d8edf0281e7a3cbfe30321cf9df2b4f87d8839ec1c901568d918f5011e8fc
- External validation:
sha256:b5b38e7b5d6013a66581be1b1db819b1850351d585312e7b1d509786ca815e3c
- Model-admitted engine receipt:
sha256:2e47ee8465ef950f9ad2d30eb331453e764abf2567a6fc83ed0ec99b9fceda41
- Receipt path: receipts/engine/model_admitted/SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001-2e47ee8465ef950f.json

79. Derivation 72: Fold locality law

Claim: SFT-COMP-DIST-LOCALITY-001

79.1 Scientific necessity and exact theorem

Fold locality law is required to derive bounded dependency neighborhoods without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

79.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for bounded dependency neighborhoods.

All generated finite bounded dependency neighborhoods structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Locality is relative to the registered network and round relation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

79.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is

source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

79.4 Operational laws and consequences

- Carrier law: network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

79.5 Depth-independent closure

Base: The structural One supplies one canonical bounded dependency neighborhoods form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bounded dependency neighborhoods form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

79.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-LOCALITY-001; result: PASS.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-LOCALITY-001; result: PASS.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-LOCALITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Locality is relative to the registered network and round relation.	PASS

79.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: local computation, distributed locality.

Locality is relative to the registered network and round relation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Locality is relative to the registered network and round relation.

79.8 Exact evidence identities

- Source manifest: sha256:b6ffc79d6147f815689d6f7de439921326a5c66486d30f4c4cf1810403dc681d
- Complete-census certificate:
sha256:1af96f4623e9adf8de58fc51a08eb1f5fbf438ce2311b18767e099810f072429
- Derivation seal: sha256:52733a46f531bfb6cf359e04411633d64b8daa8d79e70c6bcb2e114d7e98c69
- Independent implementation:
sha256:f480454682af4bc3c9fcd5ff8cf7586bd356b42f0b30c571d68b11d485510478
- Independent certificate:
sha256:7b519f5972fa40355453ff053370f91fa7cfe1e4c57bf309b3d21b93b0877892
- External validation:
sha256:e049568757b31066d538d4633f44dc5ea8ec7822f55180090561079a7b29c8e8
- Model-admitted engine receipt:
sha256:d75dcc0dc6937266743f829559bfe4b59dfe31fc0d2b186ccbd7979b13cba9fa
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-DIST-LOCALITY-001-d75dcc0dc6937266.json

80. Derivation 73: Fold network-computation law

Claim: SFT-COMP-DIST-NETWORK-COMPUTATION-001

80.1 Scientific necessity and exact theorem

Fold network-computation law is required to derive computation over generated communication graphs without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-LOCALITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001

- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

80.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for computation over generated communication graphs.

All generated finite computation over generated communication graphs structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Operating systems, cloud platforms and physical networks are validation domains only.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

80.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

80.4 Operational laws and consequences

- Carrier law: network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

80.5 Depth-independent closure

Base: The structural One supplies one canonical computation over generated communication graphs form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation over generated communication graphs form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

80.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - `distributed_computation` exact operational witness 1 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: PASS.
- `distributed_computation-witness-2` - `distributed_computation` exact operational witness 2 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: PASS.

- distributed_computation-witness-3 - distributed_computation exact operational witness 3 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Operating systems, cloud platforms and physical networks are validation domains only.	PASS

80.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: network algorithm, distributed network.

Operating systems, cloud platforms and physical networks are validation domains only.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Operating systems, cloud platforms and physical networks are validation domains only.

80.8 Exact evidence identities

- Source manifest: sha256:bde1392dce05056d4f751038f72b017f66a8616275d580e9c0b652193d5547d6
- Complete-census certificate:
sha256:07b3b59bf2f17b9085383298005d3a0ff8a1030c811ceca62683ab91b00e5027
- Derivation seal: sha256:e5de17ce510c5dee17b619932baeff26f47fb39a45be46d95529ea1148b9cba6
- Independent implementation:
sha256:40e6bd0f4a4cddad95e591f8e345388b6093afba2cdfd10fcd854215af618fce
- Independent certificate:
sha256:f96f76e03ecacbf6a6075d5fb3e6f25f8a0d814c7c994e2e0009644bf52d2e121
- External validation:
sha256:0ff1e2f18949b5be3afa6a81b5dfb16479d840183d39010f58363b7a5433ae6c
- Model-admitted engine receipt:
sha256:504861c17b0bc318c7dc6cfbfa43377ac42819e316735ab1b066b4dd11bb1951

- Receipt path: receipts/engine/model_admitted/SFT-COMP-DIST-NETWORK-COMPUTATION-001-504861c17b0bc318.json

81. Derivation 74: Fold one-wayness law

Claim: SFT-COMP-SEC-ONE-WAYNESS-001

81.1 Scientific necessity and exact theorem

Fold one-wayness law is required to derive forward computation versus inverse search without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-DIST-NETWORK-COMPUTATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

81.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for forward computation versus inverse search.

All generated finite forward computation versus inverse search structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Unbounded finite inversion is not impossible; one-wayness is always resource-relative.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

81.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

81.4 Operational laws and consequences

- Carrier law: finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

81.5 Depth-independent closure

Base: The structural One supplies one canonical forward computation versus inverse search form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated forward computation versus inverse search form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

81.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-ONE-WAYNESS-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-ONE-WAYNESS-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-ONE-WAYNESS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unbounded finite inversion is not impossible; one-wayness is always resource-relative.	PASS

81.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: one-way function, inversion hardness.

Unbounded finite inversion is not impossible; one-wayness is always resource-relative.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Unbounded finite inversion is not impossible; one-wayness is always resource-relative.

81.8 Exact evidence identities

- Source manifest: sha256:6dd95b0fb4b12cd256da8c4cd50b0bf3e5a72be91dfc7c67216cfb4a09affb8a
- Complete-census certificate:
sha256:43fb5c990476119271dd46fea841ed62af9eba3ed1b953da0cc908d649d04618
- Derivation seal: sha256:ddfec37a4c433a5afac7108f7c9efb94d851d621edc8c48b728cbcac61d14e40
- Independent implementation:
sha256:17638c5f2b4650aa4143670864dcecd5cb1ff24031a8b29ddc170fb24655957b
- Independent certificate:
sha256:e6bbfdf570e26533617e51b323bcb81bd4ba9cf746b583b62b02998fe61e35cd
- External validation:
sha256:72270df40ffff869ade128e4de351c353a0ad9fcac4088f77270223812c1bf960
- Model-admitted engine receipt:
sha256:a4de8e0ec3e624835896831a4545adc63b5aa37cb7b2cecb26404bda0175eb62
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-ONE-WAYNESS-001-a4de8e0ec3e62483.json

82. Derivation 75: Fold secrecy law

Claim: SFT-COMP-SEC-SECREC-001

82.1 Scientific necessity and exact theorem

Fold secrecy law is required to derive observation-relative message indistinguishability without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-ONE-WAYNESS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

82.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for observation-relative message indistinguishability.

All generated finite observation-relative message indistinguishability structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Security requires exact adversary, resource and support definitions.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

82.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.

Eliminated candidates	Exact first-failure reason
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

82.4 Operational laws and consequences

- Carrier law: message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

82.5 Depth-independent closure

Base: The structural One supplies one canonical observation-relative message indistinguishability form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated observation-relative message indistinguishability form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

82.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-SECREC-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-SECREC-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-SECREC-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS

Control	Expected	Observed	Result
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires exact adversary, resource and support definitions.	PASS

82.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: secrecy, confidentiality.

Security requires exact adversary, resource and support definitions.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Security requires exact adversary, resource and support definitions.

82.8 Exact evidence identities

- Source manifest: sha256:3ca8513b002c13e4a5f82ae192cdee76c0b56faf2a8190d0f762c5ab0b64d7f3
- Complete-census certificate:
sha256:2d1544ac0c7261dedc525c5954e448b785f6c1c7f01785d091ce411d425d5be4
- Derivation seal: sha256:59ae9300995bd5a4e1ed134ea927bca7428ef8dcad953c194d96c85b25ec9aba
- Independent implementation:
sha256:0be3b5dfc3878312f12f0414d1c476063532fd88932fe3be91f8b421f3fafdde
- Independent certificate:
sha256:930976772bdb29fedd4075e465f45c4d9d4b626b8976852a9766ed754b60887e
- External validation:
sha256:d5508adf2fe5b04b4e7189e2ca1a773a785d6cb98bffe668fdb6bea9758950c9
- Model-admitted engine receipt:
sha256:73a1685ef3e59a0b4cf8b8b67a1590f8a3b94e1dd5ace2d7eff0a6bdd5183b74
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-SECREC-001-73a1685ef3e59a0b.json

83. Derivation 76: Fold integrity law

Claim: SFT-COMP-SEC-INTEGRITY-001

83.1 Scientific necessity and exact theorem

Fold integrity law is required to derive detection of unauthorized record change without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-SECREC-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

83.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for detection of unauthorized record change.

All generated finite detection of unauthorized record change structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Integrity does not imply secrecy or authorship.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

83.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

83.4 Operational laws and consequences

- Carrier law: message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

83.5 Depth-independent closure

Base: The structural One supplies one canonical detection of unauthorized record change form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated detection of unauthorized record change form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

83.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-INTEGRITY-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-INTEGRITY-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-INTEGRITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Integrity does not imply secrecy or authorship.	PASS

83.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: integrity, tamper detection.

Integrity does not imply secrecy or authorship.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model

- Integrity does not imply secrecy or authorship.

83.8 Exact evidence identities

- Source manifest: sha256:2aeffa26e51414ad5f858b6994cd5efb7eed6ab1ecb17e9f4f4fa6d594c5171b
- Complete-census certificate:
sha256:2559677b0e6803955ab30583cf2e86884658249ad059c01b2cc74d85420073c3
- Derivation seal: sha256:986a5cc25662f593e3440f93aa672ba5b6464b76a10caceee4f535d930938358
- Independent implementation:
sha256:e50a46ec67178635ac4d6df9cde7330745719c7302ad7a655378325da782a6cb
- Independent certificate:
sha256:3c39a2ab4febb4ae372aef6e5ab87cb65c2d9591e3bd22c49c7e123e49d63b76
- External validation:
sha256:d85d50f86707d889c10663fa0bcd985d2ad7fefaf363d25289c143515903b7
- Model-admitted engine receipt:
sha256:54e959203614f553ff7ac956f549af5e057a240855b4175f237fb7363e080943
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-INTEGRITY-001-54e959203614f553.json

84. Derivation 77: Fold authentication law

Claim: SFT-COMP-SEC-AUTHENTICATION-001

84.1 Scientific necessity and exact theorem

Fold authentication law is required to derive source-identity verification without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-INTEGRITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

84.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for source-identity verification.

All generated finite source-identity verification structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Identity claims are only as strong as the registered credential and adversary model.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

84.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

84.4 Operational laws and consequences

- Carrier law: principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

84.5 Depth-independent closure

Base: The structural One supplies one canonical source-identity verification form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated source-identity verification form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

84.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-AUTHENTICATION-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-AUTHENTICATION-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-AUTHENTICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Identity claims are only as strong as the registered credential and adversary model.	PASS

84.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: authentication, challenge-response.

Identity claims are only as strong as the registered credential and adversary model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Identity claims are only as strong as the registered credential and adversary model.

84.8 Exact evidence identities

- Source manifest: sha256:0732acf36566e352c6679fee9ac8fe34183a7bcda30e89be6796be5fd15da9cf
- Complete-census certificate:
sha256:5b6775e9c64e49a395b06f8d2261fa33fd08bc087f620e9877b112909970f3cd
- Derivation seal: sha256:19ab81842953854a3cb2d3354bb9670af318dac19fdbeaec14a93aeab64bfb82
- Independent implementation:
sha256:8c06c672907c71606cb536667777f63d84e0add9a28cdaaf7d80c7e9206b3efb
- Independent certificate:
sha256:30945f38ecef4be19638d405dbe7af6020ed8663aa260f2012d54eee4e231fe5
- External validation:
sha256:cf6bfc6b8ab765bdc9f84ee432e122e03ea34764d788da1e97efcd71c6bfe5c1
- Model-admitted engine receipt:
sha256:2c39303d9b2dd557650fe04f8acb2f132510ca3af8258dc10792a276796cb29c
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-AUTHENTICATION-001-2c39303d9b2dd557.json

85. Derivation 78: Fold hashing law

Claim: SFT-COMP-SEC-HASHING-001

85.1 Scientific necessity and exact theorem

Fold hashing law is required to derive fixed-interface many-to-one record mapping without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-AUTHENTICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

85.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for fixed-interface many-to-one record mapping.

All generated finite fixed-interface many-to-one record mapping structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

85.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

85.4 Operational laws and consequences

- Carrier law: message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

85.5 Depth-independent closure

Base: The structural One supplies one canonical fixed-interface many-to-one record mapping form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated fixed-interface many-to-one record mapping form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

85.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-HASHING-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-HASHING-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-HASHING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.	PASS

85.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: hash function, collision resistance.

Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

85.8 Exact evidence identities

- Source manifest: sha256:3813266588daf1bad1e5cfb7490c5eeb712a7db71bdb3bfdea7e31f8b8f3a246
- Complete-census certificate:
sha256:2e1ce8ae91707798d1b54be7dcec8e6464f43c4802f7f8b293f0e643b0e32f5b
- Derivation seal: sha256:01617a3c0bf0134dc0c4fa284d583d22153e56db32bc9ba49acc70880b1282d9
- Independent implementation:
sha256:fd20652f052b8a1d5d18b9d12fdb3ee66fd12331f1075360660cff6aa0b45e23
- Independent certificate:
sha256:3e486f817fe2b3d7bdee5c000d0aac1c12fc77145d7d9e5a0ee989dc62551e4e
- External validation:
sha256:0436e07b628c57057b9ba0f7cb4a2a0051a444fd9275d02bbf0243ef7daee657
- Model-admitted engine receipt:
sha256:30adbf79b53883ef1c3266d37eabc5389dc245bd879ff5be7c3c81df62767984
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-HASHING-001-30adbf79b53883ef.json

86. Derivation 79: Fold commitment law

Claim: SFT-COMP-SEC-COMMITMENT-001

86.1 Scientific necessity and exact theorem

Fold commitment law is required to derive binding and hiding record phases without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-HASHING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

86.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for binding and hiding record phases.

All generated finite binding and hiding record phases structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Perfect hiding and perfect

binding claims require compatible exact supports and may be mutually limited.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

86.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

86.4 Operational laws and consequences

- Carrier law: message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

86.5 Depth-independent closure

Base: The structural One supplies one canonical binding and hiding record phases form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated binding and hiding record phases form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

86.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-COMMITMENT-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-COMMITMENT-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-COMMITMENT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.	PASS

86.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: commitment scheme, binding and hiding.

Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.

86.8 Exact evidence identities

- Source manifest: sha256 : 275b5aa4fbf532130c43d18e07f458a89f5a9125d26371830ea59649ff25696b
- Complete-census certificate:
sha256 : 7c012f93d8e4a33696460bc6443c46923f978d2d539c74ee59bd3c44224f30e3
- Derivation seal: sha256 : 8d9f534c3964bff3c525e06a7a83a4dc6be41b3ec1ee7c57bcd60df2c6158686
- Independent implementation:
sha256 : 784fc2f024c0611721529092f94ec3afc5d735b7d812c1d0b1805b7a2e27e078
- Independent certificate:
sha256 : 7d125b12815107bbf048d8978fe647c04a510fe9c3f27c68e99cd5818c74015f
- External validation:
sha256 : 33264f3ce62b46b936a0731aa0e0de8a6df818298318d342ac47e10b4816bd54
- Model-admitted engine receipt:
sha256 : 402789a36ddf90549527f52c193fbb0d0065ed548efaeb8178b629a367c59e5f
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-COMMITMENT-001-402789a36ddf9054.json

87. Derivation 80: Fold signature law

Claim: SFT-COMP-SEC-SIGNATURE-001

87.1 Scientific necessity and exact theorem

Fold signature law is required to derive publicly checkable source authorization without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-COMMITMENT-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

87.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for publicly checkable source authorization.

All generated finite publicly checkable source authorization structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

87.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

87.4 Operational laws and consequences

- Carrier law: message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

87.5 Depth-independent closure

Base: The structural One supplies one canonical publicly checkable source authorization form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated publicly checkable source authorization form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

87.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-SIGNATURE-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-SIGNATURE-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-SIGNATURE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.	PASS

87.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: digital signature, unforgeability.

No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.

87.8 Exact evidence identities

- Source manifest: sha256:a051961ecca2e1bd80888f5d7d856ba23b3fb70a8a9848bdeb59b764f76771ca
- Complete-census certificate:
sha256:8356a945a268513005817384c3da5f46b86ff408f57011cef2297a2e8dd0eb86
- Derivation seal: sha256:f6c665845c4cb731e7abc6523af0082873c08a343c7e4c7974205bc696fd63a9
- Independent implementation:
sha256:9e1a001d6e4ab6299ffc425aaf2d7c3912bb1545278bc11e70e34efe3c45b0ef
- Independent certificate:
sha256:ac4c20be5d8a6615688ca4fcd27541f4b7a0358fd7751b355dbf4822ffabce62
- External validation:
sha256:bdd57b0d7eb4417ceebfbd4c9b257631e20f1f7bebf266b2a4d1be764c09dfb7
- Model-admitted engine receipt:
sha256:1e80fcc3c81615ba2f131e83a91759abecf9a02f7a23b0a19b191a612d820bce
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-SIGNATURE-001-1e80fcc3c81615ba.json

88. Derivation 81: Fold proof-of-knowledge law

Claim: SFT-COMP-SEC-PROOF-KNOWLEDGE-001

88.1 Scientific necessity and exact theorem

Fold proof-of-knowledge law is required to derive extractable witness-bearing interaction without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-SIGNATURE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

88.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for extractable witness-bearing interaction.

All generated finite extractable witness-bearing interaction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Knowledge means existence

of an exact extraction relation in the registered grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-messages-or-keys	sampled-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

88.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

88.4 Operational laws and consequences

- Carrier law: statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

88.5 Depth-independent closure

Base: The structural One supplies one canonical extractable witness-bearing interaction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated extractable witness-bearing interaction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

88.6 Executed witnesses and adverse controls

- cryptography_security-witness-1 - cryptography_security exact operational witness 1 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: PASS.
- cryptography_security-witness-2 - cryptography_security exact operational witness 2 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: PASS.
- cryptography_security-witness-3 - cryptography_security exact operational witness 3 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge means existence of an exact extraction relation in the registered grammar.	PASS

88.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: proof of knowledge, witness extraction.

Knowledge means existence of an exact extraction relation in the registered grammar.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Knowledge means existence of an exact extraction relation in the registered grammar.

88.8 Exact evidence identities

- Source manifest: sha256:d6129fbbe4e42e045f47e98d50fb7c69bc2b466dcb694ace1e513e217ba28b99
- Complete-census certificate:
sha256:47f0d1f03b6c7d4ad9bcd6e3014e8bed0a2f28465bce0f86e9495b71a72cc451
- Derivation seal: sha256:096e1be819c3cbbc16da33d8aee4a9ec53e1f5d88153ea987802509c2c2bed13
- Independent implementation:
sha256:f5004305f90756de4a02683a31cabda06491ba72784c1ce3036f03e625ea3523
- Independent certificate:
sha256:240f016841b14142ad7671c8a07814cdaa2920e5aaf78f57fb3a092a895b5f68
- External validation:
sha256:6651ed591513d864d58c47fc68f4c0a5b16042cc0776045deb5e71b793ed5227
- Model-admitted engine receipt:
sha256:30375e18292a970fd81f642586376049976d61928fff9bff79bb0ee0692f0eb6
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-PROOF-KNOWLEDGE-001-30375e18292a970f.json

89. Derivation 82: Fold zero-knowledge law

Claim: SFT-COMP-SEC-ZERO-KNOWLEDGE-001

89.1 Scientific necessity and exact theorem

Fold zero-knowledge law is required to derive verification without additional witness distinction without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-PROOF-KNOWLEDGE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

89.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for verification without additional witness distinction.

All generated finite verification without additional witness distinction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Zero knowledge is relative to the registered verifier observation and simulator support.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

89.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

89.4 Operational laws and consequences

- Carrier law: statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

89.5 Depth-independent closure

Base: The structural One supplies one canonical verification without additional witness distinction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated verification without additional witness distinction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

89.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Zero knowledge is relative to the registered verifier observation and simulator support.	PASS

89.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: zero-knowledge proof, simulation.

Zero knowledge is relative to the registered verifier observation and simulator support.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Zero knowledge is relative to the registered verifier observation and simulator support.

89.8 Exact evidence identities

- Source manifest: sha256:620bbad2881acb7c41f3224a48f16c5ae2c8aa8fde8fd162e9bb90f0268debc4
- Complete-census certificate:
sha256:bf2e1d87dcd07838f14b6cec85d7487f0592f4bdd78710ae736cd2b8853b0274
- Derivation seal: sha256:5be0fa434efed7312395f86a9afb749199774bb3b538534566c3b6a783f37f8c
- Independent implementation:
sha256:24b62399d900fdd7fe64843cd05cf1e6f3d765489dcecc3f0941281fa57dfffb9
- Independent certificate:
sha256:a730450b5720e8b9b620c379e228ea634d9317646f7b52e43cbaf4f97fc7720f
- External validation:
sha256:c46233d93c746e6cf701fd5f16b696caf5f0f2188fd8307c103102101cfd87b
- Model-admitted engine receipt:
sha256:060ee3e0322b39395749298b995a9f4d67c148ade8eeba389c05906219d720ed
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-ZERO-KNOWLEDGE-001-060ee3e0322b3939.json

90. Derivation 83: Fold multiparty-computation law

Claim: SFT-COMP-SEC-MULTIPARTY-001

90.1 Scientific necessity and exact theorem

Fold multiparty-computation law is required to derive joint function evaluation with local privacy without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-ZERO-KNOWLEDGE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

90.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for joint function evaluation with local privacy.

All generated finite joint function evaluation with local privacy structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Security requires

an explicit corruption and communication model.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

90.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

90.4 Operational laws and consequences

- Carrier law: party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

90.5 Depth-independent closure

Base: The structural One supplies one canonical joint function evaluation with local privacy form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated joint function evaluation with local privacy form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

90.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-MULTIPARTY-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-MULTIPARTY-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-MULTIPARTY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires an explicit corruption and communication model.	PASS

90.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: secure multiparty computation, MPC.

Security requires an explicit corruption and communication model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Security requires an explicit corruption and communication model.

90.8 Exact evidence identities

- Source manifest: sha256:f99a5d1d82db730ede50111e4923fbdf63981e58c2187b81d9e5e7a3a0398dc4
- Complete-census certificate:
sha256:a2e3bc1c437dcf70d4a3becf015e03c0f5bd17d3efa7803076c8f1d2178fdd0b
- Derivation seal: sha256:e439ad17a32ce5c851815d3b14ab76da6da98235dfef93ce7e5a634a6a21d85e
- Independent implementation:
sha256:540ddd1dc71c191068a84d648aecaa8c87dca286b22f8eba3a22a2525d5c5bc4
- Independent certificate:
sha256:d2d42553baf0939f1eb080df51ff421d84598397699c3fe6debdfd601ddcc1b5
- External validation:
sha256:1914804774000141c84f200cb7b301bd9c7c524f9c2825cda44fb861df601e2d
- Model-admitted engine receipt:
sha256:211b27cbbbc88f0e82c4441ed0d47afe01703b5dd89892a3e35759025c5f29eb
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-MULTIPARTY-001-211b27cbbbc88f0e.json

91. Derivation 84: Fold adversarial-computation law

Claim: SFT-COMP-SEC-ADVERSARIAL-001

91.1 Scientific necessity and exact theorem

Fold adversarial-computation law is required to derive computation against generated hostile strategies without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-MULTIPARTY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

91.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for computation against generated hostile strategies.

All generated finite computation against generated hostile strategies structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An undefined adversary cannot support a security theorem.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

91.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

91.4 Operational laws and consequences

- Carrier law: system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

91.5 Depth-independent closure

Base: The structural One supplies one canonical computation against generated hostile strategies form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation against generated hostile strategies form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

91.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-ADVERSARIAL-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-ADVERSARIAL-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-ADVERSARIAL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An undefined adversary cannot support a security theorem.	PASS

91.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: adversary model, security game.

An undefined adversary cannot support a security theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An undefined adversary cannot support a security theorem.

91.8 Exact evidence identities

- Source manifest: `sha256:be1520c84305b1e364a98fef71e3a03a2b35b42e3131c3722cfe77c1c24208b6`

- Complete-census certificate:
sha256: ff38ea69df3de315e9a0da0ac142dfec4255ba3459637f6ea73eaffb7c850e73
- Derivation seal: sha256: 131a0d4c23230de9c1973b0660172a74c795f138add9a30cea8056abaa5f377f
- Independent implementation:
sha256: 667c35f6442a6883878e24e51c08894ee8f86106c7937bb824eca4df7a94d595
- Independent certificate:
sha256: 7b5cda64ba1d06a7f27e9723db24054715d8763d5e9b25d6a5087b88cbb06fe4
- External validation:
sha256: bc5047e436959279dfd58647b6a20fae0e479e20fd3f6b94507402aacd616d4a
- Model-admitted engine receipt:
sha256: feaa06a2fbc1430f8930075dada167efdfcb60df06ed5e21424557ce89b81b78
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SEC-ADVERSARIAL-001-feaa06a2fbc1430f.json

92. Derivation 85: Fold information-theoretic and computational-security law

Claim: SFT-COMP-SEC-SECURITY-DEFINITION-001

92.1 Scientific necessity and exact theorem

Fold information-theoretic and computational-security law is required to derive exact separation of unconditional and resource-bounded security without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-ADVERSARIAL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

92.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for exact separation of unconditional and resource-bounded security.

All generated finite exact separation of unconditional and resource-bounded security structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science

forms. Information-theoretic and computational claims must never be conflated.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-messages-or-keys	sampld-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

92.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

92.4 Operational laws and consequences

- Carrier law: secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

92.5 Depth-independent closure

Base: The structural One supplies one canonical exact separation of unconditional and resource-bounded security form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact separation of unconditional and resource-bounded security form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

92.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Information-theoretic and computational claims must never be conflated.	PASS

92.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: information-theoretic security, computational security.

Information-theoretic and computational claims must never be conflated.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Information-theoretic and computational claims must never be conflated.

92.8 Exact evidence identities

- Source manifest: sha256:41a6b4267c2f2ae8babbb09d61629ef19a5f4aad0cc3b512c5655628ea2e4d003
- Complete-census certificate:
sha256:92b6f656b84f07e9eb480e65d3a87489c1916aca3bcfe79e44278a2cf78e846b
- Derivation seal: sha256:6b21cf6168cb2b3e44e495511de413b364806a7f4f525932ee7469ac1ba28d2a
- Independent implementation:
sha256:c7d7f2d2ad3488af844772f361a82d489dd4775d704d8ac1f78a139c8736d496
- Independent certificate:
sha256:fb830648518dacbf88f79aec34306fa7f19a7791d9a649d6e97483bfb7658a22
- External validation:
sha256:8a7e1c759e576b62fa7c4db35de79ac8630f90be53a4d7b9d7faaf7ddf7af458
- Model-admitted engine receipt:
sha256:e4af2a902c19997a8a81d6df7fe8ebd8246a2900aac642a6003b287d4bde7007
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEC-SECURITY-DEFINITION-001-e4af2a902c19997a.json

93. Derivation 86: Fold post-quantum security-boundary law

Claim: SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001

93.1 Scientific necessity and exact theorem

Fold post-quantum security-boundary law is required to derive classical claims under an enlarged adversary interface without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-SECURITY-DEFINITION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

93.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for classical claims under an enlarged adversary interface.

All generated finite classical claims under an enlarged adversary interface structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-messages-or-keys	sampled-messages-or-keys removes a required exact support condition.	complete-generated-challenge-support	complete-generated-challenge-support retains the complete registered support condition.
adversary	undefined-adversary	undefined-adversary removes a required exact adversary condition.	exact-adversary-process-grammar	exact-adversary-process-grammar retains the complete registered adversary condition.
interaction	outcome-only	outcome-only removes a required exact interaction condition.	complete-interaction-trace	complete-interaction-trace retains the complete registered interaction condition.
observation	implicit-leakage	implicit-leakage removes a required exact observation condition.	exact-adversary-observation	exact-adversary-observation retains the complete registered observation condition.
security	informal-security	informal-security removes a required exact security condition.	retained-closed-distinction-property	retained-closed-distinction-property retains the complete registered security condition.
resources	unbounded-claim	unbounded-claim removes a required exact resources condition.	declared-information-or-resource-bound	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	challenge-examples	challenge-examples removes a required exact generality condition.	support-successor-certificate	support-successor-certificate retains the complete registered generality condition.
addition	imported-hardness-assumption	imported-hardness-assumption removes a required exact addition condition.	no-extra-security-premise	no-extra-security-premise retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

93.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

93.4 Operational laws and consequences

- Carrier law: classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

93.5 Depth-independent closure

Base: The structural One supplies one canonical classical claims under an enlarged adversary interface form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated classical claims under an enlarged adversary interface form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

93.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - `cryptography_security` exact operational witness 1 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: PASS.
- `cryptography_security-witness-2` - `cryptography_security` exact operational witness 2 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: PASS.
- `cryptography_security-witness-3` - `cryptography_security` exact operational witness 3 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.	PASS

93.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: post-quantum cryptography, quantum adversary.

No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.

93.8 Exact evidence identities

- Source manifest: sha256:db51baede6272149653383aac66a7f35aa42bc3af23f7a03f111bd71035fa022
- Complete-census certificate:
sha256:473ff558f791c67050eed2c25afaa5e8f1f8fc2532903b6554acf8be82a63cc5
- Derivation seal: sha256:8134279bbf4e56299f7ac9f4a9473bcc5dd141f070698642f5fdac6b8cfc53e9
- Independent implementation:
sha256:2dda32b30a46541bb6bb86c5ebe17d4883594404de13200d4bd086e1656e6ec1
- Independent certificate:
sha256:a5a2e61fb892deaa58864f2562348e0e83b69771cb5249932797ab1d3f01e591
- External validation:
sha256:76bac704135f577f50d1b819bfc5395d19aa7490015fd3032691a1b592b23a6
- Model-admitted engine receipt:
sha256:4e450e475eb74f6968064e4ba58530226f4a58f8a6c3fff12176306a4d028c4d
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001-4e450e475eb74f69.json

94. Derivation 87: Fold inference law

Claim: SFT-COMP-LEARN-INFERENCE-001

94.1 Scientific necessity and exact theorem

Fold inference law is required to derive selection among generated hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

94.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for selection among generated hypotheses.

All generated finite selection among generated hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Inference cannot import a

prior distribution or pretrained weight.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-hypotheses	sampled-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

94.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

94.4 Operational laws and consequences

- Carrier law: hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

94.5 Depth-independent closure

Base: The structural One supplies one canonical selection among generated hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated selection among generated hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

94.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INFERENCE-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INFERENCE-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INFERENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Inference cannot import a prior distribution or pretrained weight.	PASS

94.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: inference, hypothesis selection.

Inference cannot import a prior distribution or pretrained weight.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Inference cannot import a prior distribution or pretrained weight.

94.8 Exact evidence identities

- Source manifest: sha256 : b205f5d2bcffba31b2a8517eb32f9478c8a4305e3dcf85ae9ed9082971a96036
- Complete-census certificate:
sha256 : b1bd2070f66348ce7b4214d62727122c435e9d6980474d4c5b391600df55da9c
- Derivation seal: sha256 : 86235ff5fef7bf72b8dce90c3aed7ffbd387d6ac9aec4bcf541f310352df4cb5
- Independent implementation:
sha256 : 17c8d94beaa2ae8bd3ef6b1ee8007444129b98b82b09740194612e4434583399
- Independent certificate:
sha256 : bb6159b3a5f1ea8b49e4db61ce525e7b0c7f719296f9d966b1120bd369774de8
- External validation:
sha256 : 4521c49947e66c92bb50422ef3747db8aadb7f6885d35a0b5cfd3fb40a8dee8d
- Model-admitted engine receipt:
sha256 : 483c9c1cb618e3c57985fbbcdad9d4c3450b6dc50874c3f86a5a09fd4a8e54160
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-LEARN-INFERENCE-001-483c9c1cb618e3c5.json

95. Derivation 88: Fold classification and prediction law

Claim: SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001

95.1 Scientific necessity and exact theorem

Fold classification and prediction law is required to derive sealed mapping from inputs to held labels without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-INFERENCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

95.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for sealed mapping from inputs to held labels.

All generated finite sealed mapping from inputs to held labels structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Natural-data performance requires the separate empirical constitution.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampl ed-hypotheses	sampl ed-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasur e	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

95.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

95.4 Operational laws and consequences

- Carrier law: input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

95.5 Depth-independent closure

Base: The structural One supplies one canonical sealed mapping from inputs to held labels form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated sealed mapping from inputs to held labels form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

95.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Natural-data performance requires the separate empirical constitution.	PASS

95.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classification, prediction.

Natural-data performance requires the separate empirical constitution.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Natural-data performance requires the separate empirical constitution.

95.8 Exact evidence identities

- Source manifest: sha256:c49f4676c62a1e29209b8955a86b21670b0e44f10a575b8eb3400fff94bc92d8
- Complete-census certificate:
sha256:e2e8663b301ea772432ae0984bf1568ec622319f150a53914e7d6d9d6dc07214
- Derivation seal: sha256:69d028c417c15a65f8d335b120de039ccbebe7c4c70c8b316ef553a049d63c42
- Independent implementation:
sha256:2e499ab68732d073e51c68bd9ccadff6a9dd19b2fb20ccba2853b66ade014b24
- Independent certificate:
sha256:cf48193afa6527e13bf45359d46ac6145cca308770d6a52628430a328afcd0a4
- External validation:
sha256:aeb1268d3d337b6596c4f33c8af49d6220db7e674a724796260844808d75f041
- Model-admitted engine receipt:
sha256:2902c0a1035fc27073a7a1be9ad39806838e05a1ef7fb2cbf8e045f1705be080
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001-2902c0a1035fc270.json

96. Derivation 89: Fold representation-learning boundary law

Claim: SFT-COMP-LEARN-REPRESENTATION-001

96.1 Scientific necessity and exact theorem

Fold representation-learning boundary law is required to derive generated encodings preserving declared distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

96.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for generated encodings preserving declared distinctions.

All generated finite generated encodings preserving declared distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No opaque learned representation is admitted as a law without an exact transformation trace.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

96.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__`

retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter.

The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

96.4 Operational laws and consequences

- Carrier law: source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

96.5 Depth-independent closure

Base: The structural One supplies one canonical generated encodings preserving declared distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated generated encodings preserving declared distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

96.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-REPRESENTATION-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-REPRESENTATION-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-REPRESENTATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No opaque learned representation is admitted as a law without an exact transformation trace.	PASS

96.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: representation learning, feature representation.

No opaque learned representation is admitted as a law without an exact transformation trace.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No opaque learned representation is admitted as a law without an exact transformation trace.

96.8 Exact evidence identities

- Source manifest: sha256:b137149be197d5b38e3b24cb5cb468d8ecf855fe881d1ee3a8dbcc617d7ab390
- Complete-census certificate:
sha256:b55adedd6d5e18635cbcf6e75d74882a409d34581283f7b8117b0f845d921eca
- Derivation seal: sha256:9eaa78f002935ed855587d3be65f3de944d0a04472cd30757259663d777c07fe
- Independent implementation:
sha256:acf5b1a98cf9f1c0bef4dcd2b233d8e23c0861dc2f8d2e18a2f03e859204ce54
- Independent certificate:
sha256:a9d9123e1ce01feb9bba7752a76a7b7b8f72cab588a060badfcca338a18b8b2a
- External validation:
sha256:3ae7189b33dfec76df2087ba9c398ab5d3aecdfac84eed0ab446e85e6db9c763
- Model-admitted engine receipt:
sha256:aaf1bb06300eddd353f33d549cc9ee8ace48e5f131d530ff9bdb7976c1cd0061
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-REPRESENTATION-001-aaf1bb06300eddd3.json

97. Derivation 90: Fold generalization law

Claim: SFT-COMP-LEARN-GENERALIZATION-001

97.1 Scientific necessity and exact theorem

Fold generalization law is required to derive extension from registered evidence to generated unseen support without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

97.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for extension from registered evidence to generated unseen support.

All generated finite extension from registered evidence to generated unseen support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Generalization is empirical unless a structural successor proof exhausts the target family.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

97.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

97.4 Operational laws and consequences

- Carrier law: hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

97.5 Depth-independent closure

Base: The structural One supplies one canonical extension from registered evidence to generated unseen support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated extension from registered evidence to generated unseen support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

97.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-GENERALIZATION-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-GENERALIZATION-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-GENERALIZATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Generalization is empirical unless a structural successor proof exhausts the target family.	PASS

97.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: generalization, out-of-sample prediction.

Generalization is empirical unless a structural successor proof exhausts the target family.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Generalization is empirical unless a structural successor proof exhausts the target family.

97.8 Exact evidence identities

- Source manifest: sha256:8941fcdacbdb992fdffa7704d92aee58549b81bed0ed5f99bc7c358a1bb2340c
- Complete-census certificate:
sha256:0a392f50c6824a6fab3cda881923c4aa02bb8438374791476c134a66dd0b7454
- Derivation seal: sha256:a9bc47b89cf7b5320b47b847a995c2a156143272e9219c67f0ea6f256ad0411e
- Independent implementation:
sha256:eb08a6b81d8f850328b67873f31028e849338accc1bf929d01b1506aa763cb80
- Independent certificate:
sha256:53095ce3818727b136a2ca6f4699d6f376ebc7219ea632a98129ffb4b4c18d2d
- External validation:
sha256:73e4b939d6ae084174bf079d91dc2df84b6465f61b132169c263a04257f189f1
- Model-admitted engine receipt:
sha256:c99db1b2646ffad5f54e060c0f1ae1b32a9d8ebcdc719d36a6e3aec10abf446d
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-GENERALIZATION-001-c99db1b2646ffad5.json

98. Derivation 91: Fold sample-complexity law

Claim: SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001

98.1 Scientific necessity and exact theorem

Fold sample-complexity law is required to derive evidence support required to close hypothesis distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-GENERALIZATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

98.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for evidence support required to close hypothesis distinctions.

All generated finite evidence support required to close hypothesis distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No

asymptotic formula is imported; each family requires a generated certificate.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampl ed-hypotheses	sampl ed-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

98.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampl ed-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

98.4 Operational laws and consequences

- Carrier law: hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

98.5 Depth-independent closure

Base: The structural One supplies one canonical evidence support required to close hypothesis distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated evidence support required to close hypothesis distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

98.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No asymptotic formula is imported; each family requires a generated certificate.	PASS

98.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: sample complexity, teaching dimension.

No asymptotic formula is imported; each family requires a generated certificate.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No asymptotic formula is imported; each family requires a generated certificate.

98.8 Exact evidence identities

- Source manifest: sha256 : a8c2a967ac606c5ee071bfab9ff0ea1676861940591718fa3a472127d3c28ddb
- Complete-census certificate:
sha256 : caa25ef318a02f773c9460e8ea08be4d447b317355cfc0b6285dff580c94c89b
- Derivation seal: sha256 : 9fddabe49ce98bba0f880bb1cff97015501842d31fe376af0da5c9096627a420
- Independent implementation:
sha256 : de4a1e8920b9a571c0477fbe9a05bec0fdf78829f65ce150603fcfa5c161ad9f
- Independent certificate:
sha256 : 027768d0addb63d237cba645555e463eb1f01c6b030c8a672fe5f335c660ea1
- External validation:
sha256 : 10fed92e8c01ab8b517e587549341509dfe734924381421434671751f5b6b5e8
- Model-admitted engine receipt:
sha256 : d4b13522a727a382b31bddb83eda909f6706c4f7b9bc48633b8e1fd28c133092
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001-d4b13522a727a382.json

99. Derivation 92: Fold optimization-in-learning law

Claim: SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001

99.1 Scientific necessity and exact theorem

Fold optimization-in-learning law is required to derive exact objective selection over hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

99.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for exact objective selection over hypotheses.

All generated finite exact objective selection over hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Gradient, differentiability and fitted parameter spaces are not assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

99.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

99.4 Operational laws and consequences

- Carrier law: hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

99.5 Depth-independent closure

Base: The structural One supplies one canonical exact objective selection over hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact objective selection over hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

99.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Gradient, differentiability and fitted parameter spaces are not assumed.	PASS

99.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: learning optimization, empirical risk.

Gradient, differentiability and fitted parameter spaces are not assumed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Gradient, differentiability and fitted parameter spaces are not assumed.

99.8 Exact evidence identities

- Source manifest: sha256:12dce3bb852954f394ecb4d67d026ccda30f3923c6851bfb0eb8c95930ed8698
- Complete-census certificate:
sha256:3059d8e36f4a965707ebcc0a5a61f9030a6050f7ff64c49d0117e19826507bc8
- Derivation seal: sha256:db572a877ce20e6fa9c85144aa3abc6588031199d66e6f967b6d744fe908f9bc
- Independent implementation:
sha256:a7e21f4888ff04407aee3540f61a11ac7d327f6d70bd97a48943ed830e819339
- Independent certificate:
sha256:da8458ec44ab9e55677ace1d413bc4be984ddb2033182648add409771da500af
- External validation:
sha256:086de46d3035bc9d85c7be6b357dcca7180725c96eb1502478ba96d1ce65d1fe
- Model-admitted engine receipt:
sha256:543256016991fee945bc9d01d128355d0a9777a814dc7428ff9146ac93a8a398
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001-543256016991fee9.json

100. Derivation 93: Fold computational-induction law

Claim: SFT-COMP-LEARN-INDUCTION-001

100.1 Scientific necessity and exact theorem

Fold computational-induction law is required to derive rule construction from finite evidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

100.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for rule construction from finite evidence.

All generated finite rule construction from finite evidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Finite agreement alone never

proves unrestricted continuation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

100.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

100.4 Operational laws and consequences

- Carrier law: observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

100.5 Depth-independent closure

Base: The structural One supplies one canonical rule construction from finite evidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated rule construction from finite evidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

100.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INDUCTION-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INDUCTION-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INDUCTION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Finite agreement alone never proves unrestricted continuation.	PASS

100.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: induction, rule learning.

Finite agreement alone never proves unrestricted continuation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Finite agreement alone never proves unrestricted continuation.

100.8 Exact evidence identities

- Source manifest: sha256:216c47067c713bcb4c79907da10663b2217e48cb311d6bf71983d40aa42baf90
- Complete-census certificate:
sha256:565d863a25e8bdc9278071a9cc418c11c3fd46daece418426847aaba0820ebb4
- Derivation seal: sha256:3efc6894220eb6df93411e220a210816dfefa95a8195b2835ddc002134ed63ab
- Independent implementation:
sha256:2a7ec40d95f8877c0c4985d214c3a60c199fec4ff536133fecb7776cf01da5c2
- Independent certificate:
sha256:fb353e73890cb1f146d4612722f473d27a53ae6b55e398ede969035e4a2fc80c
- External validation:
sha256:c1989a33f208cd8840a0743d971659c41fff7acfa4df1fe38b105401638050eb
- Model-admitted engine receipt:
sha256:919a71359ffd05922865e02e03ef9013abb551bc0ccb5b0b95363aee5cd998ed
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-LEARN-INDUCTION-001-919a71359ffd0592.json

101. Derivation 94: Fold search and planning law

Claim: SFT-COMP-LEARN-SEARCH-PLANNING-001

101.1 Scientific necessity and exact theorem

Fold search and planning law is required to derive action-sequence construction toward exact goals without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-INDUCTION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

101.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for action-sequence construction toward exact goals.

All generated finite action-sequence construction toward exact goals structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only the generated environment model is covered.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-hypotheses	sampled-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-eraser	single-winner-eraser removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

101.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

101.4 Operational laws and consequences

- Carrier law: state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

101.5 Depth-independent closure

Base: The structural One supplies one canonical action-sequence construction toward exact goals form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated action-sequence construction toward exact goals form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

101.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only the generated environment model is covered.	PASS

101.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: planning, state-space search.

Only the generated environment model is covered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only the generated environment model is covered.

101.8 Exact evidence identities

- Source manifest: `sha256:fd1e4c6bce5884546b46d3b8bdb487c34e5f0d9fc144467cbbd255f53ec2627b`
- Complete-census certificate:
`sha256:94004478e61e1ab8efbf1281203fc61c897cf44e38c8db4ad02b52365c7019ad`

- Derivation seal: sha256:15ffc8f3e5def5649d23e98ea4fdc3424e78eaa8f8b6d73c82bb60159dcfcf57
- Independent implementation:
sha256:261e951d99ce2cda0b9fd6e33cada770b98f997f344ef9ab5d2983b8eae2f571
- Independent certificate:
sha256:aa5ddf2ede9accea8029726d20fa5375af6ea31049796f13c29f611c8715ad5f
- External validation:
sha256:4d3e3ee3ed6bcd4c5d13258a994ffbc068e989dfd5dd8c1472d0bdc2ac2979e6
- Model-admitted engine receipt:
sha256:fae483c37bc6bdafa9b29553a47c1620a9038fb736fd8cf2af77fe96535e780c7
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-SEARCH-PLANNING-001-fae483c37bc6bdafa.json

102. Derivation 95: Fold reinforcement-computation law

Claim: SFT-COMP-LEARN-REINFORCEMENT-001

102.1 Scientific necessity and exact theorem

Fold reinforcement-computation law is required to derive policy evaluation through generated interaction traces without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-SEARCH-PLANNING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

102.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for policy evaluation through generated interaction traces.

All generated finite policy evaluation through generated interaction traces structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No stochastic cause, fitted value function or external reward scale is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

102.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

102.4 Operational laws and consequences

- Carrier law: states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

102.5 Depth-independent closure

Base: The structural One supplies one canonical policy evaluation through generated interaction traces form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated policy evaluation through generated interaction traces form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

102.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-REINFORCEMENT-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-REINFORCEMENT-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-REINFORCEMENT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No stochastic cause, fitted value function or external reward scale is imported.	PASS

102.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reinforcement learning, policy.

No stochastic cause, fitted value function or external reward scale is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No stochastic cause, fitted value function or external reward scale is imported.

102.8 Exact evidence identities

- Source manifest: sha256:6ddc0f26308bb7a994c49435c6b65a7e566854bd9a4777a0edb8505c18911acd
- Complete-census certificate:
sha256:be3e82a9b47bfbde120ee0ed38771bbf9f844742494adc945dc3cb9b83edd28b
- Derivation seal: sha256:922810d79cb65be09158633bb008b07b816501fc5b47e20d5527bf4422f350fd
- Independent implementation:
sha256:3e6c09be62e44db9928e0f51ac60f3c0a21d392640b02961341d29aa93b546bd
- Independent certificate:
sha256:c3c5e2c21db8d0448d9c1f5fac9dfc1978a2e85db163404b0754b3a6be9d25e2
- External validation:
sha256:656c0d9cc8004254483b20cb731012f582f72db6e28a241e4a7edc710c99ca83
- Model-admitted engine receipt:
sha256:0cb38ecfea78976c1452c6d9974fff417f27c4cb05a6721698d9d8c1c4328a1a
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-LEARN-REINFORCEMENT-001-0cb38ecfea78976c.json

103. Derivation 96: Fold multi-agent computation law

Claim: SFT-COMP-LEARN-MULTIAGENT-001

103.1 Scientific necessity and exact theorem

Fold multi-agent computation law is required to derive interacting adaptive processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-REINFORCEMENT-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

103.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for interacting adaptive processes.

All generated finite interacting adaptive processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Game-theoretic correspondence is downstream and requires exact preference relations.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

103.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

103.4 Operational laws and consequences

- Carrier law: agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

103.5 Depth-independent closure

Base: The structural One supplies one canonical interacting adaptive processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated interacting adaptive processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

103.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-MULTIAGENT-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-MULTIAGENT-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-MULTIAGENT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Game-theoretic correspondence is downstream and requires exact preference relations.	PASS

103.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: multi-agent systems, agent interaction.

Game-theoretic correspondence is downstream and requires exact preference relations.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Game-theoretic correspondence is downstream and requires exact preference relations.

103.8 Exact evidence identities

- Source manifest: sha256:f62816b2440c54f007826324e7acdc51362bb01ebec449195d02bc2171ed0fb0
- Complete-census certificate:
sha256:8377eb399e70932b70ca84cd87750e53ceb42a24143a5cb38048300085926098
- Derivation seal: sha256:fb345343a53247874bd2912ed6f77073446b826010268bc6afc8fae76c14a47f
- Independent implementation:
sha256:74bd7c749f6e5716fd7a91e3f9b6bc247a333b6ab385d4d268519368534aaf10
- Independent certificate:
sha256:d32be77b1c6b6a3bb80e9210bedabe9bd0248c382462bc8404e6de95610ffbfc
- External validation:
sha256:9b57258598dd95151a00eb00580d24a060d532ee02a7914cd1e6ac36a5b8b443
- Model-admitted engine receipt:
sha256:9d95cd6493bea3f2e926a8beb970becf6593a0aad4c08b4ea2f999e6760202a2
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-LEARN-MULTIAGENT-001-9d95cd6493bea3f2.json

104. Derivation 97: Fold adaptation law

Claim: SFT-COMP-LEARN-ADAPTATION-001

104.1 Scientific necessity and exact theorem

Fold adaptation law is required to derive trace-bound change of computational rules or state without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-MULTIAGENT-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

104.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for trace-bound change of computational rules or state.

All generated finite trace-bound change of computational rules or state structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Adaptation

cannot silently alter the constitutional engine or claim law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-hypotheses	sampled-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

104.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

104.4 Operational laws and consequences

- Carrier law: initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

104.5 Depth-independent closure

Base: The structural One supplies one canonical trace-bound change of computational rules or state form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated trace-bound change of computational rules or state form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

104.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-ADAPTATION-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-ADAPTATION-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-ADAPTATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Adaptation cannot silently alter the constitutional engine or claim law.	PASS

104.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: adaptation, online learning.

Adaptation cannot silently alter the constitutional engine or claim law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Adaptation cannot silently alter the constitutional engine or claim law.

104.8 Exact evidence identities

- Source manifest: sha256:d86cf331ef2b3a2f54cc9d847ad323ae0a0637c68ba19ea2401be1abc075ea95
- Complete-census certificate:
sha256:686d736a609a82cd61ffc1446cf5a08c81eeeb199a17e1300bb9908515862c0e
- Derivation seal: sha256:fbfe8663647aa11d4c95ce51175bb01a3c0ed27da3ae8148bf20d8313d646fd8
- Independent implementation:
sha256:bd4698fccdf68c28d5cbeea2976f776b0f15dd2377b2ce00ca4714fa140b836a
- Independent certificate:
sha256:1edd726dbaeaf10b6bdd6f7e816f7073bbaf8f3533e1013670f69d938117b268
- External validation:
sha256:4de8725dbfb89714aadda6c0fc76ce9dbc2bb2534f36911908c3ddb74d32f174
- Model-admitted engine receipt:
sha256:64ec3907b0e627e9e06f675e15a8ee10818781609c850b20948008691aa9d1cb
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-LEARN-ADAPTATION-001-64ec3907b0e627e9.json

105. Derivation 98: Fold computational limits-of-learning law

Claim: SFT-COMP-LEARN-LEARNING-LIMITS-001

105.1 Scientific necessity and exact theorem

Fold computational limits-of-learning law is required to derive indistinguishable evidence and unidentifiable hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT - COMP - LEARN - ADAPTATION - 001
- SFT - FOUNDATION - FORM - ENFORCEMENT - 001
- SFT - MATH - DISCRETE - 001
- SFT - MATH - GRAPH - NETWORK - 001
- SFT - MATH - LOGIC - PROOF - 001
- SFT - MATH - CATEGORY - TYPE - COMPOSITION - 001
- SFT - INFO - ENCODING - DECODING - 001
- SFT - INFO - CONSERVATION - LOSS - 001

105.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for indistinguishable evidence and unidentifiable hypotheses.

All generated finite indistinguishable evidence and unidentifiable hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Limits are relative to the exact evidence, hypothesis and target grammars.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampled-hypotheses	sampled-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

105.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

105.4 Operational laws and consequences

- Carrier law: hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

105.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable evidence and unidentifiable hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable evidence and unidentifiable hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

105.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Limits are relative to the exact evidence, hypothesis and target grammars.	PASS

105.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: no-free-lunch, identifiability.

Limits are relative to the exact evidence, hypothesis and target grammars.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Limits are relative to the exact evidence, hypothesis and target grammars.

105.8 Exact evidence identities

- Source manifest: `sha256:55152f2e7369fb034b347faa7c8d385705344969ce55f80e29d0311b2912fda0`

- Complete-census certificate:
sha256:33009e7095b23df40e776554f188351b95bd11b85b677f3d070747836fc131ab
- Derivation seal: sha256:f5e5533666d646c7ef7bf627324974f71f2c66942cc4a1bfcf76e931fba31ea4
- Independent implementation:
sha256:6f0b680771420d91f20a198e0be71505ba6bf4d1857ec9b562906dea65c7cadf
- Independent certificate:
sha256:2ce5d5a80478e0d65c29ff54fe9ab22de89e9c7b86b381f3b2eee2fd2919ced4
- External validation:
sha256:8c965863ce40503dee10d74d35ebe003bcbb75400e84133ab01119fc3d1c62fd
- Model-admitted engine receipt:
sha256:969046a8a85fc0802d0269d6a4a46e706d278b71450e8b7c7dbd1188fe21bc90
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-LIMITS-001-969046a8a85fc080.json

106. Derivation 99: Fold interpretability and verification law

Claim: SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001

106.1 Scientific necessity and exact theorem

Fold interpretability and verification law is required to derive traceable relation from inputs to decisions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-LEARNING-LIMITS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

106.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for traceable relation from inputs to decisions.

All generated finite traceable relation from inputs to decisions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A narrative explanation or score cannot replace the exact trace.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

106.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampld-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

106.4 Operational laws and consequences

- Carrier law: model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

106.5 Depth-independent closure

Base: The structural One supplies one canonical traceable relation from inputs to decisions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated traceable relation from inputs to decisions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

106.6 Executed witnesses and adverse controls

- learning_intelligence-witness-1 - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: PASS.
- learning_intelligence-witness-2 - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: PASS.
- learning_intelligence-witness-3 - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A narrative explanation or score cannot replace the exact trace.	PASS

106.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: interpretability, model verification.

A narrative explanation or score cannot replace the exact trace.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A narrative explanation or score cannot replace the exact trace.

106.8 Exact evidence identities

- Source manifest: sha256:57953fb29c34d5ba8d75378b797de3cfeebe71f875ffa2c17dbfd0fde75d2ed4
- Complete-census certificate:
sha256:b1f510b5bcd5dea8b61574693d523e057df0e57df49061a5cb052128bc70ad19
- Derivation seal: sha256:c8db64a84f070d98e4c50cdc3282e18181699e08907e6325c40ce7f8d0b3c710
- Independent implementation:
sha256:56cf69782fa0d5db9f55475a297018e3351f9e65f24512b0a88548f3d1b67943
- Independent certificate:
sha256:c9831ba4b85a410b2432b135d2d024421a92478a2ebe70c18e559d9e6ee863b6
- External validation:
sha256:4983de7e26afc28ec869589a8285b7cac9c6a31bd71dfbc23b2f890b22223f45
- Model-admitted engine receipt:
sha256:c0246290a461042d4797f427dfd27adaebf0954e5372091899b896b3418a1424
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001-c0246290a461042d.json

107. Derivation 100: Fold classical-learning model law

Claim: SFT-COMP-LEARN-CLASSICAL-LEARNING-001

107.1 Scientific necessity and exact theorem

Fold classical-learning model law is required to derive complete classical finite learning processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

107.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for complete classical finite learning processes.

All generated finite complete classical finite learning processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Quantum learning operations remain entirely outside this branch.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	sampld-hypotheses	sampld-hypotheses removes a required exact support condition.	complete-generated-hypothesis-support	complete-generated-hypothesis-support retains the complete registered support condition.
representation	opaque-representation	opaque-representation removes a required exact representation condition.	traceable-canonical-representation	traceable-canonical-representation retains the complete registered representation condition.
inference	pretrained-answer	pretrained-answer removes a required exact inference condition.	exact-evidence-to-hypothesis-relation	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
evaluation	development-score	development-score removes a required exact evaluation condition.	sealed-independent-evaluation	sealed-independent-evaluation retains the complete registered evaluation condition.
uncertainty	single-winner-erasure	single-winner-erasure removes a required exact uncertainty condition.	retained-alternative-ledger	retained-alternative-ledger retains the complete registered uncertainty condition.
resources	unreported-search	unreported-search removes a required exact resources condition.	complete-learning-resource-trace	complete-learning-resource-trace retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	training-examples-only	training-examples-only removes a required exact generality condition.	structural-or-empirical-target-boundary	structural-or-empirical-target-boundary retains the complete registered generality condition.
addition	fitted-weight-or-prior	fitted-weight-or-prior removes a required exact addition condition.	no-extra-learning-parameter	no-extra-learning-parameter retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

107.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

107.4 Operational laws and consequences

- Carrier law: examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

107.5 Depth-independent closure

Base: The structural One supplies one canonical complete classical finite learning processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete classical finite learning processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

107.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - `learning_intelligence` exact operational witness 1 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: PASS.
- `learning_intelligence-witness-2` - `learning_intelligence` exact operational witness 2 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: PASS.
- `learning_intelligence-witness-3` - `learning_intelligence` exact operational witness 3 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: PASS.

Control	Expected	Observed	Result
<code>false_premise</code>	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	PASS
<code>tampered_source</code>	reject a changed official source identity	changed source identity differs	PASS
<code>tampered_artifact</code>	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
<code>boundary</code>	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Quantum learning operations remain entirely outside this branch.	PASS

107.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classical learning, machine learning theory.

Quantum learning operations remain entirely outside this branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Quantum learning operations remain entirely outside this branch.

107.8 Exact evidence identities

- Source manifest: sha256:84c7423aac0c5dd42a0da2038331ca17fc561cf85c820db66054604dbf5a44e
- Complete-census certificate:
sha256:de855f8a0d135b04ae3c2ba92de145a816d1607b24bf17c6bdd175b404fc2638
- Derivation seal: sha256:d0f21b1802e74a2e9b58544c4d87f8c3934cf28aa6628b44b863a14c4df9a29f
- Independent implementation:
sha256:c009e53b2ed811511621e85d68a3f422e8467044881b0c670d98c74a365f57d0
- Independent certificate:
sha256:17f40ce955c28ee371c3107d4e114ce125b9fd68dfe1ebdb760af5621159658e
- External validation:
sha256:7f34c30e760456e4653d516838ec65c0f449d819f1a4fc2e2878ade0222383a2
- Model-admitted engine receipt:
sha256:86cdfa5fb4a647883b4b1300ea87c83dddd01b8f34f96c3738fec0baa8f8a8ce
- Receipt path: receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSICAL-LEARNING-001-86cdfa5fb4a64788.json

108. Derivation 101: Fold exact and approximate calculation law

Claim: SFT-COMP-SCI-EXACT-APPROXIMATE-001

108.1 Scientific necessity and exact theorem

Fold exact and approximate calculation law is required to derive separation of proof values and bounded approximations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

108.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for separation of proof values and bounded approximations.

*All generated finite separation of proof values and bounded approximations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms.
Approximate values cannot flow backward to select an exact law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

108.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

108.4 Operational laws and consequences

- Carrier law: exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

108.5 Depth-independent closure

Base: The structural One supplies one canonical separation of proof values and bounded approximations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated separation of proof values and bounded approximations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

108.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Approximate values cannot flow backward to select an exact law.	PASS

108.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: exact computation, approximate computation.

Approximate values cannot flow backward to select an exact law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Approximate values cannot flow backward to select an exact law.

108.8 Exact evidence identities

- Source manifest: sha256 : fb2986b63e83b23133a3823d055f0c3beb020e6b928274281935b3055ecfa40d
- Complete-census certificate:
sha256 : 8ab424ccb156e7c2e467452c86467587b3e5cb0a0a1f61b5390ae0d8f6365401
- Derivation seal: sha256 : 8d29ccecdb3e151f4ca0c51c27e60e628a21c6471db567add4c18e7e8f8f53a3
- Independent implementation:
sha256 : 5a5450ce9d27d00c9c63c225d46ecb4cb8ac2a8f713dc63ac331245ee27549c2
- Independent certificate:
sha256 : a873faf2411c2484210b6488ffa52a8d7f4f564c06d6caab2a38e3524b1304d1
- External validation:
sha256 : 3dc1f3c153b7c6420f71d1f912eef13516f2952441df895cd2f246e0b8dc640a
- Model-admitted engine receipt:
sha256 : d04dba138ab4ed09a8fba0c3751fc8588a7ad3b34790ac1f536b44b013b1a905
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-EXACT-APPROXIMATE-001-d04dba138ab4ed09.json

109. Derivation 102: Fold numerical-stability law

Claim: SFT-COMP-SCI-STABILITY-001

109.1 Scientific necessity and exact theorem

Fold numerical-stability law is required to derive output distinction under input or operation perturbation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-EXACT-APPROXIMATE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

109.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for output distinction under input or operation perturbation.

All generated finite output distinction under input or operation perturbation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No floating epsilon or norm is imported without derivation.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

109.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

109.4 Operational laws and consequences

- Carrier law: exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

109.5 Depth-independent closure

Base: The structural One supplies one canonical output distinction under input or operation perturbation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated output distinction under input or operation perturbation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

109.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-STABILITY-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-STABILITY-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-STABILITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating epsilon or norm is imported without derivation.	PASS

109.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: numerical stability, conditioning.

No floating epsilon or norm is imported without derivation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No floating epsilon or norm is imported without derivation.

109.8 Exact evidence identities

- Source manifest: `sha256:d5cf72d09f2ddd68644ff7db62e2a034d814b2c271bb3a73e8348e6ee7825924`
- Complete-census certificate:
`sha256:c936bf0f2a0fecc107d19c73caec2ed4c0aa05e8cb379920574a4cb536a8867a`
- Derivation seal: `sha256:a43ca827851cfe1c7db7b09f951670311b50ebcca5938c0c5d205e9e7c9a7f21`

- Independent implementation:
sha256:ed227d1ffe6b9cca5e1bd78b18ea37b96420ec366c45fc0886fd956d6f69dae0
- Independent certificate:
sha256:f8a39efdb4e40d985b773a578ac4cbb790f0491b94fdf2f9a1b2e5e5dce5d6f
- External validation:
sha256:bd975ace29c4d474e3faf48d0b6399acf9f3e6cee4cd2f9f7178be691565b923
- Model-admitted engine receipt:
sha256:f50a038a131659cc3156af54451f66c24d7562bb4284ab1ad6a71b6e97cc6980
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-STABILITY-001-f50a038a131659cc.json

110. Derivation 103: Fold error-propagation law

Claim: SFT-COMP-SCI-ERROR-PROPAGATION-001

110.1 Scientific necessity and exact theorem

Fold error-propagation law is required to derive transport of exact approximation distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-STABILITY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

110.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for transport of exact approximation distinctions.

All generated finite transport of exact approximation distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Signed arithmetic is represented by held orientation labels, not negative proof values.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

110.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

110.4 Operational laws and consequences

- Carrier law: local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

110.5 Depth-independent closure

Base: The structural One supplies one canonical transport of exact approximation distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated transport of exact approximation distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

110.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Signed arithmetic is represented by held orientation labels, not negative proof values.	PASS

110.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: error propagation, uncertainty propagation.

Signed arithmetic is represented by held orientation labels, not negative proof values.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Signed arithmetic is represented by held orientation labels, not negative proof values.

110.8 Exact evidence identities

- Source manifest: sha256:715d0ace314d6fe5a0307bef26e63634bcf52fb83416a5a01f6ccbb9cea1fa62
- Complete-census certificate:
sha256:7a39d937f82bc63a6bb808c8deb7e81842764dfcfe753d000e76a9b4b29efc5a
- Derivation seal: sha256:3c1e26188805f325b58f26f59dd2b777a2b71a8294e318eb6a679564baab529d
- Independent implementation:
sha256:ea298bac8ba3d6d280b08d831a3da7564bed420acc69dfff8630e933bfb791f0
- Independent certificate:
sha256:8ff2ad3dbafe41f907f100ace8d23fb0c2b177a1e8720b8a1b6aec3c2c11e8b6
- External validation:
sha256:10b7ab2f6ebc21135b80a940a993f81e0ab0a0120ca0db765d795507dc542d68
- Model-admitted engine receipt:
sha256:083dc87700741205332c48c4a74aa5658b9f205141bcb39a13bb88288354230b
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-ERROR-PROPAGATION-001-083dc87700741205.json

111. Derivation 104: Fold discretization law

Claim: SFT-COMP-SCI-DISCRETIZATION-001

111.1 Scientific necessity and exact theorem

Fold discretization law is required to derive finite generated representation of a declared model without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-ERROR-PROPAGATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

111.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for finite generated representation of a declared model.

All generated finite finite generated representation of a declared model structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An ungenerated continuum is never the derivational premise.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-re-certificate	base-successor-closure-re-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

111.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.

Eliminated candidates	Exact first-failure reason
64	sampld-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

111.4 Operational laws and consequences

- Carrier law: source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

111.5 Depth-independent closure

Base: The structural One supplies one canonical finite generated representation of a declared model form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite generated representation of a declared model form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

111.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-DISCRETIZATION-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-DISCRETIZATION-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-DISCRETIZATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An ungenerated continuum is never the derivational premise.	PASS

111.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: discretization, mesh.

An ungenerated continuum is never the derivational premise.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An ungenerated continuum is never the derivational premise.

111.8 Exact evidence identities

- Source manifest: sha256 : fb2079496ebb17f2dfb5cf163b7b360fee2dcbaaa0ad198198a4ee65af700d0e
- Complete-census certificate:
sha256 : 57e7908787684a30402fccfb4ae3e6ac8e27c0f1f734dbf09ab2d77c10b6344e
- Derivation seal: sha256 : 65fec14eb6a66e240b026fbc893525962285ed8e6dae678f503d42de9b2da781
- Independent implementation:
sha256 : 5c232f884af76629b5fe61f5338cdb2bd73cbb4a41d4b6890d01c357442e9d67
- Independent certificate:
sha256 : 72050853d92aed0efab89da8ae0097f4801c8609a7dcfc27edc23fc7b8b59b91
- External validation:
sha256 : 8a903c90d8dd50d73ddbba422eab2bfe78126a414ee94f93c2ab812b78d70b12
- Model-admitted engine receipt:
sha256 : bcf98ba0535b49f76382780ec0f1e22b9f184249bb3ccb596463f48d0532080c
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-DISCRETIZATION-001-bcf98ba0535b49f7.json

112. Derivation 105: Fold convergence-certificate law

Claim: SFT-COMP-SCI-CONVERGENCE-001

112.1 Scientific necessity and exact theorem

Fold convergence-certificate law is required to derive successor refinement preserving and closing error distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-DISCRETIZATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

112.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for successor refinement preserving and closing error distinctions.

All generated finite successor refinement preserving and closing error distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closure is depth-independent only when the successor certificate proves the registered descent property.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-re-certificate	base-successor-closure-re-certificate retains the complete registered closure condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

112.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

112.4 Operational laws and consequences

- Carrier law: refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

112.5 Depth-independent closure

Base: The structural One supplies one canonical successor refinement preserving and closing error distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated successor refinement preserving and closing error distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

112.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-CONVERGENCE-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-CONVERGENCE-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-CONVERGENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closure is depth-independent only when the successor certificate proves the registered descent property.	PASS

112.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: convergence, convergence proof.

Closure is depth-independent only when the successor certificate proves the registered descent property.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Closure is depth-independent only when the successor certificate proves the registered descent property.

112.8 Exact evidence identities

- Source manifest: sha256:119a67a6a0e8a1a61cf2acd01d62f4a442ffa9bcc4efd6a8bc93b1ff439a6d42
- Complete-census certificate:
sha256:a7f115d89adefcff54bae05486e6de093d175a54c2007af4913f5a12b5784d46
- Derivation seal: sha256:ed6247aa7d641db78d7d340bf3c5b3d14aa9fbab778506d4ba1aba8dd53d24ca
- Independent implementation:
sha256:a1a4741ce2299d736fe56a7891feadc4e14e17b620a17b2356a08d5e6bb89598
- Independent certificate:
sha256:75712ca92daab71c4ed80079d561ccaae73c8f058914420ba5740326759e65df
- External validation:
sha256:ce2f9d06a52c46ff06ffb04608e4f3146f2c152ca69836601707192be0df483a
- Model-admitted engine receipt:
sha256:a3ae509175198d515e8ae636ad4b669d942524409b96f97c4ce26a94a98c03b5
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-CONVERGENCE-001-a3ae509175198d51.json

113. Derivation 106: Fold scientific symbolic-computation law

Claim: SFT-COMP-SCI-SYMBOLIC-001

113.1 Scientific necessity and exact theorem

Fold scientific symbolic-computation law is required to derive exact model-expression transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-CONVERGENCE-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

113.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model syntax; exact rewrite rules; dimensional or type interfaces;

normalization; identity proof; evaluator correspondence; result trace.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for exact model-expression transformation.

All generated finite exact model-expression transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical interpretation remains outside formal symbolic closure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

113.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

113.4 Operational laws and consequences

- Carrier law: model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

113.5 Depth-independent closure

Base: The structural One supplies one canonical exact model-expression transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact model-expression transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

113.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-SYMBOLIC-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-SYMBOLIC-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-SYMBOLIC-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical interpretation remains outside formal symbolic closure.	PASS

113.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: symbolic scientific computing, computer algebra.

Physical interpretation remains outside formal symbolic closure.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical interpretation remains outside formal symbolic closure.

113.8 Exact evidence identities

- Source manifest: sha256:8f3afcab73070c32caa7f7a244e2d3381cfe0c4c10361d97d213528b7a67cfa9
- Complete-census certificate:
sha256:86c189a3ff38a24347c519207a6a20cdd6c35bed586bc3e768760803fea23466
- Derivation seal: sha256:ccdce6568fba17556bb00419d7c10978c1938ff74de2d20f214d126f993351ac
- Independent implementation:
sha256:a47d9f806660a188331934953c9585f113b3b60beeea1e994b4878c5172007fc
- Independent certificate:
sha256:f08802aacd47de59d3f84730e1e10748406d5e76f0e8b2d2ee36a2cd40dca71f
- External validation:
sha256:a1d97bead16859a3ac163ad75406876848eb51cdbc35ae06549bdfdf4827a341e
- Model-admitted engine receipt:
sha256:484049210c99c3cf945baa6cf8f64f14beb56db6f50fe8ede3d322dee49c41b8
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-SYMBOLIC-001-484049210c99c3cf.json

114. Derivation 107: Fold simulation law

Claim: SFT-COMP-SCI-SIMULATION-001

114.1 Scientific necessity and exact theorem

Fold simulation law is required to derive execution of a registered model relation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-SYMBOLIC-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

114.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for execution of a registered model relation.

All generated finite execution of a registered model relation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A simulation validates consequences of its model; it cannot establish the model as a natural law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

114.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

114.4 Operational laws and consequences

- Carrier law: model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

114.5 Depth-independent closure

Base: The structural One supplies one canonical execution of a registered model relation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated execution of a registered model relation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

114.6 Executed witnesses and adverse controls

- scientific_computation-witness-1 - scientific_computation exact operational witness 1 for SFT-COMP-SCI-SIMULATION-001; result: PASS.
- scientific_computation-witness-2 - scientific_computation exact operational witness 2 for SFT-COMP-SCI-SIMULATION-001; result: PASS.
- scientific_computation-witness-3 - scientific_computation exact operational witness 3 for SFT-COMP-SCI-SIMULATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A simulation validates consequences of its model; it cannot establish the model as a natural law.	PASS

114.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: simulation, computational model.

A simulation validates consequences of its model; it cannot establish the model as a natural law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A simulation validates consequences of its model; it cannot establish the model as a natural law.

114.8 Exact evidence identities

- Source manifest: sha256 : c8e683c02a890d93d37979d9946e396fba5e64c5789dbdd9d1f08c65f97c7e45
- Complete-census certificate:
sha256 : 433f57b352285b1a23fa05ce1bbfb89e9cc7a615c6e12252d08396d5d5e54c1e
- Derivation seal: sha256 : b04cfb756fbbb395c383e325234af2577760c79c32252a1f80f6ba23a85a6cee

- Independent implementation:
sha256:b93c4dfbe30b3dbd96c4865fc0ef9b9ad242cd0c0c0e1376087d48dbc6bb9712
- Independent certificate:
sha256:1d19485b5b1604d13a29197016920d51da71b365201df597af35d2305911649e
- External validation:
sha256:0f028f2611fc2b19447d3b2f57a2515829a4b95d8743d165903cb7ec50945962
- Model-admitted engine receipt:
sha256:b06dd069154f18ea84aadfe5c47b070e0ec017113e01945e55690303f274e293
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-SIMULATION-001-b06dd069154f18ea.json

115. Derivation 108: Fold computational-dynamics law

Claim: SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001

115.1 Scientific necessity and exact theorem

Fold computational-dynamics law is required to derive iterated exact finite state transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-SIMULATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

115.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for iterated exact finite state transformation.

All generated finite iterated exact finite state transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical time and continuous dynamics require later empirical correspondence.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

115.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

115.4 Operational laws and consequences

- Carrier law: state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

115.5 Depth-independent closure

Base: The structural One supplies one canonical iterated exact finite state transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated iterated exact finite state transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

115.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical time and continuous dynamics require later empirical correspondence.	PASS

115.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: computational dynamics, dynamical simulation.

Physical time and continuous dynamics require later empirical correspondence.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical time and continuous dynamics require later empirical correspondence.

115.8 Exact evidence identities

- Source manifest: sha256:fb8bacdb9b1b4ec965b144d3d65d72b86dd81919c721a370e797b55c99b236f1
- Complete-census certificate:
sha256:2a564ff077594a99ced8cdb01d265feb58a6b2af6e0c87afa27d231badda5904
- Derivation seal: sha256:9c0a4479e8a6808361dfac8eaa8d54c89067ff7aad1cf3294f8ff62c9e35ec9c
- Independent implementation:
sha256:e89c6e112f56650b644a04ff5b68e2ec9f48826e68f71975b1375d4beda1e1b3
- Independent certificate:
sha256:c10c1dfce8f8e6f23c60190fb1800adee24259093c8d851aaf5fc5cd9f0b4160
- External validation:
sha256:b4faf647cc85a39c4437bb902767c84e3556daf5bb17f7265166feb6b6881346
- Model-admitted engine receipt:
sha256:9f40f7a193ca6e09bbc51fd61d7a526dc81c9224a6797bac0c0c57e8005ae0de
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001-9f40f7a193ca6e09.json

116. Derivation 109: Fold inverse-problem law

Claim: SFT-COMP-SCI-INVERSE-PROBLEM-001

116.1 Scientific necessity and exact theorem

Fold inverse-problem law is required to derive reconstruction of sources from observations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

116.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for reconstruction of sources from observations.

All generated finite reconstruction of sources from observations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No fitted regularizer or prior may select among unresolved fibres unless separately registered.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-re-certificate	base-successor-closure-re-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains 256 candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are 1 survivors and 255 eliminations.

116.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.

Eliminated candidates	Exact first-failure reason
64	sampld-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

116.4 Operational laws and consequences

- Carrier law: source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

116.5 Depth-independent closure

Base: The structural One supplies one canonical reconstruction of sources from observations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated reconstruction of sources from observations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

116.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No fitted regularizer or prior may select among unresolved fibres unless separately registered.	PASS

116.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: inverse problem, identifiability.

No fitted regularizer or prior may select among unresolved fibres unless separately registered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No fitted regularizer or prior may select among unresolved fibres unless separately registered.

116.8 Exact evidence identities

- Source manifest: sha256:ebbe50d370311564302b91465ef3dcd85324bc6b44c2e4865abf130155e60ad0
- Complete-census certificate:
sha256:69c171260b2d27e2ccd68e8868e52e07e63886eead5baf1257ac06a080cd05ed
- Derivation seal: sha256:9aedf19b7d236a4710da983fbd678c9f179bccb617188f080fa9540c02385310
- Independent implementation:
sha256:631220d2e521292b1f4000a76e228380d7f613100c10076d8ebf18a09aac1162
- Independent certificate:
sha256:6ac9ca23bb7c37bb14f2795f7e768b25108c651af84a744950cdccf4c7b67ee0
- External validation:
sha256:7c54dcae327e865ff3d257363c532cb2016fdbf4cdf42f6b21375c9bd8c3112d
- Model-admitted engine receipt:
sha256:41ca83901e2ba93484c943df39462f4196c046678f4d9cb17afafb5277cacdde
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-INVERSE-PROBLEM-001-41ca83901e2ba934.json

117. Derivation 110: Fold computational-statistics law

Claim: SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001

117.1 Scientific necessity and exact theorem

Fold computational-statistics law is required to derive exact support summaries and inference algorithms without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-INVERSE-PROBLEM-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

117.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for exact support summaries and inference algorithms.

All generated finite exact support summaries and inference algorithms structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Probability remains deterministic-support bookkeeping unless an external distribution is measured.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-re-certificate	base-successor-closure-re-certificate retains the complete registered closure condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

117.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampld-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

117.4 Operational laws and consequences

- Carrier law: complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

117.5 Depth-independent closure

Base: The structural One supplies one canonical exact support summaries and inference algorithms form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact support summaries and inference algorithms form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

117.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Probability remains deterministic-support bookkeeping unless an external distribution is measured.	PASS

117.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: computational statistics, statistical computing.

Probability remains deterministic-support bookkeeping unless an external distribution is measured.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Probability remains deterministic-support bookkeeping unless an external distribution is measured.

117.8 Exact evidence identities

- Source manifest: sha256:89565510d8bf65cc4168e57422355f8a2cfeccdc09807867ac85716e37f2a1cb
- Complete-census certificate:
sha256:a57c05e5723cfa9ddafce937b5a6c078fe7ada36e42f5a62ed58b3019d506b6a
- Derivation seal: sha256:acc45ba06d6b9dfffc0e01169f023269669bdfc1c720e15db7c5feb630c26096
- Independent implementation:
sha256:c6bb9adaf5e746b4ca7bed2d6a76b2f0183231070804981ed063b9f5eb8fa179
- Independent certificate:
sha256:bdf581ea46da78d86afa92203678ff3422cc8bc86070705f8cd5e304c31c275a
- External validation:
sha256:641f049a78a531c8dd32e5f49d3e31ed2788db1ba097dbbf3d0fd941eb2dc193
- Model-admitted engine receipt:
sha256:af3c12e592f489613af6f00e0bed93b948f0ccc0a7f975ec89a55fce888ceffa
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001-af3c12e592f48961.json

118. Derivation 111: Fold high-dimensional computation law

Claim: SFT-COMP-SCI-HIGH-DIMENSIONAL-001

118.1 Scientific necessity and exact theorem

Fold high-dimensional computation law is required to derive product-support computation with retained coordinate structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

118.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated coordinate carriers; complete product support; sparse or

factored relation; algorithm trace; resource ledger; lost-interaction boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for product-support computation with retained coordinate structure.

All generated finite product-support computation with retained coordinate structure structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Dimension is a generated coordinate trace, not a completed infinite vector space.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

118.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

118.4 Operational laws and consequences

- Carrier law: generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

118.5 Depth-independent closure

Base: The structural One supplies one canonical product-support computation with retained coordinate structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated product-support computation with retained coordinate structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

118.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Dimension is a generated coordinate trace, not a completed infinite vector space.	PASS

118.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: high-dimensional computing, curse of dimensionality.

Dimension is a generated coordinate trace, not a completed infinite vector space.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Dimension is a generated coordinate trace, not a completed infinite vector space.

118.8 Exact evidence identities

- Source manifest: sha256:e515d03a970bf5b51a7661b975738c624a4fc11224ae3597d6991b19c55eece3
- Complete-census certificate:
sha256:8084ac3a48b17cb0a6a629d032ca3e3ba98dc2e4738b7ee73b9d6cec56af8921
- Derivation seal: sha256:ef184706f84eb5d737e955ed436ae8913d9b5cf2db1cafa541a94c0cc4165f79
- Independent implementation:
sha256:66bba91ba3902bc2caf8dac70ef204e8cbad12d8f0f59634772f6f4f34d4cc04
- Independent certificate:
sha256:db5a4afe687da48685c9b4f295bf78d0c71745c3db90405d81abab15632101f0
- External validation:
sha256:4c20be1f568135bdfa36bb34a1ca13ebee60bb55f9d19c208a26bf65cb119b06
- Model-admitted engine receipt:
sha256:08b4870cfa97a44e956720e0bf6ccfef2216790ad6ffcc0261c127f46625f84d
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-HIGH-DIMENSIONAL-001-08b4870cfa97a44e.json

119. Derivation 112: Fold many-body computation law

Claim: SFT-COMP-SCI-MANY-BODY-001

119.1 Scientific necessity and exact theorem

Fold many-body computation law is required to derive composed local forms and interaction support without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-HIGH-DIMENSIONAL-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

119.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for composed local forms and interaction support.

All generated finite composed local forms and interaction support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical many-body law or quantum amplitude is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

119.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

119.4 Operational laws and consequences

- Carrier law: body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

119.5 Depth-independent closure

Base: The structural One supplies one canonical composed local forms and interaction support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated composed local forms and interaction support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

119.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-MANY-BODY-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-MANY-BODY-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-MANY-BODY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical many-body law or quantum amplitude is imported.	PASS

119.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: many-body computation, interaction system.

No physical many-body law or quantum amplitude is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No physical many-body law or quantum amplitude is imported.

119.8 Exact evidence identities

- Source manifest: `sha256:1836ffc81b8d6aedb59f67ef78d14e2a426b903858e85b7e47472d933b2c1e94`
- Complete-census certificate:
`sha256:e8b0d3c3872d072d24e85200a62182589347d59df0a5cef00b81711e56d64c7e`
- Derivation seal: `sha256:9150974b48199453faca9a3ec42c5774fd3ea6dbd947b91fe3fcbb4a634421ab`

- Independent implementation:
sha256:1d2047ea1a1f1b8bf3b16ff8e2608a6d87b0df913c0a975134ffca4e7b842534
- Independent certificate:
sha256:b478fd50145544b87303e4c72b0ed5f442a50fca78cdebbdf0fa26609d28b8d8
- External validation:
sha256:3c8b8b90e8e91e724142f784b5c7d34340401fab6d2fabcdc2b6c40547f84649
- Model-admitted engine receipt:
sha256:4e4db714accde19f571063b6d1c1c64c9f07650378a797107cb424bea54de4ca
- Receipt path:
receipts/engine/model_admitted/SFT-COMP-SCI-MANY-BODY-001-4e4db714accde19f.json

120. Derivation 113: Fold mathematical-modelling law

Claim: SFT-COMP-SCI-MATHEMATICAL-MODELLING-001

120.1 Scientific necessity and exact theorem

Fold mathematical-modelling law is required to derive explicit relation between formal structures and observations without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-MANY-BODY-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

120.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for explicit relation between formal structures and observations.

All generated finite explicit relation between formal structures and observations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A successful fit or opaque prediction does not force the model law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
model	unstated-model	unstated-model removes a required exact model condition.	registered-exact-model-relation	registered-exact-model-relation retains the complete registered model condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
inputs	sampled-or-fitted-inputs	sampled-or-fitted-inputs removes a required exact inputs condition.	complete-declared-input-support	complete-declared-input-support retains the complete registered inputs condition.
operation	opaque-calculation	opaque-calculation removes a required exact operation condition.	exact-executable-operation-trace	exact-executable-operation-trace retains the complete registered operation condition.
error	unsigned-error-scalar	unsigned-error-scalar removes a required exact error condition.	orientation-labelled-error-ledger	orientation-labelled-error-ledger retains the complete registered error condition.
closure	visual-convergence	visual-convergence removes a required exact closure condition.	base-successor-closure-certificate	base-successor-closure-certificate retains the complete registered closure condition.
validation	fit-only	fit-only removes a required exact validation condition.	sealed-falsifiable-comparison-boundary	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
generality	single-resolution	single-resolution removes a required exact generality condition.	generated-refinement-successor	generated-refinement-successor retains the complete registered generality condition.
addition	imported-equation-or-parameter	imported-equation-or-parameter removes a required exact addition condition.	no-extra-scientific-prior	no-extra-scientific-prior retains the complete registered addition condition.

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

120.3 Exhaustion, eliminations and unique survivor

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-t-successor__no-extra-scientific-prior`.

The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

120.4 Operational laws and consequences

- Carrier law: phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

120.5 Depth-independent closure

Base: The structural One supplies one canonical explicit relation between formal structures and observations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated explicit relation between formal structures and observations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

120.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - `scientific_computation` exact operational witness 1 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: PASS.
- `scientific_computation-witness-2` - `scientific_computation` exact operational witness 2 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: PASS.
- `scientific_computation-witness-3` - `scientific_computation` exact operational witness 3 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	PASS
tampered_source	reject a changed official source identity	changed source identity differs	PASS
tampered_artifact	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A successful fit or opaque prediction does not force the model law.	PASS

120.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: mathematical modelling, scientific model.

A successful fit or opaque prediction does not force the model law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A successful fit or opaque prediction does not force the model law.

120.8 Exact evidence identities

- Source manifest: sha256:e2da80d2e34c9e77c9c70e6fa18a7209bc39d59b1b69c4d7ab4f22c142f69eb4
- Complete-census certificate:
sha256:c3bab5094675d4dc9b452804f521d6e149289070180ba9353b2590e635621d90
- Derivation seal: sha256:bf1cfe19e9f1107fc446eeb94708330d2a1f69e6593d9d1a18fcb2ef7938d7ec
- Independent implementation:
sha256:5be00110e135b7023427d7d71b21f44efbe3ace7848a73e02c41a286eb0a78de
- Independent certificate:
sha256:5599d7b3549eba1bb43cd7545c725f9fd7ddabf910f916c3ca1aeb0dc254e621
- External validation:
sha256:852cfe82540a75b5e540d04049e107608bf2ddc7c663eca6626a69868ae516aa
- Model-admitted engine receipt:
sha256:465c067c8c543effc5ba03cf8677a5caa2386f4695d1f3905b6e652060ba7ed4
- Receipt path: receipts/engine/model_admitted/SFT-COMP-SCI-MATHEMATICAL-MODELLING-001-465c067c8c543eff.json

121. Branch synthesis and consequences

The 113 laws form one dependency chain from exact state change to models, computability, resources, algorithms, meaning, distributed knowledge, adversaries, learning and scientific calculation. The common invariant is complete provenance: every input belongs to a generated support, every operation names its relation, every execution retains a trace, every resource count names the representation, every loss names closed distinctions and every claimed generality supplies a successor certificate or an explicit finite boundary.

122. Reproduction, criticism and falsification

Run `python3 -m sft verify-all` on macOS/Linux or `py -m sft verify-all` on Windows. Reviewers may invalidate a claim by producing an omitted coordinate inside its declared grammar, a second survivor, a simpler preserving form, an induction counterexample, an operational counterexample, a source mismatch or a failed adverse control. Hashes identify artifacts; they do not immunize the grammar from criticism.

123. Scope and next branch

The next dependency branch is Reversible and Quantum Computation. It may use these classical receipts but must independently derive branch support, phase, interference, entanglement, measurement, gates, quantum circuits, algorithms, communication, correction and limits without importing a conventional quantum formalism.

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